BIOLOGY GRADE 10: CELL STRUCTURE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.1 (12 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Biology
Strand	Strand 1.0: Cell Biology and Biodiversity
Sub-Strand	Sub-Strand 1.3: Cell Structure and Specialisation
Total Duration	12 lessons × 40 minutes = 480 minutes total
Sub-Strand Content	– Cell as the basic unit of life – Types of cells: prokaryotic and eukaryotic cells – Structure and ultrastructure of the plant cell (cell wall, cell membrane, nucleus, chloroplast, vacuole, mitochondria, endoplasmic reticulum, ribosomes, Golgi apparatus) – Structure and ultrastructure of the animal cell (cell membrane, nucleus, mitochondria, endoplasmic reticulum, ribosomes, Golgi apparatus, lysosomes, centrioles) – Comparison of plant and animal cells (similarities and differences) – Comparison of prokaryotic and eukaryotic cells – Cell specialisation: meaning and significance – Specialised animal cells (red blood cells, white blood cells, sperm cells, egg cells, nerve cells, muscle cells, ciliated epithelial cells) – Specialised plant cells (root hair cells, guard cells, xylem vessels, phloem sieve tubes, palisade mesophyll cells) – Relationship between cell structure and function – Cell organisation: cells → tissues → organs → organ systems → organism – Preparation and use of temporary slides to observe cell structure under a light microscope
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) describe the structure of prokaryotic and eukaryotic cells, b) distinguish between plant and animal cells based on their structures, c) explain the relationship between the structure and function of specialised cells, d) prepare temporary slides and observe cell structures using a light microscope, e) appreciate the importance of cell specialisation in the functioning of living organisms.
Core Competencies	– Communication and Collaboration: The learner works in groups to prepare temporary slides, share microscope observations, and present comparisons of cell types to peers using labelled diagrams and discussion. – Critical Thinking and Problem Solving: The learner analyses the structural features of specialised cells and constructs logical arguments linking organelle structure to cell function, e.g. why a red blood cell lacks a nucleus. – Imagination and Creativity: The learner creates annotated models or diagrams of plant and animal cells using locally available materials (e.g. clay, fruit, bottle caps) to represent organelles and their functions.

	– Self-efficacy: The learner develops confidence in laboratory skills by independently preparing and observing temporary slides of onion epidermal cells and cheek cells, and recording accurate observations.
Core Values	– Responsibility: The learner handles microscopes and laboratory materials with care, and clears and cleans the workspace after practical activities. – Respect: The learner appreciates diverse contributions from group members during collaborative model-building and peer presentations on cell structure. – Unity: The learner works harmoniously with peers from different backgrounds during group practicals, recognising that diverse perspectives strengthen sensemaking about cell organisation. – Social Justice: The learner acknowledges that all body cells, though structurally different, contribute equally to the survival of the organism — drawing a parallel to the equal value of all individuals in society.
Science & Engineering Practices	– Developing and Using Models (SEP 2): The learner constructs and progressively revises a 2D or 3D model of a generalised cell, adding organelles as new learning accumulates across lessons, and uses the model to explain the anchoring phenomenon of wound healing. – Planning and Carrying Out Investigations (SEP 3): The learner plans and conducts a microscopy practical to prepare temporary slides of onion epidermal cells and human cheek cells, recording observations systematically with labelled drawings. – Analysing and Interpreting Data (SEP 4): The learner analyses microscope images and diagrams to compare structural features of prokaryotic vs. eukaryotic cells and plant vs. animal cells, identifying patterns and anomalies. – Constructing Explanations (SEP 6): The learner constructs written and oral explanations linking the specialised structure of cells (e.g. biconcave disc shape of red blood cells, long axon of nerve cells) to their specific functions in the context of wound healing. – Obtaining, Evaluating, and Communicating Information (SEP 8): The learner researches and critically evaluates information from textbooks, print media, and (where available) internet sources about cell structure, then communicates findings through labelled diagrams, presentations, and Driving Question Board updates.
Pertinent & Contemporary Issues (PCIs)	– Health Education: The learner connects cell specialisation to human health by exploring how specific cells (white blood cells, platelets, skin cells) coordinate during wound healing and immune response — relating this to hygiene practices and first-aid awareness relevant to Kenyan households and schools. – Environmental Conservation: The learner uses locally and sustainably sourced materials (onion skins from Wakulima Market, cheek swabs, leaf sections from school gardens) for microscopy practicals, minimising waste and appreciating the value of local biological resources. – Safety and Security: The learner observes proper laboratory safety protocols when handling glass slides, coverslips, iodine stain, and methylene blue during cell preparation practicals. – Life Skills Education: The learner reflects on the concept of specialisation and interdependence of cells as a metaphor for teamwork and individual roles within families, schools, and communities in Kenya.
Career Connections	– Medicine and Surgery: Understanding cell structure and specialisation is foundational for learners pursuing medicine, nursing, or surgery at institutions such as the University of Nairobi School of Medicine or Kenyatta National Hospital. – Biomedical Laboratory Science: Learners who study cell morphology and microscopy skills are prepared for careers as medical laboratory technologists who examine blood cells and tissue samples. – Biotechnology and Genetic Engineering: Knowledge of eukaryotic cell ultrastructure underpins careers in Kenya's growing biotechnology sector, including work at KEMRI (Kenya Medical Research Institute) and KARI (Kenya Agricultural Research Institute). – Teaching and Education: Learners may pursue careers as Biology teachers or university lecturers, transmitting cell biology

	knowledge in Kenyan secondary schools and tertiary institutions. – Agriculture and Agronomy: Understanding plant cell structure (guard cells, xylem, phloem) supports careers in crop science, horticulture, and plant breeding — critical for food security in Kenya's agricultural economy. – Environmental Science: Knowledge of cell biology supports careers in ecology, conservation biology, and wildlife research relevant to Kenya's national parks and biodiversity hotspots.
Focus for Lessons	This unit anchors the study of cell structure and specialisation in the concrete, puzzling phenomenon of wound healing — a familiar experience for every Kenyan learner — to drive genuine curiosity about why the body deploys different cell types with radically different shapes and functions. Over 12 lessons, learners progress from macroscopic observation of a healing cut to microscopic investigation of cell organelles, building an increasingly detailed and accurate model of both plant and animal cells. The unit foregrounds laboratory skills (microscopy, slide preparation), collaborative model-building, and the disciplinary practice of constructing explanations that link form to function, culminating in a comprehensive understanding of how cell specialisation enables multicellular life.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: "Why do cells in our body look so different from each other — and how does the shape of a cell determine the job it can do?" KICD KEY INQUIRY QUESTIONS: 1. What are the structural differences between plant and animal cells? 2. How does cell specialisation enable organisms to carry out complex life functions?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Healing Cut: A Form 1 student in Nairobi cuts their finger while slicing onions for supper. Within seconds, bleeding slows and a dark clot begins to form. Over the next few days, the wound seals, the scab falls away, and new pink skin appears — the finger looks almost exactly as it did before. This sequence of events is deeply familiar yet profoundly puzzling: How does the body "know" what to do? How does it send exactly the right response — stopping bleeding, fighting infection, and rebuilding skin — all at once? Closer examination reveals that this is not one process but many, each carried out by a different type of cell: platelets aggregate and release clotting proteins, neutrophils and macrophages engulf bacteria, fibroblasts secrete collagen to rebuild tissue, and epidermal cells divide and migrate to close the surface. Each of these cells looks completely different under a microscope — some are round, some elongated, some have no nucleus at all — yet all of them originate from the same fundamental unit of life. The puzzle that launches this unit: If all cells come from the same organism and carry the same DNA, why do they look so different, and how does their shape make them so perfectly suited to their specific job?
Storyline Thread	Lesson 1 — Phenomenon Launch & Prior Knowledge Elicitation: Students observe a short narrated photo sequence of a healing cut (or teacher's live description of the Nairobi onion-cutting scenario). In a Think-Pair-Share, learners predict: "What do you think is happening inside the skin right now?" Students record initial ideas in their science notebooks. Teacher introduces the Driving Question Board (DQB) — a large class poster — and students post their first "What do we notice?" and "What do we wonder?" sticky notes. Teacher introduces the driving question. No answers given yet. Lesson 2 — What Is a Cell? Introduction to Cell Theory: Learners search for and discuss information on the cell as the basic unit of life. Teacher uses the analogy of a ugali-making kitchen in Nairobi: every kitchen (cell) has the same basic tools but is set up differently for different tasks. Students are introduced to cell theory (Schleiden, Schwann, Virchow) and the concept of prokaryotic vs. eukaryotic cells using labelled diagrams. Learners sketch and label a generalised cell — this becomes Version 1 of their cumulative cell model. Lesson 3 — Prokaryotic vs. Eukaryotic Cells: Learners analyse and compare diagrams of a bacterium (prokaryote, e.g. E. coli

	causing food poisoning in poorly stored nyama choma) and a human cheek cell (eukaryote). Using a structured comparison table, learners identify: presence/absence of nucleus, membrane-bound organelles, size, and cell wall composition. Cold-call questioning (minimum 5 students) consolidates understanding. DQB updated: "We now know prokaryotes and eukaryotes differ in… but we still wonder…" Lesson 4 — Plant Cell Ultrastructure (Teacher-Led with Diagram Analysis): Using large labelled diagrams and (where available) projected electron micrograph images, learners identify and describe the function of plant cell organelles: cell wall, cell membrane, nucleus, chloroplasts, large central vacuole, mitochondria, endoplasmic reticulum, ribosomes, Golgi apparatus. Teacher uses the metaphor of a maize plant from the Rift Valley farm: every organelle has a specific farming job. Learners add plant cell organelles to their cumulative model (Version 2 — Plant Cell). Lesson 5 — Animal Cell Ultrastructure (Teacher-Led with Diagram Analysis): Learners compare the animal cell structure with the plant cell from Lesson 4. They identify organelles present in animal cells but not plant cells (lysosomes, centrioles) and those present in plant cells but not animal cells (chloroplasts, cell wall, large vacuole). Learners update their cumulative model to show both cell types side by side (Version 3 — Plant vs. Animal). Group presentations of annotated diagrams with peer feedback. Lesson 6 — Microscopy Practical 1 (Plant Cells — Onion Epidermal Cells): Learners prepare temporary slides of onion epidermal cells using iodine stain. Teacher demonstrates technique first, then groups of 3–4 carry out the practical. Learners draw and label observed cells at low and high power, noting cell wall, vacuole, and nucleus. Observation: "Why are there no chloroplasts in onion skin cells?" — linked back to the supporting phenomenon. Safety protocols enforced throughout. Results recorded in structured practical books. Lesson 7 — Microscopy Practical 2 (Animal Cells — Human Cheek Cells): Learners prepare temporary slides of cheek (buccal) cells using methylene blue stain. Learners draw and label observed cells, noting cell membrane, cytoplasm, and nucleus. Direct comparison with onion cell drawings from Lesson 6. Learners articulate at least two structural differences they can directly observe. DQB updated with new evidence: "We observed that animal cells have no cell wall — this tells us…" Lesson 8 — Cell Specialisation: Meaning, Significance, and Specialised Animal Cells: Teacher re-anchors to the healing-cut phenomenon: "We now know cells have organelles — but why do blood cells look so different from skin cells?" Learners explore specialised animal cells: red blood cells (biconcave, no nucleus — why?), white blood cells (irregular shape, many lysosomes — why?), nerve cells (long axon), sperm cells (flagellum, many mitochondria), muscle cells (many mitochondria, myofilaments). Each cell is connected back to its role in wound healing. Learners use the sickle-cell supporting phenomenon to consolidate structure-function links. Lesson 9 — Specialised Plant Cells: Learners examine specialised plant cells: root hair cells (large surface area for absorption), guard cells (control stomatal opening — relevant to water loss in dry Turkana vs. humid Kisumu), xylem vessels (hollow, lignified — for water transport), phloem sieve tubes (perforated end plates — for sugar transport), palisade mesophyll cells (packed with chloroplasts — for photosynthesis). Learners connect each cell to the bean seedling supporting phenomenon. Structured group activity: match-the-card game linking cell diagrams to functions. Lesson 10 — Cell Organisation: Cells → Tissues → Organs → Organ Systems → Organism: Learners trace the organisational hierarchy using the wound-healing context: skin cells form epithelial tissue → tissue forms the skin organ → skin is part of the integumentary system → the system protects the whole organism. A parallel is drawn to a Kenyan secondary school: individual students (cells) → class (tissue) → school (organ) → county education system (organ system) → Kenyan society (organism).

	Learners construct a hierarchy diagram and update their cumulative model to show cell organisation (Version 4). Lesson 11 — Synthesis and Model Revision: Learners return to their original predictions from Lesson 1 and compare them with current understanding. In groups, learners produce a final annotated poster-model showing: (a) a labelled plant cell, (b) a labelled animal cell, (c) three specialised cells from the wound-healing phenomenon with structure-function annotations, and (d) the cell organisation hierarchy. Groups present posters; peers use a structured feedback protocol (1 glow, 1 grow). DQB is reviewed — all original "wonder" questions are addressed or re-categorised as "questions for future learning." Lesson 12 — Consolidation, DQB Closure, and Reflection: Teacher facilitates a whole-class return to the anchoring phenomenon: "We started with a puzzling cut on a finger. Using everything we now know about cell structure and specialisation, let's fully explain what happens during wound healing — cell by cell." Students contribute orally in a structured round (cold-call, minimum 8 students). Learners write individual exit reflections: "One thing I now understand that I did not understand in Lesson 1 is… because…" Final DQB update: questions are marked ANSWERED, PARTIALLY ANSWERED, or BEYOND THIS UNIT. Sub-strand assessment administered.
Supporting Phenomena	– The Sickle-Cell Mosquito Net: In western Kenya near Lake Victoria, some children inherit sickle-cell anaemia — their red blood cells are curved like a sickle rather than the normal biconcave disc. These children suffer painful crises because the abnormal shape prevents efficient oxygen transport. This phenomenon supports the concept that a tiny change in cell shape has dramatic consequences for function. – The Onion Skin Under the Microscope: When a thin layer of onion epidermal cells from a Nakuru market onion is placed under a light microscope, learners see a neat brick-like grid of cells, each with a visible cell wall, cell membrane, and large central vacuole — but no chloroplasts. This supports the comparison of plant vs. animal cells and raises the question: why do onion cells lack chloroplasts if they are plant cells? – The Bean Seedling in the Dark: A bean seed (njugu mawe) germinated in a dark box in Kisumu produces a tall, pale yellow seedling with no green colour. When moved into sunlight, it turns green within days. This supports the concept of chloroplast development and the relationship between organelle function and environmental conditions. – The Amoeba and the Nerve Cell: A projected image or diagram comparing a single-celled amoeba (a prokaryote-like free-living cell) with a human motor nerve cell — one is a blob that does everything alone, the other is a metre-long, highly specialised cell that only transmits signals — illustrates the extreme range of cell specialisation and introduces the prokaryote vs. eukaryote distinction.

LESSON 1 (40 minutes): The Healing Cut: Launching Our Investigation Into Why Cells Look Different	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon and surface learners prior ideas about cells, setting up the driving question that will guide the entire sub-strand
Knowledge	Learners will recall that the human body responds to injury in multiple ways simultaneously, and will begin to articulate that different responses may involve different types of cells
Skills	Learners will practise scientific observation, prediction, and recording of initial ideas in their science notebooks using Think-Pair-Share and sticky-note protocols
Attitudes	Learners will demonstrate curiosity and openness by sharing tentative ideas without fear of being wrong, and will value the contributions of peers during collaborative discussion
Key Inquiry Question	How does cell specialisation enable organisms to carry out complex life functions?
Purpose in Storyline	This is the opening lesson that introduces the anchoring phenomenon — a healing cut on a finger — and elicits prior knowledge. No content is delivered yet. The lesson exists to create intellectual need: learners feel puzzled, post genuine questions, and are motivated to investigate. The Driving Question Board is established and will grow across all 12 lessons.
Safety Notes	No hazardous materials are used in this lesson. If a teacher uses a real photograph of a wound, ensure images are not graphic or distressing. Remind learners that anyone who feels uncomfortable viewing injury images may look away and the teacher will describe the image verbally instead.

B. LESSON OVERVIEW

The teacher narrates the Nairobi onion-cutting scenario vividly, then guides learners through a structured Think-Pair-Share to surface prior ideas about what happens inside the body during wound healing. Learners record predictions in science notebooks, then post sticky notes on the class Driving Question Board under two columns: What do we notice? and What do we wonder? The teacher closes by formally introducing the driving question. No answers are provided — the lesson ends in productive uncertainty.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [The force of tension] Source: Khan Academy (English - US curriculum)	Learners listen with eyes closed as the teacher narrates the scenario of a Form 1 student in Nairobi — Amina — who is helping her mother prepare supper and cuts her finger while slicing onions. The teacher describes the immediate	Teacher narrates slowly and with deliberate detail: Amina is in the kitchen in Eastleigh. She is slicing onions — the smell is sharp, her eyes are beginning to water. Then the knife slips. She looks down and sees a thin line of red appear	Independent written prediction activates prior knowledge from primary science and everyday lived experience. Writing before talking prevents dominant voices from shaping peer responses and ensures every learner has a	Teacher circulates during writing time and reads over shoulders without commenting, mentally noting the range of ideas present — for example, whether learners mention blood clotting, skin repair, bacteria, specific cells, or only

🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	events: the sting, the appearance of blood, and then the slowing of bleeding. Learners are then asked to open their science notebooks and write for 2 minutes in response to the prompt: What do you think is happening inside Amina finger right now? Write anything that comes to mind — there are no wrong answers today. Learners write independently without consulting neighbours.	on her fingertip. She presses it with her thumb. After about a minute, the bleeding is already slowing down. A dark, sticky material begins to form. What is happening inside that finger right now? After narrating, teacher says: I want everyone to write in your notebook. Do not talk yet. Do not look at your neighbour. Just write what you think. You have 2 full minutes. WAIT TIME: Teacher sets a visible timer for 2 minutes and remains silent, walking slowly between rows to signal that thinking time is genuine and protected. Teacher does not prompt or hint during this period.	personal record of their starting point, which will be compared with final understanding in Lesson 12.	describe the wound in surface terms. This informs how the teacher frames the Think-Pair-Share debrief.
Observe Phase 📹 **VIDEO:** The force of tension Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners participate in a Think-Pair-Share. First, pairs share their notebook predictions with each other for 90 seconds. Then the teacher displays or describes a four-image narrated photo sequence on the board — or narrates it verbally if projection is unavailable — showing: (1) a freshly cut finger with blood, (2) a dark clot forming, (3) a scab on healing skin after two days, and (4) new pink skin with the scab gone. For each image the teacher pauses and asks learners to observe carefully and add anything new to their notebook predictions. After the sequence, three to four volunteers share their pair discussion with the class. Teacher records key phrases on the board verbatim without correcting or affirming them yet — for example: the blood goes hard, white cells come, skin grows back, something inside stops the bleeding.	During pair sharing, teacher circulates and listens for interesting or surprising ideas to highlight later. When displaying or describing each image, teacher uses language such as: Look carefully. What do you notice? What changed between image one and image two? After pairs share with the class, teacher scribes on the board and says: I am writing your ideas exactly as you said them. I am not telling you yet whether they are right or not. Every idea here is valuable data — it tells us what we already know and what we still need to find out. Teacher deliberately calls on learners who are not usually first to raise their hand. WAIT TIME: After asking What do you notice? teacher waits a full 5 seconds before taking responses, signalling that observation requires thought.	The four-image sequence (or vivid verbal narration) grounds the lesson in a concrete, observable event rather than abstract content. Scribing learner language without correction models scientific practice — all observations are recorded before interpretation begins. This also builds psychological safety for sharing incomplete ideas.	The board list of learner ideas serves as a public prior-knowledge map. Teacher checks whether any learner has spontaneously mentioned cell types (platelets, white blood cells, skin cells) — if so, this is noted as a bridge for Lesson 8. The absence of cell-level thinking confirms the need for the full unit sequence.
Explain Phase 📹 **VIDEO:**	Teacher leads a brief whole-class discussion using three targeted	Teacher uses Socratic questioning — responding to learner answers	The three-question sequence is designed to move learners from	Teacher listens for whether learners can distinguish between a

[The force of tension](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	questions to help learners begin to articulate the complexity of what they have observed. The questions are asked one at a time with think time between each. Learners respond orally and may also add to their notebook. Question 1: You said the bleeding slows — what do you think causes that? Question 2: You mentioned the skin grows back — do you think one thing does that job, or many things? Question 3: If many different things are doing different jobs inside the wound, what might those things look like — are they all the same or different? After responses, the teacher introduces a bridging statement — not an answer, but a reframe: Scientists who looked at healing wounds under a microscope found something remarkable. They found that dozens of different types of tiny structures were all working at the same time — each doing a completely different job, and each looking completely different from the others. That raises a very big question, which we are going to investigate for the next several weeks.	with follow-up probes rather than evaluation. For example, if a learner says the blood goes hard, teacher responds: Interesting — what do you think makes it go hard? Is that a chemical reaction? A physical change? Something else? Teacher does not confirm or deny. For Question 3, teacher says: Think about this for 10 seconds before you answer. WAIT TIME: A full 10 seconds of silence is enforced after Question 3 because it is the most cognitively demanding. Teacher counts silently and resists the urge to rephrase or prompt. After the bridging statement, teacher pauses and asks: Does that surprise you? Why? allowing 2 or 3 spontaneous responses.	surface description (bleeding slows) to causal reasoning (what causes it) to systems thinking (multiple components doing different jobs). The bridging statement creates cognitive dissonance — if many different cell types are involved, the question of why they look different becomes genuinely interesting rather than merely academic.	single-cause explanation and a multi-component explanation. Learners who describe multiple simultaneous processes without being prompted are showing readiness for the cell specialisation concept. Learners who provide only single-cause answers will need more scaffolding in Lessons 8 and 9.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [The force of tension](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)](#)	Each learner receives two sticky notes of different colours — for example, yellow for What do we notice? and blue for What do we wonder? Learners write one observation on the yellow note and one genuine question on the blue note. They are given 3 minutes to write both. Learners then come forward in groups of six to eight and place their sticky notes on the large class poster (the DQB) under the two column headings. The teacher reads aloud a selection of the wonder questions and groups	Before distributing sticky notes, teacher models the task using a think-aloud: I notice in the images that the wound looks completely healed after just a few days — that is my notice note. I wonder how the body knows exactly what shape to rebuild the skin in — that is my wonder note. Teacher writes these on the board as examples. Teacher emphasises: Your wonder note should be a real question you actually have — not a question you think I want to hear. After learners post notes, teacher does	The two-colour sticky-note protocol separates observation (what we notice) from inquiry (what we wonder), teaching a fundamental scientific distinction. Public posting of questions democratises knowledge construction — every learner contributes to the shared class investigation. The driving question is introduced after learner questions are posted, so learners can see that their own questions connect to it.	Teacher photographs the DQB at the end of the lesson for reference. The wonder notes are analysed to identify: (a) questions that will be answered within this unit, (b) questions that reveal misconceptions to address, and (c) questions that show sophisticated thinking. This analysis shapes the emphasis in Lessons 2 through 5.

Source: CK-12 🔍 Search ARES for similar readings	similar ones together on the board, narrating: I can see several of you are wondering why the skin grows back so perfectly. Others are wondering what is actually inside blood. These are exactly the kinds of questions scientists ask. Then the teacher formally introduces the driving question by reading it aloud from the top of the DQB poster: Why do cells in our body look so different from each other — and how does the shape of a cell determine the job it can do? Teacher explains that this question will be answered piece by piece across the unit and that the DQB will be updated at the end of every lesson.	not evaluate or rank the questions. Instead, teacher says: Every single question on this board is legitimate. Some we will answer in this unit. Some might take years of science to answer. All of them matter. WAIT TIME: After reading the driving question aloud, teacher waits 8 seconds in silence before asking: What is your very first reaction to this question? allowing 2 to 3 brief spontaneous responses before closing the phase.		
Model Building Phase 📹 **VIDEO:** The force of tension Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are told that throughout this unit they will build a cumulative cell model — a diagram that grows and improves as they learn more. In this first lesson, Version 0 of the model is a simple drawn response to a single prompt: Draw what you think a cell looks like. Learners spend 3 minutes drawing in their science notebooks, labelling anything they think might be inside a cell. There is no right answer at this stage. Learners are reminded that this drawing will be compared with their Version 4 model in Lesson 11, and that seeing how their model changes is part of the learning. Teacher closes the lesson by returning to Amina: In Lesson 2, we will begin to look at what a cell actually is and why cell theory is one of the most important ideas in all of biology. But today, you have already done something scientists do every single time they start an investigation — you have observed carefully, asked good	Teacher distributes no diagrams or reference materials during the drawing task — learners must draw from memory and imagination only. Teacher says: This is not an art competition and it is not a test. It is a snapshot of what you already know. Draw whatever comes to mind — a box, a circle, dots inside, labels — anything at all. While learners draw, teacher circulates and asks individual learners: What made you draw it that way? or What is this part doing? These brief conversations surface reasoning without altering the drawing. Teacher does not correct any drawing at this stage. WAIT TIME: Teacher gives a genuine 3 minutes for drawing — not less — and uses a visible timer. At closure, teacher says the sentence about Amina slowly and deliberately, pausing after the word predictions to allow the connection to sink in.	The Version 0 model serves as a baseline artefact that makes learner thinking visible and traceable across the unit. Drawing from imagination is less threatening than a written test and more revealing of intuitive mental models. The explicit framing that the model will grow removes the pressure to be correct and positions revision as the goal of learning rather than an admission of failure.	Teacher collects science notebooks briefly at the end of class — or photographs selected drawings — to document baseline mental models. Common patterns to look for include: cells drawn as simple circles with no internal structure, cells drawn with a nucleus only, cells drawn with several named organelles (showing prior exposure), and cells drawn to look like bricks or boxes (influenced by the word cell). These patterns guide differentiation in Lesson 2.

	questions, and made your first predictions. Hold onto those predictions. We will need them again.			

D. TEACHER REFLECTION
After the lesson, consider: Did learners write genuine wonder questions or did they write what they thought was expected? Were learners willing to share incomplete ideas publicly — if not, what might have inhibited them? Did the narration of Amina scenario create genuine curiosity or did it feel contrived — if the latter, how could the delivery be made more vivid next time? Were there any learners who already mentioned specific cell types such as platelets or white blood cells — how will their prior knowledge be channelled productively in later lessons without making peers feel behind? Was the 8-second wait time after the driving question genuinely observed — silence is often harder for the teacher than for the learners. Finally, are the wonder questions on the DQB rich enough to sustain 12 lessons of inquiry, or do they need to be supplemented with teacher-generated questions in Lesson 2?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed a sequence of changes that happen when a finger is cut — bleeding slows, a clot forms, a scab develops, and new skin appears over several days
What did I learn?	We learned that this familiar process is actually made up of many simultaneous events, and that scientists have found many different types of tiny structures working together during wound healing
How does this explain the phenomenon?	We cannot yet explain why different structures look different or how each one knows what job to do — these are exactly the questions our Driving Question Board will help us answer over the next eleven lessons

LESSON 2 (80 minutes): What Is a Cell? Introduction to Cell Theory	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce learners to the cell as the basic unit of life, establish the three tenets of cell theory, and distinguish between prokaryotic and eukaryotic cells through labelled diagrams and model sketching.
Knowledge	Learners will know: (a) the definition of a cell and its role as the basic unit of structure and function in all living organisms; (b) the three tenets of cell theory as developed by Schleiden, Schwann, and Virchow; (c) the distinguishing features of prokaryotic and eukaryotic cells.
Skills	Learners will be able to: (a) search for and discuss information on the cell as the basic unit of life; (b) sketch and label a generalised plant and animal cell diagram; (c) compare prokaryotic and eukaryotic cells using observable structural features.
Attitudes	Learners will: (a) appreciate that foundational scientific knowledge is built incrementally by many contributors across cultures and time; (b) develop curiosity about how microscopic structures underpin the complex healing events observed in the anchoring phenomenon.
Key Inquiry Question	What are the structural differences between plant and animal cells?
Purpose in Storyline	In Lesson 1, learners were introduced to the anchoring phenomenon — the healing cut — and began asking questions about why different cells look and behave differently. Lesson 2 establishes the conceptual foundation: every cell, no matter how specialised, is governed by the same cell theory. By building Version 1 of their cumulative cell model today, learners create a baseline they will revise in every subsequent lesson as they discover how structure drives function.
Safety Notes	No hazardous materials are used in this lesson. If learners use scissors or sharp tools for any cut-and-paste diagram activity, remind them to handle these carefully — connecting back to the anchoring phenomenon (the finger cut) is appropriate and reinforces safety awareness. Ensure the classroom internet (if used for research) is filtered and safe.

B. LESSON OVERVIEW

Learners investigate the cell as the basic unit of life by searching for information, engaging with the ugali-kitchen analogy, and studying labelled diagrams of prokaryotic and eukaryotic cells. They are formally introduced to cell theory (Schleiden, Schwann, Virchow) and produce Version 1 of their cumulative cell model — a sketched and labelled generalised cell. The lesson keeps the anchoring phenomenon (the healing cut) alive by asking: "All these different cells that healed Amina's finger — what do they all have in common at the most basic level?" Every activity drives toward that question.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Prokaryotic and eukaryotic cells] Source: Khan Academy (English	Learners are presented with the Phenomenon Recap prompt on the board: 'Yesterday we met Amina's healing finger. Platelets, neutrophils, fibroblasts, and skin cells all rushed to help — and they	Display the phenomenon image (Amina's healing finger, cross-section diagram from Lesson 1) on the board before learners enter. As learners settle, say: 'Before we learn anything new today, I want	Think-Pair-Share Prediction Harvest: By surfacing prior conceptions before instruction, the teacher identifies misconceptions (e.g., 'all cells have a nucleus') that the lesson will either confirm or	Observable indicator: Each learner has written at least 3 sentences in their notebook before pair-share begins. Teacher scans notebooks during circulation. Listen for: Do learners mention a membrane?

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	all look completely different. Today's question: What do ALL of those cells — no matter how different — have in common?' Learners individually write a 3-sentence prediction in their science notebooks in response to the prompt: 'I think all cells share… because…'. They then pair-share with a shoulder partner for 90 seconds.	you to predict. Look at Amina's finger. All those different cells working together — platelets, skin cells, macrophages — what do you think every single one of them has in common? Write three sentences starting with: I think all cells share… because…' Allow 3 minutes of individual silent writing. WAIT TIME: Do not speak for the full 3 minutes. After pair-share, cold-call 3 pairs: 'Pair at the window table — what did you agree on?' Record key prediction words on the board in a column labelled OUR PREDICTIONS. Do not correct errors — affirm thinking: 'Interesting — let's hold that idea and test it today.'	productively challenge. Recording predictions publicly creates cognitive accountability — learners are motivated to revisit their words.	Genetic material? A 'control centre'? Flag any learner who writes 'I don't know' for targeted support during the Observe phase.
Observe Phase 📹 **VIDEO:** Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — The Ugali Kitchen Analogy (15 min): Teacher introduces the analogy verbally and with a simple whiteboard sketch. Learners discuss in groups of four how the analogy maps onto cell structure concepts. They complete a two-column Analogy Mapping Table in their notebooks: Kitchen Feature → Cell Feature (e.g., 'The kitchen itself / the jiko boundary' → 'Cell membrane', 'The recipe book' → 'DNA/Nucleus', 'Cooking pots and utensils' → 'Organelles'). PART B — Guided Research & Diagram Study (20 min): Learners are directed to their ARES platform resource cards (or printed A3 diagram sheets) showing: (i) a labelled generalised animal cell, (ii) a labelled generalised plant cell, (iii) a labelled prokaryotic cell (e.g., E. coli). They read the embedded text on cell theory and complete a Cell Theory Evidence Table: Scientist | Country/Period | Contribution | How it connects to Amina's	PART A: Draw a simple kitchen outline on the board — label: 'door = cell membrane', 'recipe book on shelf = nucleus/DNA', 'jiko (stove) = mitochondria', 'sufuria, mwiko, chopping board = organelles'. Say: 'Every kitchen in Nairobi — whether in a Kibera home or a Westlands restaurant — has these basics. But the kitchen in a hospital canteen is set up differently from a home kitchen in Kericho. Same tools, different arrangement, different job. That is exactly what we will discover about cells.' Pause for 10 seconds. Ask: 'What part of the kitchen would you say is like the cell membrane — the thing that controls what comes in and out?' WAIT TIME: 12 seconds. Cold-call 2 learners. PART B: Say: 'You will now read about three scientists — Schleiden, Schwann, and Virchow — who each added one piece to the cell theory puzzle. As you read, fill in your Evidence Table. Most importantly: for each	Analogy Mapping (Ugali Kitchen): Anchors abstract cell vocabulary to a concrete, culturally familiar Kenyan context, reducing cognitive load for new terminology. Card Sort (Prokaryote/Eukaryote): A structured academic controversy strategy — physical manipulation of feature cards externalises thinking, exposes disagreements within groups, and generates productive talk. Evidence Table with Phenomenon Link: Forces learners to connect historical scientific knowledge directly to the anchoring phenomenon, preventing isolated memorisation.	Observable indicators: (1) Analogy Mapping Table has at least 5 correct mappings — teacher checks two tables per group during circulation. (2) Cell Theory Evidence Table: each scientist row has a completed 'Phenomenon Connection' sentence — a blank row signals the learner cannot yet connect theory to observation. (3) Card Sort: Teacher notes any card consistently misplaced across groups (e.g., 'Has ribosomes' placed only under Eukaryote) — this informs the Explain phase mini-lecture focus.

	healing. PART C — Prokaryote vs. Eukaryote Sort (10 min): Groups receive a set of 12 feature cards (e.g., 'Has a nucleus', 'DNA floats freely', 'Has mitochondria', 'Found in bacteria'). They physically sort cards into two columns on their desk: PROKARYOTE / EUKARYOTE. Groups photograph or sketch their sorted table.	scientist, write one sentence connecting their idea to Amina's healing finger.' Circulate and prompt: 'Virchow said all cells come from pre-existing cells — how does that explain where the new skin cells on Amina's finger came from?' WAIT TIME: 10 seconds after each prompt before assisting. PART C: Distribute feature cards. Say: 'Sort these into Prokaryote or Eukaryote. You have 8 minutes. Be ready to justify every card.' After 8 minutes, cold-call one group per disputed card: 'Group by the door — where did you put Has no membrane-bound nucleus and why?'		
Explain Phase 📹 **VIDEO:** Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher delivers a focused 12-minute direct instruction sequence consolidating cell theory and the prokaryote/eukaryote distinction, explicitly correcting the most common card-sort errors observed. Learners annotate their ARES diagram resource cards using a colour-coding system: GREEN = present in both prokaryotes and eukaryotes, BLUE = eukaryotes only, RED = plant eukaryotes only. Learners then write a 4-sentence Claim–Evidence–Reasoning (CER) paragraph in their notebooks responding to: 'Use cell theory to explain how Amina's body was able to produce new skin cells to close the wound on her finger.'	Begin with: 'Let me address what I saw during the card sort. Ribosomes — almost every group put these under Eukaryote only. Let's look at the diagram together.' Point to prokaryote diagram. 'Ribosomes are actually present in BOTH. This matters — it means even the simplest bacteria can make proteins. That's why bacteria in an infected wound can be so dangerous — they're making proteins too.' Deliver direct instruction on: (1) Three tenets of cell theory — write verbatim on board; (2) Key structural differences — nucleus membrane, organelles, DNA organisation; (3) Brief mention of Matthias Schleiden (botanist, studied plant cells), Theodor Schwann (studied animal cells), Rudolf Virchow ('Omnis cellula e cellula' — all cells from cells). Say: 'Now, using your colour-coded diagram, write a CER paragraph. Your CLAIM answers: How did Amina's body make new skin cells? Your EVIDENCE comes from Virchow's	Colour-Coded Annotation: Transforms passive diagram study into an active categorisation task — colour coding makes structural patterns visible and supports retention of comparative features. CER Writing with Phenomenon Anchor: Forces learners to move from declarative knowledge ('Virchow said…') to explanatory knowledge ('Therefore Amina's body could…'), which is the core NGSS practice of Constructing Explanations.	Observable indicators: (1) Colour-coded diagrams — teacher checks that ribosomes are GREEN (not BLUE) and that cell wall is RED; any deviation is addressed individually. (2) CER paragraph must contain: a specific CLAIM sentence, at least one piece of EVIDENCE naming a scientist or tenet, and a REASONING sentence that uses the word 'therefore' or 'because' to connect to Amina's healing. Collect 6 notebooks (two from each ability group) at end of phase for rapid review.

		contribution. Your REASONING explains the mechanism.' WAIT TIME: Allow 8 minutes of silent writing. Circulate — if a learner is stuck, prompt: 'What did Virchow say about where new cells come from? Start there.' Cold-call 2 learners to read their CER aloud. After each, ask the class: 'Thumbs up if their Reasoning clearly connects cell theory to the phenomenon.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board displayed at the front. The anchoring driving question is re-read aloud together: 'Why do cells in our body look so different from each other — and how does the shape of a cell determine the job it can do?' Each learner writes one new sticky note to add to the board: either (a) a NEW question today's lesson raised for them, or (b) a statement of something today's lesson helped them understand that moves them closer to answering the driving question. Learners also place a star sticker next to any Lesson 1 question on the DQB that today's learning has begun to answer.	Hand out sticky notes (two colours: yellow = new question, green = growing understanding). Say: 'We've been learning about cell theory and cell structure today. Look at our Driving Question Board. Has today's lesson answered any of the questions we put up in Lesson 1? If so, put a star on it. Then write your own note — either a new question or a sentence that shows your thinking has grown.' WAIT TIME: Allow 4 minutes of silent reflection. As learners post notes, read 3–4 aloud: 'Listen to this question from the back table: If all cells come from pre-existing cells, where did the FIRST cell come from? That is a brilliant scientific question that researchers still debate.' Point to the driving question: 'We still haven't answered the big one. Hold it. We'll get there.'	DQB Iterative Update: Making thinking visible on a shared public board builds a classroom culture of scientific inquiry. Colour-coded note types (question vs. understanding) allow the teacher to track the ratio of confusion to consolidation across the class — a real-time class-level formative signal. Star-sticker annotation shows learners that science is a progressive process of question refinement, not one-shot answers.	Observable indicators: (1) Every learner posts at least one note — absence of a note flags disengagement or confusion requiring follow-up. (2) Green 'growing understanding' notes are checked for accuracy — a note reading 'I now understand that all cells are the same' is a misconception requiring correction before Lesson 3. (3) Teacher photographs the DQB at end of lesson for lesson-planning reference.
Model Building Phase 📹 **VIDEO:** Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners produce Version 1 of their Cumulative Cell Model. Using a full blank page in their science notebook, they sketch and label a generalised eukaryotic cell (they may choose plant or animal). Required labels for Version 1: cell membrane, cell wall (if plant), nucleus, cytoplasm, mitochondria, ribosomes, vacuole (if plant),	Say: 'Every scientist who studies cells uses models. Models are never perfect — but they capture what we know right now. Today you will draw Version 1 of YOUR cell model. By Lesson 12, this same page will be full of revisions, additions, and corrections. That's what science looks like.' Display the Version 1 checklist on the	Cumulative Model Building (Version 1): Establishing a physical, dated, revisable model in the learner's notebook externalises current understanding and creates a longitudinal scientific record. The Limitation Statement is a metacognitive tool — requiring learners to articulate what they do NOT yet know is a high-order	Observable indicators: (1) Model has all 8 required labels — teacher uses the checklist during circulation to identify missing labels and prompt without telling: 'Count your labels — how many do you have?' (2) Model Key sentences must include both structure AND function — 'Nucleus: oval shape, contains

📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	endoplasmic reticulum. Beneath the sketch, learners write a Model Key: one sentence per label explaining the structure's role. At the bottom of the page, they write a 'Version 1 Limitation Statement': 'This model does not yet show…' — listing at least two things they know they don't yet understand about cell structure. The page is headed VERSION 1 — [Date] and will be physically revised (not replaced) across the 12-lesson sequence.	board: 8 required labels, 1 sentence per label in the Key, 1 Limitation Statement with at least 2 gaps. Say: 'You have 12 minutes. Work silently and independently — this is YOUR scientific record.' Circulate and narrate observations: 'I see Wanjiku has labelled the mitochondria and written that it produces energy — excellent. I see Otieno has drawn a cell wall on an animal cell — let's check: do animal cells have cell walls?' WAIT TIME: After posing a correction, wait 10 seconds for the learner to self-correct before assisting. In the final 2 minutes, ask learners to share their Limitation Statement with a partner: 'Read your partner's limitation. Do you agree those are real gaps? Add one of their gaps to your own list if you hadn't thought of it.'	NGSS practice (Developing and Using Models) and prevents false closure. The version-numbering system builds anticipation for future lessons.	DNA, controls cell activities' is adequate; 'Nucleus: it's in the middle' is insufficient and requires a prompt. (3) Limitation Statement: at least 2 genuine gaps listed — accept any scientifically honest statement (e.g., 'I don't yet know how the cell membrane decides what to let in', 'I don't know how the nucleus communicates with the rest of the cell'). Collect all Version 1 models at the end of class, photograph for records, return next lesson.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your planner: (1) During the card sort, which features were most consistently misclassified — and does your Lesson 3 plan address those specific misconceptions? (2) Review the 6 collected CER paragraphs: did learners connect Virchow's tenet directly to new skin cell production, or did they write generically about cells? If the latter, plan a 5-minute CER revisit at the start of Lesson 3. (3) Count the ratio of yellow (question) to green (understanding) DQB notes: if yellow heavily outnumbers green, learners may need a consolidation activity before Lesson 3 advances to cell specialisation. (4) Scan the Version 1 models: are animal cells being drawn with cell walls? Is the nucleus consistently represented as membrane-bound? Patterns across the class should inform your Lesson 3 opening correction sequence. (5) Reflect on wait time: did you hold the full 10–12 seconds, or did you answer for learners? Longer wait time consistently produces more complete learner responses — practise holding silence even when it feels uncomfortable.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In this lesson, learners observed labelled diagrams of generalised animal cells, plant cells, and prokaryotic cells. They sorted structural feature cards into Prokaryote and Eukaryote categories and studied the ugali-kitchen analogy mapped onto cell structure components.
What did I learn?	Learners learned the three tenets of cell theory (all living things are made of cells; the cell is the basic unit of life; all cells arise from pre-existing cells) and the contributions of Schleiden, Schwann, and Virchow. They learned the key structural differences between prokaryotic and eukaryotic cells, including the presence or absence of a membrane-bound nucleus and membrane-bound

	organelles.
How does this explain the phenomenon?	Learners explained, using a Claim–Evidence–Reasoning paragraph, how Virchow's tenet ('all cells from pre-existing cells') accounts for the production of new skin cells during the healing of Amina's finger. They also explained their Version 1 cell model using a labelled diagram with a structural-function key and articulated the limitations of their current model.

LESSON 3 (40 minutes): Prokaryotic vs. Eukaryotic Cells: Two Worlds Inside One Body	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of this lesson, learners will be able to compare and contrast prokaryotic and eukaryotic cells using structural evidence, and connect this distinction to the healing-cut phenomenon.
Knowledge	Learners will identify and explain the key structural differences between prokaryotic and eukaryotic cells, including presence or absence of a membrane-bound nucleus, membrane-bound organelles, relative cell size, and cell wall composition.
Skills	Learners will construct and complete a structured comparison table, analyse labelled diagrams of E. coli and a human cheek cell, and articulate differences using precise biological vocabulary.
Attitudes	Learners will appreciate that even the smallest structural difference — such as the absence of a nucleus — has profound consequences for how an organism functions, fostering curiosity about the diversity of life.
Key Inquiry Question	What are the structural differences between plant and animal cells? (This lesson extends that inquiry by first establishing the broader distinction between prokaryotic and eukaryotic cells, which underpins all later comparisons.)
Purpose in Storyline	In Lesson 2, learners built a generalised cell model and encountered the prokaryote-eukaryote distinction briefly. Lesson 3 deepens that distinction by using a Kenya-relevant context — E. coli contamination in poorly stored nyama choma — to make prokaryotes real and consequential. This prepares learners for Lessons 4 and 5, where they will examine plant and animal eukaryotic ultrastructure in detail. The lesson also advances the driving question by establishing that not all cells are built the same way, even before specialisation begins.
Safety Notes	No hazardous materials are used in this lesson. Diagrams and tables only. Remind learners that E. coli is used as a conceptual example; no live bacterial cultures are present. General classroom conduct rules apply.

B. LESSON OVERVIEW

Learners begin by revisiting the healing-cut phenomenon and predicting whether the bacteria that might infect the wound are built the same way as the body cells fighting the infection. Through structured diagram analysis comparing E. coli (prokaryote) and a human cheek cell (eukaryote), learners complete a comparison table identifying four key structural differences. Cold-call questioning consolidates understanding across a minimum of five students. The lesson closes with a targeted Driving Question Board update.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Prokaryotic and eukaryotic cells] Source: Khan Academy (English	Teacher re-anchors the lesson to the healing-cut phenomenon. Learners are told: when Amina cut her finger while slicing onions in Nairobi, bacteria from the onion skin or blade could enter the	Teacher opens with a brief verbal callback to Lesson 2: 'Last lesson we said a cell is the basic unit of life — but are all cells built the same way? Let us think about this carefully.' Teacher writes the	Prediction with written justification activates prior knowledge from Lesson 2 and creates cognitive dissonance — learners hold a tentative idea they are motivated to test. Scribing predictions	Teacher scans written predictions during the 2-minute silent phase, noting whether learners are drawing on Lesson 2 vocabulary (nucleus, membrane, organelles) or relying only on everyday

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Prokaryote Evolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	wound. The teacher asks learners to open their science notebooks and respond individually (2 minutes, silent writing) to the prompt: 'Amina is worried about bacteria entering her cut. Do you think a bacterium like E. coli is built the same way as Amina's skin cells — or differently? Write down two reasons for your prediction.' After silent writing, learners share predictions with a shoulder partner for 1 minute. Three volunteers share with the class. Teacher scribes key prediction words on the board (no corrections yet): learners typically predict that bacteria are 'smaller,' 'simpler,' or 'have no nucleus.' Teacher validates thinking without confirming or denying: all ideas remain live for testing.	lesson focus question on the board: 'Are the bacteria that could infect Amina's cut built the same way as her body cells?' Teacher gives the silent-writing prompt verbally and writes it on the board. After 2 minutes, teacher says: 'Turn to your shoulder partner — share what you wrote. You have 1 minute.' After partner talk, teacher cold-calls three students using the class register (not volunteers only). Teacher scribes their key words in a 'Our Predictions' box on the board. Teacher then says: 'By the end of this lesson, we will have enough evidence to confirm or revise every single one of these predictions. Let us go and find out.'	publicly creates accountability and makes thinking visible for later revision.	language. This informs the depth of scaffolding needed during the Observe Phase.
Observe Phase 📹 **VIDEO:** Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Prokaryote Evolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher distributes (or projects) two large, clearly labelled diagrams side by side: (1) E. coli — a prokaryote responsible for food poisoning in poorly stored nyama choma, labelled with: cell wall (peptidoglycan), plasma membrane, cytoplasm, ribosomes, circular DNA (nucleoid region — no nuclear membrane), flagellum, approximate size 1–5 micrometres; (2) Human cheek cell — a eukaryote, labelled with: cell membrane, cytoplasm, nucleus with nuclear membrane, mitochondria, ribosomes, rough endoplasmic reticulum, approximate size 10–100 micrometres. Learners are given a structured comparison table (pre-printed or drawn in notebooks) with the following rows: Feature | E. coli (Prokaryote) | Human Cheek Cell (Eukaryote). Features listed: Presence of membrane-	Teacher introduces the diagrams by contextualising E. coli: 'E. coli is a bacterium. In Kenya, it has been linked to food poisoning cases from nyama choma that is left out too long at a roadside kiosk in Nairobi or Mombasa. It is very small — about 2 micrometres. Amina would not see it, but her white blood cells would find it. Now, let us look carefully at how it is built compared to her own cheek cell.' Teacher models how to read one row of the comparison table using the first feature only: 'I look at the E. coli diagram — I can see DNA here, but is there a membrane around it? No. So I write: no membrane-bound nucleus. Now you complete the rest.' Teacher sets a 4-minute individual work timer and circulates silently, observing tables without correcting. After 4 minutes: 'Compare your table with your	Diagram analysis with a structured comparison table is a disciplinary core practice in biology. The table format reduces cognitive load and ensures all learners engage with the same four structural features simultaneously. The individual-then-partner-then-whole-class sequence (I Do briefly, You Do, We Check) builds confidence before public sharing.	Teacher scans comparison tables during circulation. Key diagnostic questions: Are learners correctly identifying the nucleoid region as distinct from a membrane-bound nucleus? Are they noting that ribosomes appear in both cells? These are the two highest-frequency errors. Teacher makes a mental note of two or three specific misconceptions to address during cold-call review.

	bound nucleus; Membrane-bound organelles present; Approximate size; Cell wall (present or absent; composition); Genetic material location. Learners work individually for 4 minutes to complete the table using the diagrams only — no teacher input during this phase. They then compare answers with a partner for 1 minute before a whole-class review.	partner. If you disagree on any row, put a small question mark — we will sort it out together.' After 1 minute of partner comparison, teacher leads whole-class review row by row using cold-call: 'Wanjiku, what did you write for membrane-bound organelles in E. coli?' WAIT TIME: 5 seconds minimum before prompting. Teacher confirms correct answers and addresses misconceptions immediately, especially the common confusion between ribosomes (present in both, not membrane-bound) and membrane-bound organelles (absent in prokaryotes).		
Explain Phase 📹 **VIDEO:** Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Prokaryote Evolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher facilitates a cold-call consolidation round with a minimum of 5 students. Each student is asked a different question at a slightly higher level of explanation (not just recall). Questions escalate from identification to explanation to application. After each response, the class is asked to signal agreement or disagreement using thumbs (up, sideways, down). Learners then write a 3-sentence summary in their notebooks: Sentence 1 — one structural feature that makes E. coli a prokaryote; Sentence 2 — one structural feature that makes the cheek cell a eukaryote; Sentence 3 — why this difference matters for understanding the healing-cut story. Teacher concludes the explain phase by formally introducing the terms prokaryote (from Greek: pro = before, karyon = nucleus) and eukaryote (eu = true, karyon = nucleus), writing both etymologies on the board and asking learners to copy them as a	Cold-call sequence (minimum 5 students, called from the register, not self-selected): Student 1 — identification level: 'Ochieng, look at the E. coli diagram. What is the name of the region where its DNA is found, and why is it different from the nucleus in the cheek cell?' WAIT TIME: 5 seconds. If no response after 5 seconds: 'Look at the label on the diagram — what does it say near the DNA?' Student 2 — explanation level: 'Fatuma, E. coli has ribosomes AND the cheek cell has ribosomes — does that make them the same type of cell? Explain your thinking.' WAIT TIME: 7 seconds. Student 3 — comparison level: 'Kamau, give me one feature from your table that is DIFFERENT between the two cells and explain what that difference means.' WAIT TIME: 5 seconds. Student 4 — size reasoning: 'Aisha, the cheek cell is about 50 times bigger than E. coli. Why might a larger cell need membrane-bound organelles when a smaller cell does not?' WAIT	Escalating cold-call questions move learners from recall to reasoning, a deliberate Bloom progression within a single phase. Thumbs signalling makes all learners accountable for evaluating peers answers, not just the student called upon. Three-sentence written summary forces synthesis at the individual level before moving to DQB and model work.	Teacher reads over shoulders during the 3-minute writing period, checking that Sentence 3 connects to the phenomenon and is not purely definitional. Learners who write only definitions for all three sentences are identified for targeted questioning during model-building. The thumb-signal pattern also reveals whether the class as a whole has resolved the ribosome misconception.

	memory anchor.	TIME: 8 seconds — this is a higher-order question; teacher accepts reasonable reasoning, not a single right answer. Student 5 — application to phenomenon: 'Brian, Amina is worried about bacteria in her cut. Based on what we have just learned, are those bacteria prokaryotes or eukaryotes? How do you know?' WAIT TIME: 5 seconds. After each response, teacher asks the class: 'Thumbs — do you agree, partly agree, or disagree with what Brian said? ... Good. Anyango, why did you put your thumb sideways?' After cold-call round, teacher says: 'Now write three sentences in your notebook. I will give you 3 minutes. Sentence 1: one structural feature that makes E. coli a prokaryote. Sentence 2: one structural feature that makes the cheek cell a eukaryote. Sentence 3: why this difference matters for the healing-cut story.' Teacher then writes the etymologies on the board and says: 'These words carry their meaning inside them. Prokaryote — before nucleus. Eukaryote — true nucleus. Every time you see these words, let the Greek roots remind you what they mean.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Prokaryotic and eukaryotic cells] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Teacher directs learners attention to the class DQB poster (established in Lesson 1). Teacher reads aloud two or three of the original wonder-questions posted by the class that are now partially or fully addressable — for example: 'Why are some cells bigger than others?' or 'Are germs the same as cells?' Learners are asked to take a sticky note (or a small piece of paper) and write one sentence beginning with: 'We now know that prokaryotes and	Teacher stands at the DQB and says: 'In Lesson 1, some of you wondered whether bacteria and human cells are the same thing. Let us look — (reads the relevant wonder sticky note aloud). Can we now answer that? Turn to your partner — 20 seconds.' Teacher then distributes sticky notes and gives the two-sentence scaffold verbally and in writing on the board: SENTENCE A: We now know that prokaryotes and eukaryotes differ in... SENTENCE	The DQB update anchors new learning to the class narrative and makes epistemic progress visible. Separating KNOW from WONDER trains learners to distinguish evidence-based claims from open questions — a core scientific practice. The teacher reading WONDER notes aloud signals that not-yet-knowing is intellectually valued, not a failure.	Teacher quickly scans the KNOW column notes during posting. Are learners using structural vocabulary (nucleus, organelles, cell wall) or only general language (bigger, more complex)? The quality of vocabulary in KNOW notes reveals whether the comparison table exercise achieved its learning target.

Prokaryote Evolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	eukaryotes differ in...' and a second sentence beginning with: 'But we still wonder...' Learners post their notes on the DQB in two columns: WHAT WE NOW KNOW and WHAT WE STILL WONDER. Teacher facilitates a 2-minute gallery scan where three learners read aloud one note each from the WONDER column — these become flagged questions for Lessons 4 and 5.	B: But we still wonder... Teacher says: 'You have 2 minutes to write both sentences and post them on the board.' Teacher circulates and encourages specificity: 'Do not just write they are different — write HOW they are different, using vocabulary from your table.' After posting, teacher selects three WONDER notes and reads them aloud: 'These are excellent questions — and they are exactly what we will be investigating in Lessons 4 and 5, when we go inside the organelles of plant and animal cells. Hold onto these questions.'		
Model Building Phase 📹 **VIDEO:** Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Prokaryote Evolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their science notebooks to their Version 1 cumulative cell model (the generalised cell sketch from Lesson 2). Teacher instructs them to add a second panel beside or below their existing sketch, labelled Version 2: Prokaryote vs. Eukaryote. Learners draw two simple outline diagrams side by side — one for E. coli and one for a human cheek cell — and annotate each with at least four labelled features from their comparison table. They use a simple colour code: red annotation = feature present in prokaryote only; blue annotation = feature present in eukaryote only; black annotation = feature present in both. Learners add a one-sentence caption below the two diagrams: 'A prokaryote has no membrane-bound nucleus, while a eukaryote does — and this difference matters because...' Teacher circulates and asks targeted questions of individual learners to probe depth of understanding.	Teacher explains the task: 'Your cumulative cell model is growing. In Lesson 2 you drew a generalised cell. Today you add a new panel — showing both types of cells side by side, with the differences clearly annotated. This is not just copying — it is building a model that explains. Use the colour code on the board: red for prokaryote only, blue for eukaryote only, black for both.' Teacher writes the colour code on the board and the sentence stem for the caption. Teacher circulates during the 5-minute drawing period and poses probing questions to individual learners: 'You have drawn a cell wall on both — but are they made of the same material? Check your table.' or 'You have not labelled the nucleoid region on your E. coli — is that because E. coli has no DNA, or because the DNA is arranged differently?' Teacher selects two learners whose models show strong annotations to briefly share with the class in the final 1 minute: 'Wanjiru, hold up your model and	Cumulative model building is the primary learning artefact of this unit. Each lesson adds a new version, making intellectual growth tangible and visible. The colour-coding system requires learners to make comparative judgements, not just label in isolation. Sharing two models briefly at the close creates a low-stakes public standard and celebrates precision.	Teacher evaluates whether learners are correctly using the colour code — specifically, whether ribosomes appear in black (present in both) rather than red or blue. The sentence stem caption is a key formative probe: learners who can complete 'this difference matters because...' with a biologically accurate consequence (e.g. compartmentalisation allows more complex functions) are demonstrating transfer beyond recall. Learners who leave the caption blank or write only 'they are different types' are flagged for re-engagement in Lesson 4.

		tell us one annotation you are proud of and why.' Teacher closes the lesson by connecting back to the phenomenon: 'Next lesson, we go inside the eukaryotic cell — specifically the plant cell. And remember: the onion that Amina was cutting when she hurt her finger — that is a plant. Its cells will be our starting point.'		

D. TEACHER REFLECTION

After the lesson, consider the following: (1) Did the cold-call questions reveal any persistent confusion between ribosomes (present in both cell types, not membrane-bound) and membrane-bound organelles? If so, begin Lesson 4 with a brief 2-minute clarification using the ribosome row of the comparison table. (2) Were learners able to connect the E. coli example to the healing-cut phenomenon in their Sentence 3 summaries, or did they treat the science as abstract and disconnected? If the phenomenon link was weak, use the Lesson 4 opening to make that bridge explicit again. (3) Did the DQB WONDER notes reveal questions about organelle function that indicate readiness for Lesson 4, or do learners still have fundamental uncertainty about what a membrane-bound organelle is? Use this to calibrate the depth of diagram analysis in Lesson 4. (4) Were all five cold-called students drawn from different groups (gender, ability, seating zone) to ensure equitable participation? Review the register markings and adjust the call pattern in Lesson 4 accordingly.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed two labelled diagrams — E. coli (prokaryote, linked to nyama choma food poisoning) and a human cheek cell (eukaryote) — and identified structural differences including the presence or absence of a membrane-bound nucleus, membrane-bound organelles, relative size, and cell wall composition.
What did I learn?	Prokaryotic cells (such as E. coli bacteria) lack a membrane-bound nucleus and membrane-bound organelles; their DNA floats freely in a nucleoid region. Eukaryotic cells (such as human cheek cells) have a membrane-bound nucleus and membrane-bound organelles. Prokaryotes are generally much smaller (1–5 micrometres) than eukaryotes (10–100 micrometres). Both cell types contain ribosomes, but ribosomes are not membrane-bound in either.
How does this explain the phenomenon?	The bacteria that could infect Amina's cut (E. coli) are prokaryotes — structurally far simpler than her own body cells, which are eukaryotes. The presence of a membrane-bound nucleus in eukaryotic cells allows DNA to be protected and gene expression to be regulated in ways that support the complexity needed for wound healing. This structural difference explains why, even though bacteria and human cells are both alive, they behave so differently — and why Amina's white blood cells (eukaryotes) can carry out the complex task of recognising and destroying E. coli (prokaryote) cells in the wound.

LESSON 4 (80 minutes): Plant Cell Ultrastructure — Every Organelle Has a Farming Job	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To identify and describe the structure and function of plant cell organelles using labelled diagrams and electron micrograph images, and to connect organelle specialisation to the healing phenomenon.
Knowledge	Learners identify the following plant cell organelles and describe their functions: cell wall, cell membrane, nucleus, chloroplasts, large central vacuole, mitochondria, endoplasmic reticulum (rough and smooth), ribosomes, and Golgi apparatus. Learners distinguish plant cell ultrastructure from animal cell ultrastructure covered in Lesson 3.
Skills	Learners practise diagram analysis (interpreting labelled electron micrographs), scientific annotation (adding labels with function notes), comparative reasoning (plant vs. animal cell), and cumulative model revision.
Attitudes	Learners appreciate that the extraordinary precision of organelle function in a single maize leaf cell mirrors the precision seen in the wound-healing phenomenon — complexity is built from structure.
Key Inquiry Question	What are the structural differences between plant and animal cells? How does cell specialisation enable organisms to carry out complex life functions?
Purpose in Storyline	In Lessons 1–3 learners anchored the phenomenon (the healing cut), explored cell theory, and mapped animal cell organelles. Lesson 4 expands the structural vocabulary to plant cells, using a Rift Valley maize farm as a sustained metaphor. This prepares learners for Lesson 5, where they directly compare plant and animal cells and begin to explain how organelle differences drive functional differences — a key step toward answering why the fibroblast, the platelet, and the epidermal cell look so different from one another.
Safety Notes	No wet practical in this lesson. If a projector is used, ensure the power cable is secured and away from learner pathways. Printed diagram sheets should have no sharp staples exposed. Remind learners to handle printed resources with care.

B. LESSON OVERVIEW
This is a teacher-led diagram-analysis lesson in which learners examine large labelled diagrams and, where a projector is available, electron micrograph images of a plant cell. Using the extended metaphor of a maize farm in the Rift Valley, the teacher guides learners through nine key organelles, building function understanding organelle-by-organelle. Learners annotate a blank plant cell outline, participate in structured questioning rounds, and close the lesson by revising their cumulative cell model to produce Version 2 — the Plant Cell. The lesson deliberately contrasts plant and animal cell features to keep the driving question alive: if all cells share the same DNA, why do plant cells have chloroplasts and animal cells do not?

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Photosynthesis in organisms]	Learners receive a blank, unlabelled outline of a plant cell (prepared as a photocopied handout or drawn on the board for	Distribute blank plant cell outlines. Say verbatim: 'Before I show you anything, I want your brain's first draft. Write every organelle you	Predict-then-Poll (Elicitation). Learners externalise prior knowledge before instruction, creating cognitive dissonance that	Observable indicator: Every learner has at least 3 organelles written on their outline before partner share. Teacher scans

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	copying). Working individually for 4 minutes, they write the names of any organelles they think will be present and place a ★ next to those they believe are ALSO found in animal cells (from Lesson 3). They then share predictions with a shoulder partner for 2 minutes. Teacher collects a quick show-of-hands class poll: 'Hands up if you predicted chloroplasts.' 'Keep your hand up if you also marked chloroplasts as shared with animal cells.' The visual mismatch in raised hands surfaces the core conceptual tension immediately.	think a plant cell has. Put a star next to any you think it shares with the animal cell we studied last lesson.' Allow 4 minutes of silent individual work — do NOT circulate yet; let learners commit to predictions. After 4 minutes, say: 'Turn to your partner. Compare your lists. Do you agree on which organelles are shared?' Allow 2 minutes of partner talk. Conduct the show-of-hands poll. Cold-call 3 learners who kept their hand up for chloroplasts being shared: 'Akinyi, why did you think animal cells also have chloroplasts?' Apply 12-second wait time before accepting an answer. Record class predictions in two columns on the board — SHARED / PLANT ONLY — and leave it visible throughout the lesson for revision.	motivates information-seeking. The two-column board record gives learners a tangible target to confirm or revise, turning the lesson into an active self-correction exercise rather than passive reception.	written predictions during the 4-minute window. Flag any learner with a blank sheet for targeted support. The show-of-hands poll reveals class-level misconceptions about chloroplast distribution — if more than 40% of hands stay up (believing animal cells have chloroplasts), the teacher notes this for explicit address in Phase 3.
Observe Phase 📹 **VIDEO:** Photosynthesis in organisms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher reveals a large, fully labelled plant cell diagram (projected or poster-sized printed copy posted at the front). Where a projector is available, the teacher also shows three electron micrograph images in sequence: (1) a chloroplast from a maize leaf, (2) a large central vacuole in an onion epidermal cell, (3) the cell wall boundary between two adjacent plant cells. Learners observe silently for 90 seconds per image, then annotate their blank outline with labels and a one-phrase function note beside each label (e.g., 'chloroplast — captures sunlight to make food'). Learners work individually during annotation. After completing their own annotations, learners pair up and check each other's function notes against the labelled diagram, adding corrections in a different colour pen.	Post or project the fully labelled diagram. Say: 'You have 90 seconds to look — do not write yet. Just look. Notice what is familiar from our animal cell lesson and what is completely new.' Set a visible 90-second timer. After 90 seconds: 'Now annotate your outline. For every organelle you label, write a short function phrase — one line, in your own words.' Circulate during the 8-minute annotation window; do NOT correct yet. If a projector is available, show each EM image for 60 seconds before the annotation phase, saying: 'This is an actual electron micrograph — a photograph taken with a beam of electrons, not light. The chloroplast you are looking at is from a maize leaf, the same maize grown in farms around the Rift Valley.' After the annotation period, instruct peer-checking: 'Swap sheets with	Observe-Annotate-Peer Check (Evidence Recording with Immediate Peer Verification). The silent observation window before annotation prevents learners from writing without seeing. Peer-checking in a contrasting colour makes revision visible and non-threatening, building scientific habit of checking evidence against a source.	Observable indicator: Learner annotations contain both a label AND a function phrase for at least 7 of the 9 target organelles. During circulation, teacher checks that function phrases are present (not just labels). Teacher notes the 5 collected sheets: if fewer than 5/9 organelles have function phrases, the teacher briefly pauses Phase 3 to model two or three function phrases on the board.

		your partner. Using your green/red pen, tick what matches the diagram and add a correction where needed.' Allow 4 minutes for peer-checking. Collect 5 random sheets to scan function accuracy before Phase 3.		
Explain Phase 📹 **VIDEO:** [Photosynthesis in organisms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plant Cell Structures (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher delivers a structured, metaphor-anchored explanation of plant cell organelles using the Rift Valley Maize Farm metaphor. Each organelle is mapped to a farm role. Learners listen, watch the teacher build the metaphor table on the board, and add any missing information to their annotated outline. Midway through the explanation, learners complete a structured 'Organelle Job Card' matching task individually. At the end of the explanation, the teacher returns to the anchoring phenomenon: learners discuss in triads why a plant cell — unlike the epidermal cell in the healing cut — contains chloroplasts, and what this tells us about how structure is linked to the cell's environment and job.	Build the following table on the board line-by-line, revealing one organelle at a time. Read each line aloud, pause, and ask a cold-call question before writing the next line. Target: minimum 9 cold-calls across the table, one per organelle. MAIZE FARM METAPHOR TABLE: – Cell Wall → The perimeter fence and brick walls of the farm — gives shape, support, protection. – Cell Membrane → The farm gate — controls what enters and leaves. – Nucleus → The farm owner's office — holds all the plans (DNA) and gives instructions. – Chloroplast → The solar panels on the farm — capture sunlight to make food (glucose) via photosynthesis. – Large Central Vacuole → The farm's water storage tank — stores water, maintains pressure (turgor), stores nutrients and waste. – Mitochondria → The farm's generator — releases energy from glucose for all farm activities (respiration). – Rough Endoplasmic Reticulum → The farm's delivery road network with loading bays — transports	Metaphor Anchoring + Structured Cold-Call Explanation. The Rift Valley Maize Farm metaphor gives learners a concrete, culturally grounded schema onto which abstract organelle functions are mapped. The cold-call routine (predict the analogy before the teacher confirms) keeps learners cognitively active during direct instruction. The phenomenon reconnect at the close prevents the farm metaphor from becoming decorative — it is immediately used as a reasoning tool.	Observable indicator: Learners complete the Organelle Job Card with at least 8/9 correct matches. Teacher collects all Job Cards. Any learner with 5 or fewer correct is identified for a brief re-teach during the Model Building phase. The triad discussion on epidermal cells vs. plant cells is assessed by listening to 2 triads: teacher listens for the phrase 'structure matches function' or an equivalent, student-generated idea.

		and modifies proteins made by ribosomes. – Ribosomes → The farm workers — build proteins from instructions sent by the nucleus. – Golgi Apparatus → The farm's packaging and dispatch depot — sorts, modifies, and ships proteins and lipids to where they are needed. For each organelle, use this 3-step routine: 1. Point to the organelle on the diagram. 2. Say: 'In our maize farm, this organelle is like the [X] because...' — wait 12 seconds — cold-call one learner to complete the analogy before you confirm. 3. Write the confirmed metaphor and function on the board. After the full table is built, distribute the Organelle Job Card (a half-page handout with two columns: ORGANELLE and FARM JOB — all farm jobs listed, organelles blank). Learners fill it individually in 4 minutes. Return to the phenomenon: 'Think back to Kioni's cut finger. The epidermal cells that grew back over the wound — those are animal cells. They have no chloroplasts. Why not? What does that tell you?' Wait 15 seconds. Cold-call 2 learners. Guide toward: epidermal cells do not need to photosynthesise — they get energy from food — so carrying a chloroplast would be wasteful. Structure matches function. Say		

		verbatim: 'This is the heart of our driving question. Every organelle exists because the cell's job demands it. No more, no less.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Photosynthesis in organisms](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plant Cell Structures (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners individually write one sticky note (or a note on a folded card if sticky notes are unavailable) responding to ONE of the following two prompts, assigned by row: Row A — 'What does today's lesson about plant cell organelles help you understand about the driving question — why do cells look different?' Row B — 'What new question does today's lesson raise for you about cell structure or the healing cut?' Learners post their notes on the class Driving Question Board in the designated LESSON 4 section. The teacher reads 4 notes aloud (2 from each row), and the class briefly discusses whether each note represents a growing understanding or a new mystery. The teacher physically draws a line connecting two DQB notes that together suggest an emerging answer: 'Plant cells have chloroplasts because they make their own food — animal cells, including Kioni's epidermal cells, do not need to.'	Distribute sticky notes or folded cards. Say: 'You have 3 minutes — individual, silent. Write your best thinking on the prompt for your row.' After 3 minutes, direct learners to post notes on the DQB. Read 4 notes aloud. For each, ask the class: 'Is this a growing understanding — something we are beginning to explain? Or is this a new mystery we still need to investigate?' Use a thumbs-up (understanding) or question-mark gesture (new mystery) from all learners for each note. Draw the connecting line on the DQB and say: 'We are building an answer — slowly, piece by piece, just like the cells rebuilding Kioni's finger.'	Driving Question Board — Knowledge Tracking. The DQB makes the cumulative nature of the evidence-gathering sequence visible. Sorting notes into 'growing understanding' vs. 'new mystery' teaches learners that science is an iterative process — not all questions are resolved in one lesson. The physical connecting line models how evidence from different lessons can converge into an explanation.	Observable indicator: Every learner produces a written DQB note. Teacher scans for notes that are off-topic or merely restate a fact ('chloroplasts do photosynthesis') without connecting to the driving question. These learners are prompted: 'How does knowing that help explain why Kioni's skin cells look different from a leaf cell?'
Model Building Phase 📹 **VIDEO:** [Photosynthesis in organisms](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Learners open their science exercise books to their cumulative cell model — currently Version 1, the Animal Cell, drawn and revised in Lesson 3. On the facing page, learners draw Version 2 — the Plant Cell. Using their annotated outline and the board diagram as references, they draw and label the plant cell, including all 9 organelles studied today. Beside each organelle label, learners write	Say: 'Version 2 starts now. Open to the page facing your animal cell model. You are going to draw your plant cell large enough to fill the full page — we need room for all nine organelles and their farm labels.' Allow 12 minutes for drawing and labelling. Circulate and check: (1) Is the cell wall clearly outside the cell membrane? (2) Is the vacuole drawn large and central? (3) Are chloroplasts drawn	Cumulative Model Revision (Version 2 — Plant Cell). Placing the animal cell and plant cell models on facing pages in the same book creates a persistent visual comparison that learners carry forward across all 12 lessons. Writing the difference sentence forces learners to synthesise rather than merely reproduce. The farm-metaphor brackets inside the model integrate	Observable indicator: Version 2 diagrams contain all 9 labelled organelles, both a label and a farm-metaphor bracket, and 2 summary sentences. Teacher collects 6 books (two from each seating cluster) to assess overnight. Look specifically for: correct placement of cell wall outside membrane, large central vacuole, and at least one of the two summary sentences

Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	its farm-metaphor role in brackets. In the top corner of the Version 2 diagram, learners write one sentence stating the most important structural difference between their Version 1 and Version 2 models, and one sentence predicting what this difference tells us about plant vs. animal cell function. Learners who finish early add colour-coding: green for organelles unique to plant cells, blue for organelles shared with animal cells.	as distinct oval structures inside the cell? For any of these three errors, crouch beside the learner and ask: 'Look at the EM image on the board — where is the vacuole relative to the membrane?' rather than correcting directly. After 10 minutes, prompt the summary sentences: 'Pause your drawing. Write your two sentences in the top corner now — you can finish the diagram after.' Read two summary sentences aloud from volunteers and give brief whole-class feedback. Close the lesson by saying: 'Your model now has two versions. In Lesson 5, we compare them side-by-side to answer a key part of our driving question. Hold onto every version — your model is your growing understanding made visible.'	the lesson's metaphor anchor directly into the scientific record.	identifying chloroplast/vacuole as a plant-unique feature. Return with written feedback at the start of Lesson 5.

D. TEACHER REFLECTION

After the lesson, consider: Did the Rift Valley Maize Farm metaphor help learners retain function, or did it oversimplify any organelle's role? Review the Organelle Job Cards — were there patterns of error (e.g., confusion between ribosomes and Golgi apparatus as 'makers' vs. 'dispatchers')? Did the DQB notes show genuine connection to the driving question, or did learners treat the board as a vocabulary exercise? Check the 6 collected model books: are learners drawing the cell wall correctly positioned outside the membrane? If not, open Lesson 5 with a targeted correction using the EM image. Note: if fewer than 70% of learners completed Version 2 with all 9 organelles, budget 5 minutes at the start of Lesson 5 for model completion before the comparative analysis begins.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A fully labelled plant cell diagram and (where available) electron micrograph images of a maize leaf chloroplast, a central vacuole in an onion cell, and a plant cell wall boundary.
What did I learn?	Plant cells contain nine key organelles — cell wall, cell membrane, nucleus, chloroplasts, large central vacuole, mitochondria, endoplasmic reticulum, ribosomes, and Golgi apparatus — each with a specific structure-linked function, mapped to roles on a Rift Valley maize farm.
How does this explain the phenomenon?	Unlike the animal cells (e.g., epidermal cells, platelets) involved in healing Kioni's cut, plant cells have organelles such as chloroplasts and a large central vacuole that reflect their unique job: making their own food and maintaining structural pressure. Structure and function are inseparable — a principle that applies equally to the specialised cells in the wound-healing phenomenon.

LESSON 5 (80 minutes): Animal Cell Ultrastructure — Comparing Plant and Animal Cells	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To identify the ultrastructure of an animal cell, distinguish organelles unique to animal and plant cells respectively, and update the cumulative cell model to show both cell types side by side.
Knowledge	Learners identify and describe the structure and function of organelles found in animal cells (including lysosomes and centrioles) and distinguish them from organelles exclusive to plant cells (chloroplasts, cell wall, large central vacuole).
Skills	Learners annotate diagrams of animal cells, construct a comparative two-column model (plant vs. animal), and deliver peer-reviewed group presentations of annotated diagrams.
Attitudes	Learners appreciate that the precise structural differences between plant and animal cells reflect their distinct roles in life processes — and connect this to the variety of cells involved in healing the cut finger in the anchoring phenomenon.
Key Inquiry Question	What are the structural differences between plant and animal cells, and how do those differences connect to what each cell type is built to do?
Purpose in Storyline	In Lessons 1–4, learners established what a cell is, examined plant cell structure, and used the onion and elodea as evidence. Lesson 5 now asks: what is structurally different about the animal cells that are actually doing the work in the healing cut — the platelets, neutrophils, fibroblasts, and epidermal cells? By comparing animal cell ultrastructure with the plant cell already in their cumulative model, learners build Version 3 of their model (Plant vs. Animal side by side) and sharpen their answer to the driving question.
Safety Notes	No hazardous chemicals or specimens are used in this lesson. If printed diagrams are distributed, ensure scissors (if used for cutting activities) are handled safely. Remind learners to wash hands after any handling of printed or physical materials. Supervise group seating to prevent overcrowding around shared diagrams.

B. LESSON OVERVIEW

In this teacher-led diagram analysis lesson, learners receive a large unlabelled diagram of an animal cell and use prior knowledge plus guided instruction to annotate it. Through a structured comparison table, they identify organelles present in animal cells but absent in plant cells (lysosomes, centrioles) and organelles present in plant cells but absent in animal cells (chloroplasts, cell wall, large central vacuole). Shared organelles (nucleus, mitochondria, ribosomes, endoplasmic reticulum, Golgi apparatus, cell membrane) are confirmed. Groups then present their annotated diagrams with peer feedback. The lesson closes with learners revising their cumulative cell model to Version 3 — a side-by-side Plant vs. Animal diagram — and explicitly linking unique animal cell organelles to the cells involved in healing the cut finger (e.g., lysosomes in neutrophils digesting bacteria; centrioles enabling rapid cell division in epidermal repair).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:**	Learners receive a blank outline silhouette of an animal cell (irregular, no cell wall) with no	Distribute the blank animal cell outline. Say verbatim: 'You have studied the plant cell in detail.	Think-Pair-Share with a Prediction Wall. This activates prior knowledge from Lessons 3–4,	Observable indicator: Each learner has independently drawn and attempted to label at least 4

Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	labels. Working individually for 4 minutes, they draw and label as many organelles as they can recall from Lessons 3 and 4, predicting which organelles from the plant cell they expect to find here and which they expect to be missing or different. They then share predictions with a partner using a Think-Pair-Share. On a sticky note, each pair writes one organelle they are CONFIDENT will be present and one they are UNSURE about, and posts both on the class Prediction Wall.	Before I teach you anything new, I want you to predict — using only what you already know — what the inside of an animal cell looks like. Draw and label for 4 minutes. Go.' Set a visible timer. After 4 minutes, say: 'Now share with your partner. Do you agree on every organelle, or are there differences?' Allow 2 minutes of pair talk. Cold-call 4 pairs (choose mixed-ability): 'Tell me one organelle you are both confident about and explain why.' After each response, wait 12 seconds before affirming or probing. Then direct pairs to the Prediction Wall: 'Write your CONFIDENT organelle in green and your UNSURE organelle in orange. Stick them up.' Photograph the wall for formative records. Do NOT correct predictions yet — affirm the thinking process: 'These predictions are your hypotheses. We will test every one of them today.'	creates cognitive dissonance around expected differences, and gives the teacher real-time data on misconceptions before instruction begins. The Prediction Wall makes learner thinking visible and will be revisited at the end of the lesson to check and celebrate growth.	organelles on the blank outline. The Prediction Wall reveals class-wide patterns — if fewer than 60% of pairs correctly predict the nucleus and mitochondria as shared, the teacher re-activates this in the Observe phase before moving to differences.
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a fully labelled, high-quality diagram of an animal cell ultrastructure (showing: cell membrane, cytoplasm, nucleus with nuclear envelope and nucleolus, rough and smooth ER, Golgi apparatus, mitochondria, ribosomes, lysosomes, centrioles, vacuoles — small and numerous). Alongside this, they have their Lesson 4 plant cell diagram. Working in groups of 4, learners complete a three-column Comparison Table: Column A — 'Organelle', Column B — 'Present in Plant Cell? (✓/✗)', Column C — 'Present in Animal Cell? (✓/✗)'. The table lists 12 organelles. Groups must also write a one-sentence function for each	Distribute both diagrams and the blank Comparison Table. Say: 'You now have two diagrams in front of you — the plant cell you studied last lesson and today's animal cell. Your job is NOT to memorise — your job is to observe carefully and find the differences. Use the table. You have 15 minutes.' Circulate continuously. At the 5-minute mark, prompt any group still on the first 3 rows: 'You should be at least halfway through the table by now — keep moving.' At the 10-minute mark, stop the class for 60 seconds: 'Without telling me the answer, raise your hand if your group has found at least TWO organelles that appear in the animal cell but NOT the	Structured Comparison Table with dual-diagram analysis. By placing the plant cell diagram (prior lesson evidence) alongside the new animal cell diagram, learners engage in direct observational comparison — a core scientific practice (NGSS SEP 4: Analysing and Interpreting Data). The table format prevents passive looking and forces claim-level decisions (present/absent) with functional justification.	Observable indicator: Each group's Comparison Table correctly identifies lysosomes and centrioles as animal-cell-only, and chloroplasts, cell wall, and large central vacuole as plant-cell-only, with at least a partial function noted for each. Teacher spot-checks 3 tables during circulation. Red flag: if groups mark 'large vacuole' as present in animal cells (a common misconception), address this whole-class before Phase 3.

	organelle they mark as present. Groups have 15 minutes for this structured observation task.	plant cell.' Count hands publicly. Say: 'Good — hold that thought.' At 15 minutes, call time. Cold-call 3 groups to share one unique animal cell organelle each — requiring them to state both its NAME and its FUNCTION. Wait 12 seconds after each response before commenting. Affirm accuracy; where a function is vague (e.g., 'lysosomes break things'), press with: 'Break what things, specifically? Think about the cut finger — what might a lysosome be breaking down in a neutrophil?' Wait 15 seconds.		
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher delivers a focused 12-minute guided explanation connecting each unique organelle to the anchoring phenomenon. Learners annotate their diagrams in real time using a colour code: blue for organelles unique to animal cells, green for organelles unique to plant cells, and black for shared organelles. After the explanation, learners individually write a 3-sentence 'Functional Justification' for why an animal cell does NOT need a cell wall, a chloroplast, or a large central vacuole — using evidence from both the phenomenon and their diagram. Groups then consolidate one shared justification per unique organelle and prepare a 2-minute annotated diagram presentation.	Begin with a direct return to the phenomenon: 'When Amina cut her finger in Nairobi, neutrophils arrived at the wound site within minutes and began engulfing bacteria. What organelle makes that digestion possible?' Wait 12 seconds. Accept responses, then affirm: 'Lysosomes — membrane-bound sacs packed with digestive enzymes. Only animal cells have them in abundance. Ask yourself: does an onion cell need to engulf bacteria? No — so no lysosomes.' Proceed organelle by organelle: CENTRIOLES — 'When Amina's epidermal cells needed to divide rapidly to close the wound surface, centrioles organised the spindle fibres. Plant cells use a different mechanism — no centrioles.' CELL WALL — 'Animal cells are flexible and can change shape. A neutrophil squeezes through capillary walls to reach the wound. Could it do that with a rigid cell wall?' Wait 10 seconds. Cold-call 2 learners. CHLOROPLASTS — 'Amina's skin cells do not photosynthesise. They get energy from glucose in her bloodstream.	Phenomenon-Anchored Direct Instruction with Colour-Code Annotation (NGSS SEP 6: Constructing Explanations). By tethering each organelle to a specific cell in the healing cut, the teacher transforms abstract organelle lists into functional explanations grounded in observable evidence. The colour-code annotation creates a visual key that learners carry forward into model building. The individual written justification (before group consolidation) ensures every learner constructs their own explanation rather than free-riding.	Observable indicator: Each learner's 3-sentence Functional Justification correctly uses the word 'because' or 'therefore' to link structure to function for at least 2 of the 3 unique plant-cell organelles. Teacher reads at least 6 individual justifications during the 5-minute writing window. Red flag: justifications that describe what an organelle does without explaining why the animal cell does or does not need it — prompt these learners with: 'You told me WHAT it does. Now tell me WHY the animal cell doesn't need it, using the cut finger as your evidence.'

		Why carry the machinery for something you will never do?' Wait 12 seconds. LARGE VACUOLE — 'Animal cells have small, temporary vacuoles. They do not need to store large amounts of water or maintain turgor — that is a plant cell strategy.' Use the colour-coding annotation in real time on the projected diagram as you explain each point. After explanation, say: 'Now write your 3-sentence Functional Justification independently. You have 5 minutes.' Circulate and read over shoulders. After 5 minutes, direct groups to consolidate: 'As a group, agree on your best justification for each unique organelle. You will present in 5 minutes.' Cold-call 2 groups to share one justification before presentations begin.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups deliver 2-minute presentations of their annotated animal cell diagrams, focusing on unique organelles and their functional justifications. Each presenting group receives structured peer feedback using the 'WWW / EBI' protocol (What Went Well / Even Better If). After all presentations, the class returns to the Driving Question Board. Learners individually write a new sticky note addressing: 'Can you now explain part of the driving question — how does an animal cell's shape and structure connect to its job?' They post this under the DQB column 'What I can NOW explain'. The teacher facilitates a 4-minute whole-class discussion to update the board and highlight which parts of the driving question remain unanswered (cell specialisation — addressed in later lessons).	Signal the start of presentations: 'Each group has exactly 2 minutes. Show your annotated diagram, name your unique organelles, and give your best functional justification. Peers — you are filling in your WWW/EBI card as they present.' Time each group strictly. After all presentations, cold-call 2 peer feedback givers per group: 'Read your EBI aloud.' Probe any vague EBI (e.g., 'be clearer') with: 'Clearer about what specifically?' After presentations, direct attention to the DQB: 'Look at the question we started with: Why do cells look so different, and how does their shape determine their job? Write on a sticky note one part of that question you can now answer — and post it.' Allow 3 minutes of silent individual writing. Then facilitate: 'Who wants to read theirs?' Cold-call 3 learners. After each, ask the class: 'Does anyone	Structured Peer Presentation with WWW/EBI Feedback Protocol, followed by DQB Metacognitive Mapping (NGSS SEP 8: Obtaining, Evaluating, and Communicating Information). The presentation consolidates group learning through public articulation. The DQB update makes metacognitive progress visible — learners see their own growing explanatory power and identify the boundary between what they know and what still requires investigation.	Observable indicator: Each learner's DQB sticky note references at least one specific organelle by name and connects it to a function relevant to the cut finger phenomenon or to the driving question. Teacher reads all sticky notes after the lesson and sorts them into 'Strong explanation', 'Partial explanation', and 'Misconception present' piles to inform Lesson 6 planning.

		disagree with that answer, or want to add evidence?' Wait 12 seconds. Close by pointing to remaining unanswered sticky notes from earlier lessons: 'These questions about specialisation — about why a platelet has no nucleus — those are coming. Today we built the foundation.'		
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative cell model journal (begun in Lesson 2 and revised through Lessons 3 and 4). They now draw Version 3: a side-by-side diagram of a plant cell and an animal cell on a double-page spread. Both diagrams must be fully labelled with organelle names, colour-coded (blue for animal-only, green for plant-only, black for shared), and annotated with a brief function note beside each unique organelle. At the bottom of the double page, learners write a 'Model Revision Note': what changed from Version 2 (plant cell only) and why. They also circle one organelle on the animal cell diagram and write one sentence connecting it explicitly to the cut finger phenomenon. The model is signed, dated, and stored as a portfolio artefact.	Say: 'Open your model journals to a fresh double page. Today you are building Version 3 — the most detailed model you have made so far. On the LEFT, redraw your plant cell from Lesson 4 — you may improve it. On the RIGHT, draw your animal cell using today's diagram as reference. Both sides must be colour-coded and labelled. At the bottom, write your Model Revision Note — what is new, what changed, and why. Then circle one animal cell organelle and write one sentence connecting it to Amina's cut finger.' Allow 15 minutes for model building. Circulate and comment specifically: 'I can see you've drawn the lysosome — your label says it digests waste. Can you make that more specific? What waste, in what cell, in the phenomenon?' Wait 10 seconds for the learner to self-correct. With 3 minutes remaining, say: 'Finish your Model Revision Note. It should be at least 2 sentences.' At time, cold-call 3 learners to read their cut-finger sentence aloud. After each, affirm the connection or press: 'How does that organelle's structure — not just its function — connect to what that cell looks like?' Close: 'Version 3 is now your most powerful scientific model. In Lesson 6, we begin asking: how do these cells become	Cumulative Model Revision (NGSS SEP 2: Developing and Using Models). The side-by-side Version 3 model externalises comparative understanding — it is not a copy of a textbook diagram but a learner-constructed artefact that records the growth of their explanation over five lessons. The Model Revision Note develops metacognitive awareness of how scientific models evolve as evidence accumulates. The mandatory phenomenon connection sentence ensures the model remains tethered to the driving question rather than becoming a decorative exercise.	Observable indicator: Version 3 models include at minimum 10 labelled organelles across both cell types, correct colour-coding for at least 5 organelles, and a Model Revision Note of at least 2 sentences that explicitly names what is new compared to Version 2. The cut-finger sentence names a specific animal cell organelle and a specific cell type from the phenomenon (e.g., 'The lysosome in the neutrophil breaks down the engulfed bacteria using digestive enzymes, which is why the infection does not spread past the cut'). Teacher collects 6 journals at random to review overnight before Lesson 6.

		so specialised — and what controls that process?'		

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners successfully distinguish lysosomes and centrioles as animal-cell-only organelles, or did common misconceptions persist (e.g., placing a large vacuole in the animal cell)? Review the Comparison Tables collected. (2) Were the group presentations evidence-based, or did learners rely on memorised lists without functional justification? If the latter, Lesson 6 should open with a brief re-anchoring to the phenomenon before moving to specialisation. (3) Were Version 3 models genuinely comparative, or did learners simply redraw the plant cell twice? If model quality is low, consider a structured model-critique activity at the start of Lesson 6 where pairs swap journals and give written feedback. (4) Review DQB sticky notes overnight: which parts of the driving question are learners now answering with confidence, and which misconceptions about cell structure versus cell specialisation remain? Use these to calibrate the entry point for Lesson 6 (Cell Specialisation). (5) Note which learners are consistently silent during cold-calls and create a targeted participation strategy for Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners received and analysed a labelled animal cell ultrastructure diagram alongside their Lesson 4 plant cell diagram. They completed a structured Comparison Table identifying organelles as present or absent in each cell type, and delivered group presentations of annotated diagrams with peer feedback.
What did I learn?	Animal cells contain lysosomes (membrane-bound sacs of digestive enzymes) and centrioles (which organise cell division) — organelles absent in plant cells. Plant cells uniquely possess a cell wall, chloroplasts, and a large central vacuole. Nucleus, mitochondria, ribosomes, ER, Golgi apparatus, and cell membrane are shared. These structural differences reflect the different functional demands on each cell type.
How does this explain the phenomenon?	Learners constructed Functional Justifications explaining why animal cells lack cell walls (flexibility needed for movement and shape change), chloroplasts (they obtain energy from food, not photosynthesis), and large vacuoles (no need for turgor pressure maintenance). They connected organelles to specific cells in the cut finger phenomenon — e.g., lysosomes in neutrophils digesting bacteria, centrioles enabling epidermal cell division during wound closure. Version 3 of the cumulative cell model was completed: a side-by-side colour-coded Plant vs. Animal diagram with a Model Revision Note.

LESSON 6 (80 minutes): Microscopy Practical 1 — Observing Onion Epidermal Cells	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to prepare and observe temporary slides of onion epidermal cells, identify key plant cell structures, and connect direct microscopic evidence to the anchoring phenomenon of wound healing.
Knowledge	Learners will know: (a) the correct procedure for preparing a temporary wet mount slide using onion epidermal tissue and iodine stain; (b) the identifiable structures visible in onion epidermal cells under low and high power — cell wall, cell membrane, vacuole, nucleus, and cytoplasm; (c) that onion epidermal cells lack chloroplasts because onion bulbs are underground storage organs not exposed to light, distinguishing them from photosynthetic plant cells.
Skills	Learners will be able to: (a) safely peel and mount a thin layer of onion epidermis on a glass slide using forceps and a mounted needle; (b) apply iodine stain using the correct drop-and-coverslip technique; (c) focus a compound light microscope at low power (×40) and high power (×100 or ×400); (d) produce accurate, labelled biological drawings of onion epidermal cells including scale and magnification; (e) record observations in a structured practical book.
Attitudes	Learners will: (a) appreciate the value of direct empirical evidence in building scientific understanding; (b) demonstrate patience and precision as they develop practical microscopy technique; (c) show respect for shared laboratory equipment and enforce safety protocols voluntarily.
Key Inquiry Question	What are the structural differences between plant and animal cells? How does cell specialisation enable organisms to carry out complex life functions?
Purpose in Storyline	This is Lesson 6 of 12 in the evidence-gathering sequence. In Lessons 1–5, learners built conceptual models of cell structure and specialisation using diagrams, text, and discussion. Lesson 6 is the first direct empirical encounter with real cells under a microscope. By examining onion epidermal cells — the same onion that the Form 1 student in Nairobi was slicing when the anchoring phenomenon began — learners gather first-hand visual evidence of plant cell structures. The observation that there are NO chloroplasts in onion skin cells generates productive surprise, deepens understanding of cell specialisation, and plants a key question that will drive Lessons 7–8 (animal cells, specialised cells). This lesson transitions the unit from conceptual modelling to evidence-based model revision.
Safety Notes	SAFETY PROTOCOLS — enforce throughout: (1) Iodine stain is an irritant — learners must avoid contact with skin and eyes; lab coats and safety goggles must be worn during staining. (2) Glass slides and coverslips are fragile — demonstrate correct handling; broken glass to be reported immediately and disposed of in the designated sharps bin, NOT the general waste bin. (3) Scalpels/razor blades are NOT to be used by learners in this practical — onion layers are separated by hand and forceps only. (4) Microscope electrical cables must not trail across walkways. (5) Learners with known iodine allergies must notify the teacher before the lesson begins and may use plain water as an alternative mountant.

B. LESSON OVERVIEW

Learners prepare temporary wet mount slides of onion (Allium cepa) epidermal cells, stain with iodine solution, and observe under a compound light microscope at low and high power. The teacher demonstrates the full technique before groups of 3–4 carry out the practical independently. Learners produce labelled biological drawings and record structured observations. A guided class discussion connects the absence of chloroplasts in onion skin to the concept of cell specialisation, directly advancing the unit's driving question. Safety protocols are enforced throughout. All observations are recorded in structured practical books as cumulative evidence for model revision at the end of the lesson.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Derivative of sin(ln(x²))](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners examine a whole onion (the same vegetable from the anchoring phenomenon). Working in pairs, they sketch what they predict a single onion skin cell looks like under a microscope, labelling any structures they expect to see. They record two specific predictions in their practical books: (1) List every structure you expect to see in an onion skin cell. (2) Will there be chloroplasts? Explain your reasoning. Pairs then share predictions with a neighbouring pair and note agreements and disagreements before the class shares out.	Hold up a whole red or white onion and say: 'This is the same onion our Form 1 student in Nairobi was slicing when the cut happened. Today we are going to look inside ONE layer of its skin — not with our eyes, but with a microscope. Before we do: in your practical books, sketch what you predict you will see. Label every structure you expect.' Allow 4 minutes of silent individual work. Then say: 'Turn to your partner. Compare your predictions. What did you both include? What is different?' Allow 2 minutes of paired talk. Cold-call 4 pairs (choose one strong predictor, one who predicted chloroplasts, one who did not, one unsure): 'Tell me ONE structure you predicted and WHY.' Record all predicted structures on the board under the heading BEFORE — Our Predictions. Do NOT correct errors yet. Say: 'Hold your predictions. In about 20 minutes, your microscope will tell us who was right.'	Prediction Journaling with Pair-Share — learners activate prior knowledge from Lessons 1–5 (cell structure diagrams) and commit to specific, testable predictions before observing. This creates cognitive dissonance when observations differ from predictions, which is the engine of conceptual change. Recording predictions in the practical book establishes a before-and-after evidence trail.	Observable indicator: Every learner has a labelled sketch with at least 3 named structures in their practical book before the practical begins. Teacher scans books during cold-call phase. Flag any learner who cannot name any structures — seat them next to a peer who can, and revisit during the Observe phase. Note on the board how many groups predicted chloroplasts — this will be returned to explicitly in Phase 3.
Observe Phase 📹 **VIDEO:** [Derivative of sin(ln(x²))](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)](#)	PART A — Teacher Demonstration (10 min): The whole class watches the teacher prepare a slide at the front bench, narrating each step aloud. PART B — Group Practical (25 min): Groups of 3–4 prepare their own slides, focus the microscope, and record observations. Each learner individually draws and labels cells at BOTH low power (×40) and high power (×100 or ×400) in their practical books. Structured	DEMONSTRATION SCRIPT — say each step aloud as you perform it: 'Step 1 — I am peeling back one layer of the onion fleshy leaf. I use my fingernail to lift the thin, almost transparent inner epidermis. Step 2 — Using forceps, I lay the epidermis flat on the slide. No folds — flat. Step 3 — I add ONE drop of iodine stain directly onto the tissue. Iodine stains starch and makes the nucleus visible. Step 4 — I lower	Structured Scientific Drawing — learners do not photograph; they draw. The act of drawing forces sustained, analytical observation: learners must decide what is significant, where boundaries are, and how structures relate spatially. Labelling with a ruler-drawn label line requires learners to commit to naming what they see, connecting visual evidence to vocabulary from Lessons 1–5. The two-magnification requirement (low	Observable indicators: (1) Each learner's practical book contains two drawings (low and high power) with a minimum of 4 labelled structures. (2) Magnification is written below each drawing. (3) The observation 'I cannot see chloroplasts' (or equivalent) is recorded. Teacher checks practical books of at least 6 learners during circulation. If drawings are inaccurate (e.g., nucleus drawn outside the cell

Source: CK-12 🔍 Search ARES for similar readings	observation prompts are written on the board: (1) Draw 3–4 cells at low power. (2) Draw 1 cell at high power. (3) Label: cell wall, cell membrane, vacuole, nucleus, cytoplasm. (4) Write: 'I can / cannot see chloroplasts.' (5) Write the magnification used for each drawing.	the coverslip at an angle using a mounted needle — like closing a book slowly — to avoid air bubbles. Step 5 — I place the slide on the stage, clip it, and start at LOW power. I use the coarse focus wheel first, then fine focus.' After focusing, say: 'What do you notice? I want 10 seconds of silence — just look.' WAIT 10 seconds. 'Now — hands up — what can you see?' Take 3 responses. During group practical: circulate every 3 minutes. Use targeted prompts: 'Is your coverslip flat?', 'Are you on low power first?', 'Can you see a cell wall? Trace it with your finger on the eyepiece.' If a group cannot focus: demonstrate re-focusing at their microscope. With 5 minutes remaining, say: 'Everyone must now have at least ONE complete drawing with labels. Check your practical book now.' Cold-call 2 groups to describe one structure they can see.	and high power) develops understanding of scale.	wall), the teacher kneels to the learner's level and says: 'Look through the eyepiece again — can you trace the outer boundary of the cell with your eye? That boundary IS the cell wall. Now draw only what you can see.'
Explain Phase 📹 **VIDEO:** Derivative of sin(ln(x²)) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Class reconvenes. Learners compare their actual observations to their Phase 1 predictions, annotating their prediction sketch in a different colour pen: circling correct predictions and crossing out incorrect ones. The teacher leads a whole-class discussion anchored on one key anomaly: 'Chloroplasts — did anyone find them?' Learners use their observations as evidence to construct an explanation for WHY onion epidermal cells have no chloroplasts, connecting to the concept of cell specialisation. Learners write a 3-sentence explanation in their own words: (1) What I observed. (2) Why this makes sense for the onion's function. (3) How this connects to	Return to the BEFORE — Our Predictions list on the board. Say: 'Open your practical book to your prediction sketch. Take a different colour pen or pencil. Circle every prediction that your microscope confirmed. Cross out every prediction that was wrong. You have 2 minutes — silent individual work.' WAIT. After 2 minutes: 'How many people predicted chloroplasts would be present?' (Show of hands — count aloud.) 'How many of you found them?' (Hands — expected: zero.) Say: 'This is a significant scientific result. The question is WHY. I want 15 seconds of thinking time — no hands yet.' WAIT 15 seconds. Cold-call 3 learners: 'Give me your best explanation for	Claim–Evidence–Reasoning (CER) Writing — learners construct a written scientific explanation using the three-part structure: Claim (what I observed), Evidence (data from the microscope drawing), Reasoning (why this makes sense given what I know about cell specialisation). This strategy formalises the transition from 'I saw X' to 'X tells us Y about how cells work,' which is the core intellectual move of the unit.	Observable indicator: Each learner's 3-sentence CER response explicitly mentions (a) absence of chloroplasts, (b) the underground / no-light context of the onion bulb, and (c) a connection to specialisation or to the anchoring phenomenon. Teacher reads 4–5 responses during the writing time (circulate, do not collect). If a learner writes only 'onion cells have no chloroplasts because they don't need them,' prompt: 'Why don't they need them? What does a chloroplast DO? Where does the onion grow?' to scaffold deeper reasoning.

	the Nairobi student's healing finger.	why there are no chloroplasts in onion skin cells.' After 3 responses, say: 'Let's build the full explanation together.' Guide with questions: 'Where does the onion bulb grow? [Underground.] Does light reach it? [No.] What do chloroplasts do? [Photosynthesis — capture light.] So would it make sense for the cell to build chloroplasts if there is no light?' Connect explicitly: 'This is CELL SPECIALISATION. The onion epidermal cell is specialised for PROTECTION and STORAGE — not photosynthesis. Just like the cells in our Nairobi student's finger are each specialised for one job: platelets for clotting, fibroblasts for collagen, epidermal cells for closing the wound.' Instruct: 'In your practical book, write 3 sentences: what you observed, why it makes sense, and how it connects to the healing finger.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Derivative of sin(ln(x²))] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each group receives two sticky notes (different colours — one for ANSWERS, one for NEW QUESTIONS). On the ANSWER note, groups write one thing today's microscopy practical has confirmed or answered about the driving question. On the NEW QUESTION note, groups write one new question today's observation has raised. Groups post their notes on the class DQB under the headings 'Evidence We Now Have' and 'Questions We Still Need to Answer.' The teacher reads 3–4 notes aloud and the class votes (thumbs up/sideways/down) on whether each answer is well-supported by today's evidence.	Distribute sticky notes. Say: 'The Driving Question asks: Why do cells look so different — and how does shape determine the job a cell can do? Today you looked at a real cell for the first time. On your ANSWER note, write one thing you can now say with evidence — not just from the textbook, but from YOUR microscope today. On your QUESTION note, write one new question your observation raised — something you still cannot explain.' Allow 3 minutes. Circulate and prompt groups that are stuck: 'What surprised you most today?' or 'What would you need to see next to understand the healing finger better?' Invite groups to post notes on the DQB. Read 3 aloud. For each, say: 'Does today's evidence support this answer?	Driving Question Board (DQB) Progressive Mapping — the DQB functions as a public, cumulative record of the class's growing understanding. By distinguishing 'evidence we now have' from 'questions we still need to answer,' learners experience science as an iterative process. The thumbs vote introduces peer evaluation of evidence quality, a key component of scientific reasoning.	Observable indicator: Every group posts at least one answer note that references a specific observation from today's practical (e.g., 'We can see cell walls in plant cells — animal cells in the wound don't have these'). Teacher flags any answer note that is purely textbook-recalled without reference to today's observation and returns it to the group with the prompt: 'Can you re-write this using something you actually saw today?'

		Thumbs up if yes, sideways if partly, down if we're not sure yet.' Narrate: 'Look at our board — it is growing. Each lesson we add more. By Lesson 12, we should be able to answer the driving question fully.'		
Model Building Phase 📹 **VIDEO:** [Derivative of sin(ln(x²))] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the cumulative cell model they began in Lesson 1 (a hand-drawn or annotated diagram of a 'generic cell'). Using a different colour pen/pencil, they revise or add to their model based on today's direct microscopic evidence. Specific revision prompts are on the board: (1) Redraw or confirm the cell wall — is it thicker or thinner than you originally drew? (2) Add the large central vacuole if it was not in your original model. (3) Annotate your model: 'Chloroplasts — ONLY in cells exposed to light (e.g., leaf cells) — NOT in onion epidermis, NOT in human cells.' (4) Add a caption: 'This model is based on evidence from: [list sources used so far in the unit].' Learners then write one sentence predicting what will be different when they observe an animal cell in the next lesson.	Say: 'Open to your unit model — the diagram you have been building since Lesson 1. Today your microscope gave you real evidence. Use a NEW colour to update your model. You have 6 minutes.' Display the four revision prompts on the board. Circulate — specifically check that learners are annotating chloroplasts correctly. After 5 minutes, say: 'Last 60 seconds — write one sentence: what do you predict will be DIFFERENT when we look at an animal cell next lesson? Think about: platelets, neutrophils, fibroblasts from our Nairobi student's finger.' WAIT for writing. Cold-call 3 learners to read their prediction aloud. Say: 'Hold that prediction. Lesson 7 — we test it.' Close by pointing to the DQB: 'We now have real evidence from a real cell. Our model is stronger. But we still have questions. That is exactly how science works.'	Cumulative Model Revision — learners do not start a new diagram each lesson; they annotate and revise a single evolving model across all 12 lessons. The requirement to use a different colour for each lesson's additions makes the growth of understanding visually explicit. Citing evidence sources ('based on evidence from…') develops scientific epistemic habits — knowing not just WHAT we believe but WHY and on what basis.	Observable indicator: The revised model contains at least one addition or correction made in a new colour that is traceable to today's practical observation (not copied from a textbook diagram). The chloroplast annotation explicitly notes the condition under which chloroplasts are present (light-exposed cells) versus absent (onion epidermis, animal cells). The prediction sentence for Lesson 7 names at least one specific structural difference between plant and animal cells. Teacher collects practical books of one group (rotate groups each lesson) for detailed written feedback — returned before Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions before planning Lesson 7:

(1) EVIDENCE QUALITY: Did every learner produce a labelled drawing with at least 4 structures at both low and high power? If not — was this a focusing skill issue (needs more practice), a labelling knowledge issue (needs vocabulary consolidation), or a time issue (practical took too long)? Adjust Lesson 7 timing accordingly.

(2) THE CHLOROPLAST MOMENT: Did the absence of chloroplasts generate genuine surprise and productive discussion, or did learners already know this and treat it as routine? If the latter, the anomaly was not sufficiently foregrounded — consider opening Lesson 7 with a more provocative prediction prompt.

(3) CER WRITING: How many learners successfully linked their observation to the anchoring phenomenon in their 3-sentence explanation? If fewer than 60% made the explicit connection to the healing finger, begin Lesson 7 with a 3-minute re-read of the phenomenon before any new content.

(4) DQB VITALITY: Are the new questions on the DQB substantive and forward-pointing (e.g., 'Will animal cells have a cell wall?', 'How does the fibroblast know to make

collagen?') or vague and generic? Substantive questions indicate learners are genuinely engaging with the phenomenon. Vague questions may indicate the DQB routine needs re-scaffolding.

(5) SAFETY: Were all safety protocols followed without prompting? Note any groups that needed reminders about goggles or broken glass disposal — address this explicitly at the START of Lesson 7 before the second microscopy practical begins.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Onion epidermal cells prepared as temporary wet mount slides stained with iodine: rectangular cells with visible cell wall, large central vacuole, nucleus stained dark brown/yellow by iodine, cytoplasm, and cell membrane pressed against the inner cell wall. NO chloroplasts observed.
What did I learn?	Onion epidermal cells are plant cells that contain a cell wall, vacuole, and nucleus — but NO chloroplasts, because the onion bulb grows underground and is not exposed to light. This is evidence of cell specialisation: the onion epidermal cell is adapted for protection and storage, not photosynthesis. All plant cells share some features (cell wall, vacuole) but differ in which organelles they contain depending on their function.
How does this explain the phenomenon?	The absence of chloroplasts in onion skin cells is not an error — it is evidence of specialisation. Cells only build and maintain the structures they need for their specific job. Just as the cells healing the Nairobi student's cut finger are each specialised (platelets for clotting, fibroblasts for collagen, epidermal cells for surface repair), the onion epidermal cell is specialised for its role. This confirms that having the same DNA does not mean every cell builds the same structures — gene expression is selective, and structure is determined by function.

LESSON 7 (80 minutes): Microscopy Practical 2 — Human Cheek Cells: Observing Animal Cell Structure	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to prepare temporary slides of human cheek (buccal) cells, observe their structure under a microscope, and directly compare animal cell architecture with the plant cell observations made in Lesson 6.
Knowledge	Learners will know that animal cells possess a cell membrane, cytoplasm, and nucleus but lack a cell wall, chloroplasts, and a large central vacuole. Learners will know that the absence of a rigid cell wall gives animal cells a flexible, irregular shape suited to their functions.
Skills	Learners will be able to: (i) prepare a temporary wet mount of cheek cells using methylene blue stain; (ii) focus a compound light microscope to obtain a clear image of cheek cells; (iii) produce a labelled biological drawing of cheek cells to standard conventions; (iv) articulate at least two directly observed structural differences between plant (onion) and animal (cheek) cells.
Attitudes	Learners will appreciate the value of careful, systematic observation in revealing biological truths that are otherwise invisible to the naked eye, and will take personal responsibility for hygienic laboratory practice.
Key Inquiry Question	What are the structural differences between plant and animal cells, and what do those differences suggest about how each cell type functions?
Purpose in Storyline	In Lessons 1–6 learners established that the healing cut involves multiple specialised cells, and in Lesson 6 they built a reference image of a plant cell from onion epidermis. Lesson 7 now gives learners their first direct look at an animal cell — their own cheek cells — so they can gather concrete, first-hand evidence of what animal cells do and do not contain. This evidence is essential for answering the driving question: the flexible, wall-free shape of a cheek epithelial cell foreshadows why a red blood cell can be disc-shaped, why a platelet can change form during clotting, and why epidermal cells can migrate across a wound. Every structural observation today feeds directly into the class model of cell specialisation.
Safety Notes	Learners must use clean cotton swabs (one per learner — never shared) to collect cheek cells; used swabs must be disposed of immediately into a closed bin containing dilute disinfectant (1% bleach solution prepared by the teacher in advance). Methylene blue is a mild irritant — learners must avoid contact with eyes and skin and wash hands thoroughly after the practical. Microscope slides and cover slips are fragile — broken glass must be reported immediately and swept up by the teacher using a dustpan, never bare hands. No learner should perform this practical if they have an open sore or active oral infection; the teacher should identify and assign them the role of data recorder instead.

B. LESSON OVERVIEW

Learners open the lesson by revisiting their Lesson 6 annotated drawings of onion epidermal cells and making written predictions about what they expect to find — or not find — in a human cell. They then conduct a full microscopy practical: scraping their own cheek cells with a cotton swab, preparing a methylene blue wet mount, focusing the microscope, and producing a labelled biological drawing. A structured pair-comparison activity places the cheek cell drawing directly beside the onion cell drawing, and learners generate a two-column evidence table of observed similarities and differences. The lesson closes with a Driving Question Board (DQB) update in which learners add new evidence cards and revise their cumulative class model of cell structure, sharpening the contrast between plant and animal cell architecture.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** The mole and Avogadro's number Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their Lesson 6 annotated drawings of onion epidermal cells from their science notebooks. Working individually for 3 minutes, they write a 'Prediction Card' answering: 'I am about to look at a cell from the inside of my own cheek. Compared with the onion cell I drew last lesson, I predict I will see ___ and I predict I will NOT see ___, because ___.' Learners then share predictions with a shoulder partner (2 min), and the teacher cold-calls four pairs to share one agreement and one point of disagreement before the practical begins.	Display the Lesson 6 onion cell drawing on the board (use a pre-prepared exemplar or photograph a strong student drawing from Lesson 6). Say: 'Before we touch any equipment, I want your brain to do some work. Look at your onion cell drawing. That was a PLANT cell. You are about to look at YOUR OWN cell — an animal cell. On your Prediction Card, complete these two stems: I predict I WILL see… and I predict I will NOT see… Give me a reason for each.' Allow 3 minutes of silent individual writing. DO NOT accept verbal answers yet — enforce silence to protect thinking time. After 3 minutes say: 'Turn to your shoulder partner. You have 2 minutes — find ONE thing you agree on and ONE thing you disagree on.' After pair talk, cold-call exactly 4 pairs using the class register (rows 1, 3, 5, 7): 'Pair in row 1 — what did you AGREE on?' Record responses on the board under two columns: AGREED and DISAGREED. Do not correct misconceptions yet — annotate with a '?' to flag for revisiting. Close the predict phase by saying: 'Keep your Prediction Card visible — we will come back to it the moment we finish observing. Scientists always test their own predictions.'	Prediction Activation with Anchored Retrieval — learners are required to retrieve prior observational knowledge (onion cell structure from Lesson 6) and apply it as a cognitive baseline before new evidence is gathered. This creates productive cognitive dissonance when animal cell observations diverge from predictions, making the structural differences more memorable and meaningful.	Teacher circulates during individual writing and scans Prediction Cards for two indicators: (1) Does the learner correctly identify at least one plant-cell-specific structure (cell wall, chloroplasts, large vacuole) that they predict will be absent in the cheek cell? (2) Does the learner offer a reason connected to cell type (plant vs. animal), not just a guess? Cards that show neither indicator flag learners who need targeted questioning during the practical phase. Record names for follow-up.
Observe Phase 📹 **VIDEO:** The mole and Avogadro's number	Learners work in pairs at bench stations. Each learner independently prepares their own cheek cell slide following a step-by-step printed protocol card: (1) Gently scrape the inner cheek with	Before learners touch equipment, deliver a 4-minute safety and technique briefing: 'Three non-negotiables today: one swab per person — no sharing; used swabs go directly into the bleach bin —	Direct Observational Drawing with Structured Comparative Table — biological drawing is not merely an art task; requiring learners to commit to labelled lines forces active discrimination between what	Teacher uses a 'Drawing Checklist Clipboard' — a printed grid with learner names. For each pair, check: (1) Drawing is large (≥5 cm), in pencil, with smooth outlines — no shading or artistic hatching.

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Evolution of Simple Cells (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the flat end of a clean cotton swab. (2) Smear the swab in a thin circle on the centre of a clean glass slide. (3) Add ONE drop of methylene blue stain. (4) Lower a cover slip at 45° to avoid bubbles. (5) Use tissue to blot excess stain. (6) Place on microscope stage and focus under low power (×4 objective), then switch to medium power (×10), then high power (×40). (7) Select the clearest, most isolated cell visible and produce a large pencil biological drawing (minimum 5 cm diameter) in the designated box in their notebook. Learners label every structure they can positively identify with a ruled label line and a clear label word. They annotate their drawing with the magnification used and the date. Partner observes the same slide and verifies the drawing before the slide is discarded safely.	do not place them on the bench; if a slide breaks, hands OFF and call me immediately.' Demonstrate the cheek smear and staining technique live at the front bench, narrating each step aloud. Emphasise: 'The smear should be thin enough to read newsprint through it — too thick and you will not see individual cells.' Circulate continuously during the practical. At each station ask one diagnostic question — rotate through: 'What is the blue stain doing to the cell?', 'Why can you see the nucleus more clearly now?', 'What structure would you expect to see in the onion cell that you do NOT see here?'. Allow 25 minutes total for slide preparation, focusing, and drawing. At the 20-minute mark give a verbal time warning: 'Five minutes remaining for your drawing — make sure every structure you can see has a label line.' After drawings are complete, direct learners to compare their drawing side-by-side with their Lesson 6 onion cell drawing and complete a two-column evidence table in their notebooks: ONION CELL (plant) | CHEEK CELL (animal) — list every structure under each column, marking shared structures with a tick and unique structures with a star.	is and is not visible. The immediate side-by-side comparison with the Lesson 6 onion drawing transforms two individual observations into comparative evidence, which is the core epistemic move needed to answer the driving question about structural differences between cell types.	(2) At least three structures labelled (cell membrane, cytoplasm, nucleus at minimum). (3) Magnification recorded. (4) No cell wall labelled — if a learner has labelled a cell wall on the cheek cell, stop and ask: 'Show me the cell wall on your slide right now under the microscope.' This is the single most critical misconception to intercept in real time. Record any learner who cannot distinguish cell wall absence as a targeted follow-up for Phase 3.
Explain Phase 📹 **VIDEO:** The mole and Avogadro's number Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners participate in a whole-class 'Evidence Harvest' (10 min). Each pair contributes ONE observation from their comparative evidence table to a class T-chart built on the board: PRESENT IN BOTH | ONLY IN PLANT CELL (onion) | ONLY IN ANIMAL CELL (cheek). After the T-chart is populated, learners individually write a 3-sentence scientific	Facilitate the Evidence Harvest using a structured round-robin: go around the room, one pair per turn, each pair contributes ONE item to the T-chart. Say: 'I need one observation from your table — something you actually SAW, not something from the textbook.' After the T-chart is complete (allow 6–7 minutes), ask the class: 'Look at the ONLY IN PLANT CELL	Evidence Harvest T-Chart followed by Phenomenon-Anchored Sentence Stem Writing — the T-chart externalises and organises collective observational evidence so the class builds a shared knowledge base rather than 30 isolated individual observations. The phenomenon sentence stem then requires each learner to privately apply that shared	Collect and read the 3-sentence explanations and the phenomenon stem as an exit indicator. Look for three quality markers: (1) At least two specific structures named with correct terminology. (2) The word 'flexible' or an equivalent (e.g., 'can change shape', 'no rigid wall') appears in the explanation. (3) The phenomenon sentence correctly links the absence of a cell wall to

📖 **HTML:** [Evolution of Simple Cells (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	explanation in their notebooks answering: 'What structural differences between plant and animal cells did you directly observe today, and what might the absence of a cell wall mean for an animal cell's ability to change shape?' Learners then connect their explanation back to the anchoring phenomenon by completing the sentence stem: 'When the platelets in Aisha's cut finger change shape to form a clot, this is possible because animal cells…'	column. Every item in that column — what do they all have in common in terms of function?' Wait 12 seconds (count silently). Cold-call three learners. Guide toward the answer that plant-specific structures (cell wall, large vacuole, chloroplasts) are all related to the plant's fixed, non-moving lifestyle. Then say: 'Now look at the ONLY IN ANIMAL CELL column — or notice what is NOT there. What does the absence of a rigid cell wall allow an animal cell to do that a plant cell cannot easily do?' Wait 10 seconds. Cold-call two learners. After responses, explicitly connect to the phenomenon: 'Remember Aisha's platelets — tiny, irregular animal cells that flatten and extend pseudopods to patch the wound. Could they do that if they had a rigid cell wall? Turn and tell your partner — 30 seconds.' Cold-call one pair. Close the explain phase by directing learners to write their 3-sentence individual explanation and the phenomenon sentence stem. Circulate and read over shoulders — look for correct use of the terms: cell membrane, cytoplasm, nucleus, cell wall, flexibility.	knowledge back to the original puzzling event, closing the predict–observe–explain loop and reinforcing that science observations have explanatory power.	platelet function or animal cell motility. Learners who meet all three markers are ready for Phase 4. Learners who meet one or zero markers receive a targeted prompt card at the DQB phase: 'Go back to your T-chart — which column has cell wall? Is a cheek cell in that column?'
Driving Question Board (DQB) Creation 📹 **VIDEO:** [The mole and Avogadro's number] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Each learner writes one new 'Evidence Card' (on a sticky note or index card) using the sentence frame provided: 'We observed that animal cells have no cell wall — this tells us…' They post their card on the class DQB under the section labelled 'Lesson 7 Evidence'. Learners then read at least two classmates' cards and place a small tick on any card that adds an idea they had not written themselves. The class briefly	Give learners 3 minutes of silent writing time for their Evidence Card. Enforce silence: 'This is YOUR thinking — no copying from the board yet.' After cards are posted, say: 'Walk the board. Read at least two cards that are NOT yours. Tick any card that teaches you something new.' Allow 2 minutes of gallery walk. Then facilitate a 5-minute whole-class DQB debrief: 'Which card on this board most directly answers part	Collaborative Evidence Curation on a Living DQB — the DQB is not a display; it is a living epistemic record that makes the class's growing understanding visible and accountable. Requiring learners to read and tick peers' cards introduces social comparison of ideas, which research shows increases conceptual depth. The teacher's move of explicitly naming 'new questions raised by new evidence' models the iterative	Teacher photographs the DQB after the lesson for analysis. Evaluate the distribution of Evidence Cards against three tiers: Tier 1 — card simply restates observation ('We saw no cell wall'); Tier 2 — card makes an inference ('This tells us animal cells can be more flexible'); Tier 3 — card connects to the phenomenon ('This tells us platelets can change shape to plug Aisha's wound because they have no rigid wall').

📖 **HTML:** [Evolution of Simple Cells (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	discusses (5 min) which new evidence cards most directly advance the driving question: 'Why do cells look so different from each other, and how does shape determine job?'	of our driving question?' Cold-call three learners to nominate a card and explain their choice. Ask: 'What NEW question does today's evidence raise that we have not yet answered?' Record any new questions on the DQB under a 'NEW QUESTIONS — Lesson 7' strip. Likely emergent questions include: 'If animal cells have no wall, what stops them from bursting?', 'Do all animal cells look like cheek cells?', and 'Why does the nucleus look darker in some cells?' Validate all questions: 'These are exactly the questions a biologist would ask. We will need more evidence to answer them.' Point to the driving question displayed at the top of the DQB and ask: 'Can we answer it yet — fully? Partially? What piece of the puzzle are we still missing?' Accept partial answers only — the full answer requires specialisation lessons ahead.	nature of scientific inquiry and sustains intellectual curiosity across the 12-lesson sequence.	Aim: by Lesson 7, at least 50% of cards should be Tier 2 or 3. Cards that are Tier 1 only flag learners for the teacher's targeted small-group discussion at the start of Lesson 8.
Model Building Phase 📹 **VIDEO:** [The mole and Avogadro's number] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Evolution of Simple Cells (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative 'Cell Structure Model' page — a running annotated diagram that has been updated in every lesson since Lesson 5. Today they add a second diagram: a correctly proportioned, labelled animal cell (cheek cell) drawn to the same scale as their existing plant cell diagram. They then draw arrows between matching structures in the two diagrams (e.g., nucleus ↔ nucleus) and write one sentence next to each structure that is ABSENT in the animal cell, explaining what function that missing structure serves in the plant cell. Finally, learners write a 'Model Update Statement' at the bottom of the page: 'My model now shows… but does not yet	Say: 'Open to your Cell Structure Model page. You have been building this since Lesson 5. Today you are adding the animal cell — draw it next to your plant cell. Make them roughly the same size so we can compare directly.' Allow 8 minutes of independent drawing and annotation. Circulate and check: (1) Is the animal cell drawn without a cell wall? (2) Is the nucleus drawn as a distinct, membrane-bound structure inside the cytoplasm? (3) Are label lines ruled and horizontal? After 8 minutes, ask learners to write their Model Update Statement: 'Write: My model now shows… but does not yet explain…'. Say: 'The second part is just as important as the first. A scientist who knows	Cumulative Annotated Model Building with Explicit Knowledge-Boundary Tracking — requiring learners to add to a running model (rather than draw a fresh one each lesson) makes conceptual progression visible and tangible. The 'does not yet explain' clause is a metacognitive tool: it trains learners to distinguish between what the model represents and what it cannot yet account for, which is a core science and engineering practice (developing and using models, NGSS SEP 2) and directly sustains the epistemic arc of the 12-lesson sequence.	Teacher collects or photographs the Model pages at the end of the lesson. Assess against four criteria: (1) Animal cell drawn without cell wall — PASS/FAIL. (2) At least three structures correctly labelled on the animal cell. (3) At least two 'absent structure' sentences correctly link missing structure to its plant-cell function. (4) Model Update Statement names a specific, scientifically valid gap (e.g., 'does not yet explain why red blood cells have no nucleus' rather than 'does not yet explain everything'). Criterion 1 failure requires intervention at the start of Lesson 8. Criteria 3–4 inform groupings for the Lesson 8 cell specialisation collaborative task.

	explain…' to track the boundary of their current understanding.	what their model CANNOT explain yet is a scientist who knows what to investigate next.' Cold-call three learners to read their 'does not yet explain' clause aloud. Record these on the board as 'Open Questions for Our Model' — they become the bridge to Lesson 8 on cell specialisation. Close the lesson: 'You have just looked at your own cells under a microscope — the same type of cell that lines the inside of the wound when Aisha cut her finger. Next lesson, we ask: if all these cells started the same way, how did they end up so different?'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. PRACTICAL SAFETY & HYGIENE — Was the bleach disposal bin used correctly by all learners? Were there any incidents of swab sharing or unsafe glass handling? What adjustments will I make to the safety briefing before the next practical lesson?

2. DRAWING QUALITY — How many learners produced drawings that met the biological drawing standard (large, pencil, labelled, magnification recorded)? Which aspects of technique (label lines, outline smoothness, proportions) were most commonly weak? Do I need to model biological drawing conventions again at the start of Lesson 8?

3. CRITICAL MISCONCEPTION — Did any learners label a cell wall on their cheek cell drawing? If yes, how many, and did my in-the-moment questioning resolve the misconception? Will I need to address this again in Lesson 8 with a targeted comparison activity?

4. DQB EVIDENCE CARD QUALITY — What proportion of Evidence Cards reached Tier 2 or Tier 3 (inference or phenomenon connection)? What does this tell me about learners' readiness to engage with cell specialisation in Lesson 8?

5. PHENOMENON CONNECTION — During the Explain phase, did learners spontaneously connect the flexibility of the animal cell to platelet function in the healing cut, or did they need heavy scaffolding from me? If scaffolding was heavy, I should revisit the phenomenon narrative at the start of Lesson 8 before introducing new content.

6. EQUITY & INCLUSION — Were there any learners who could not participate in the cheek swab practical (oral sores, personal discomfort)? Were alternative roles (data recorder, drawing analyst) genuinely meaningful for those learners, or did they feel excluded? How can I design Lesson 8 to ensure full re-engagement?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners prepared temporary wet mounts of their own cheek (buccal) cells using methylene blue stain, focused them under a compound light microscope, and produced labelled biological drawings. They directly observed a cell membrane, cytoplasm, and nucleus in each cheek cell.
What did I learn?	Animal cells (such as human cheek epithelial cells) contain a cell membrane, cytoplasm, and a membrane-bound nucleus, but lack a cell wall, chloroplasts, and a large central vacuole — structures present in the plant (onion) cells observed in Lesson 6. The

	absence of a rigid cell wall allows animal cells to adopt flexible, irregular shapes.
How does this explain the phenomenon?	The structural differences between plant and animal cells are not random — they reflect the different needs and lifestyles of each cell type. The absence of a cell wall in animal cells allows them to change shape, which is directly relevant to the anchoring phenomenon: cells such as platelets and epidermal cells can flatten, extend, and migrate during wound healing precisely because they are not constrained by a rigid outer wall. This is one piece of the larger puzzle of how cell shape determines cell function.

LESSON 8 (80 minutes): Cell Specialisation: Meaning, Significance, and Specialised Animal Cells	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to explain the concept of cell specialisation and relate the structural adaptations of specialised animal cells to their specific functions, using the healing-cut phenomenon as the anchoring context.
Knowledge	Learners will know: (1) the meaning and significance of cell specialisation; (2) the structural features of red blood cells, white blood cells, nerve cells, sperm cells, and muscle cells; (3) how each structural feature directly enables the cell's function; (4) how sickle-cell anaemia illustrates the consequences of disrupted structure-function relationships.
Skills	Learners will be able to: (1) identify and describe specialised animal cells from diagrams and micrographs; (2) construct annotated diagrams of at least three specialised animal cells; (3) logically link each structural adaptation to its functional advantage; (4) use evidence from the sickle-cell supporting phenomenon to argue that cell shape determines cell function.
Attitudes	Learners will: (1) appreciate how microscopic cell differences underpin complex body processes such as wound healing; (2) show curiosity and open-mindedness when confronting unfamiliar cell shapes; (3) value collaborative sensemaking and respect peers' reasoning during discussions.
Key Inquiry Question	How does cell specialisation enable organisms to carry out complex life functions?
Purpose in Storyline	Lesson 8 is the pivot point of the evidence-gathering sequence. Learners have spent Lessons 1–7 building knowledge of cell structure, organelles, and the distinction between plant and animal cells. Now they must answer the central puzzle of the anchoring phenomenon: if all cells share the same DNA, why do platelets, neutrophils, fibroblasts, and epidermal cells look so radically different? This lesson introduces cell specialisation as the explanatory framework, anchors five key specialised animal cells to their roles in wound healing, and uses sickle-cell anaemia as a powerful supporting phenomenon that proves structure determines function. By the end, learners can update their cumulative model with a new layer: cells are not generic — they are shaped by function.
Safety Notes	No hazardous materials are used in this lesson. If printed micrograph cards are handled, ensure hands are washed before and after. Remind learners that the sickle-cell discussion involves a medical condition — maintain sensitivity and respectful language at all times. If any learner discloses personal or family experience with sickle-cell disease, acknowledge with care and redirect focus to the science.

B. LESSON OVERVIEW

In this 80-minute lesson, learners revisit the anchoring phenomenon — the healing cut — and confront the question of why cells involved in healing look so different from one another despite originating from the same organism. The teacher re-anchors the class by displaying the phenomenon image and posing a targeted challenge question. Learners then move through a structured 5-phase cycle: they predict which structural features different animal cells might need; gather evidence by examining annotated diagrams and micrographs of five specialised animal cells (red blood cells, white blood cells, nerve cells, sperm cells, and muscle cells); construct structure-function explanations using a graphic organiser; update the Driving Question Board; and revise their cumulative cell model. The sickle-cell anaemia supporting phenomenon is woven through the Explain phase to consolidate the argument that shape is not cosmetic — it is functional. All five cells are explicitly connected to their roles in the healing-cut sequence. Lesson 8 of 12 in the evidence-gathering sequence.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Professional antigen presenting cells (APC) and MHC II complexes] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the phenomenon anchor image (a close-up of a healing finger cut) and a split-screen showing five unlabelled micrograph outlines of specialised animal cells. Working individually for 3 minutes, each learner records predictions in their science notebook under the prompt: 'Each of these cells plays a role in healing this cut. Sketch what structural features you think each cell might need — and explain why.' Learners then share predictions with a shoulder partner for 2 minutes before a whole-class cold-call share-out. Predictions are displayed on a sticky-note cluster on the Driving Question Board under the header 'What we THINK we know — Lesson 8'.	Display the phenomenon image and the five cell outlines side by side. Say verbatim: 'You have all seen a cut heal. You now know cells have organelles. Here is the puzzle for today — these five cells all live in your body right now, and each one helped heal that cut on the finger. But they look completely different. Before I tell you anything, I want you to predict: what structural features would each cell NEED to do its job?' Distribute the prediction sheet. Allow 3 minutes of individual silent thinking — do NOT circulate or prompt during this time (enforce wait time). After 3 minutes, say: 'Turn to your shoulder partner. You have 2 minutes. Compare your sketches — where do you agree? Where do you disagree?' After partner talk, cold-call 5 different learners (one per cell outline) — say: 'Amina, tell me what feature you predicted for the round cell and why.' Record key predictions verbatim on the board in a 'Predictions' column. Do not confirm or correct — say: 'We will come back to every one of these predictions.'	Activation of Prior Knowledge + Phenomenon Re-anchoring. By forcing learners to predict before instruction, the teacher surfaces existing mental models (including misconceptions) and creates a cognitive need — a 'prediction gap' — that motivates the Observe phase. Connecting predictions explicitly to the healing-cut phenomenon ensures the lesson question feels authentic rather than abstract.	Observable indicator: Each learner has at least one annotated sketch in their notebook before partner talk begins. During cold-call, listen for whether learners connect structural features to function (e.g., 'it needs to be flat to carry more oxygen') versus purely visual guesses (e.g., 'it looks round'). Record names of learners who show functional reasoning — these learners can be used as peer resources in Phase 3. Flag learners who leave the prediction blank for targeted support in Phase 2.
Observe Phase 📹 **VIDEO:** [Professional antigen presenting cells (APC) and MHC II complexes] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners work in groups of four to examine an evidence pack containing: (a) labelled, annotated diagrams of five specialised animal cells — red blood cell (biconcave disc, no nucleus, packed with haemoglobin), white blood cell/neutrophil (irregular lobed nucleus, many lysosomes, flexible membrane), nerve cell/neuron (long axon, myelin sheath, dendrites, synaptic knobs), sperm cell (streamlined head, midpiece	Distribute evidence packs and say: 'Your job right now is NOT to explain — your job is to observe carefully and record what you see. Use the observation table. For every structural feature you notice, write it down. For features that are absent — like a nucleus that isn't there — write that down too. Absences are just as important as presences.' Set a visible 15-minute timer. Circulate and narrate observations without explaining:	Structured Observation with Noticing Frames. The observation table scaffolds learners to separate raw observation from interpretation — a core NGSS Science and Engineering Practice (Planning and Carrying Out Investigations; Analysing and Interpreting Data). Requiring learners to record absences (e.g., no nucleus in RBCs) is a deliberate strategy: absence as evidence is a higher-order	Observable indicator: Each group's observation table has entries in all four columns for at least 4 of the 5 cells within 15 minutes. Circulate and check that learners are recording specific structural features (e.g., 'long axon', 'no nucleus') rather than vague descriptors (e.g., 'looks different'). If a group's 'Questions Raised' column is empty, prompt: 'If you were that red blood cell squeezed through a capillary thinner than

📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	packed with mitochondria, long flagellum), and muscle cell/myocyte (many mitochondria, parallel myofilaments, multinucleate); (b) two printed micrograph images (one showing normal red blood cells vs. sickle cells; one showing a nerve cell stained with methylene blue); and (c) a data card stating: 'A red blood cell lives for ~120 days and makes ~170,000 circuits of the body. A neutrophil lives for ~6 hours during active infection.' Each group completes a structured observation table with columns: Cell Name | Structural Features Observed | Features Absent | Questions Raised. Groups are given 15 minutes. A gallery of enlarged cell diagrams is posted around the classroom walls — groups may rotate to examine them at any point during this phase.	e.g., say 'I can see your group has noticed the biconcave shape — keep going, what else?' Do NOT answer 'why' questions yet — redirect with: 'Write that question in the Questions Raised column — that is exactly what Phase 3 is for.' At the 8-minute mark, do a mid-observation check: cold-call one group: 'Group 3 — read me one feature you noticed about the nerve cell that surprised you.' At the 14-minute mark, give a 1-minute warning. After 15 minutes, say: 'Pens down. Every group should have at least 3 structural features per cell recorded.'	observation skill that directly sets up the structure-function argument in Phase 3. The sickle-cell micrograph creates a visual contrast that primes the supporting phenomenon discussion.	itself, what question would YOU have?' Use the data card responses as a quick check — groups that can connect the 120-day lifespan to the absence of a nucleus (no energy wasted on DNA replication) are demonstrating advanced reasoning to flag for Phase 3 cold-calls.
Explain Phase 📹 **VIDEO:** [Professional antigen presenting cells (APC) and MHC II complexes] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners remain in their groups and receive a Structure-Function Graphic Organiser — a table with columns: Cell | Key Structural Feature | Why That Feature Matters (Function) | Connection to the Healing Cut | Evidence from Sickle-Cell Phenomenon. Groups spend 20 minutes completing the organiser, constructing written explanations that directly link each structural feature to its functional advantage. The teacher then leads a whole-class share-out (10 minutes) using a completed version of the organiser projected on the board, cold-calling groups to contribute one cell at a time. The sickle-cell anaemia supporting phenomenon is introduced at this point: the teacher displays the micrograph comparison (normal	Distribute the graphic organiser and say: 'Now we move from observing to explaining. For every structural feature you recorded, I want you to answer: WHY does that feature exist? What job does it help the cell do? And — critically — how does that connect to what happened the day that Form 1 student in Nairobi cut their finger?' Allow 20 minutes for group work. Circulate and use targeted prompts: For RBCs — 'You said no nucleus. What does a nucleus do? So what happens to all that space inside the cell?' For nerve cells — 'You said long axon. How far does a pain signal need to travel from a fingertip to the brain? What does length make possible?' For muscle cells — 'You said many mitochondria. What do	Claim-Evidence-Reasoning (CER) Framework + Supporting Phenomenon Integration. CER is an NGSS-aligned sensemaking strategy (Constructing Explanations and Designing Solutions; Engaging in Argument from Evidence) that requires learners to move beyond description into causal explanation. Using sickle-cell anaemia as the supporting phenomenon is deliberately chosen because it provides a natural experiment: same molecule (haemoglobin), different shape, catastrophically different function. This makes the structure-function argument falsifiable and evidence-based rather than assertion-based. The graphic organiser bridges group thinking	Observable indicator: Each group's graphic organiser has a complete CER for at least 3 cells, with the 'Connection to the Healing Cut' column populated. During cold-calls, listen specifically for causal language: words like 'because', 'therefore', 'which means', 'as a result' indicate explanatory reasoning rather than simple description. A learner who says 'the red blood cell has no nucleus because that leaves more room for haemoglobin, which means it can carry more oxygen to the healing tissue' is demonstrating target-level reasoning. A learner who says 'the red blood cell has no nucleus and carries oxygen' is demonstrating a lower level — record for follow-up. The sickle-cell CER responses are the highest-

	vs. sickle RBCs) and poses the key argument question: 'Sickle-cell red blood cells have haemoglobin — so why do people with sickle-cell disease still suffer from anaemia and blocked vessels?' Learners argue from evidence in pairs for 3 minutes before a whole-class cold-call. The teacher uses the Claim-Evidence-Reasoning (CER) frame to scaffold responses: 'Make a CLAIM about what the problem is. Give EVIDENCE from the micrograph. Give REASONING connecting shape to function.'	mitochondria produce? Why would a muscle cell need a LOT of that?' After 20 minutes, begin whole-class share-out. Cold-call one group per cell — say: 'Group 2, give me your CER for the red blood cell — Claim, Evidence, Reasoning.' After each response, ask the class: 'Does any group want to add or challenge?' For sickle-cell: display the micrograph and say — 'Look at the shape of the sickle cell. Here is the argument question: the sickle cell still contains haemoglobin. So why does the shape matter?' Allow 10 seconds of wait time (count silently). Cold-call 3 learners. Listen for the key reasoning: the abnormal shape prevents the cell from passing through small blood vessels and reduces its ability to carry oxygen efficiently. Confirm and extend: 'This is your most powerful piece of evidence yet that SHAPE IS FUNCTION. Shape is not decoration.'	and public argumentation.	stakes formative checkpoint of the lesson: if fewer than 60% of cold-called learners can articulate the shape-blockage-function link, pause and re-teach using the blood vessel diameter analogy before moving to Phase 4.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Professional antigen presenting cells (APC) and MHC II complexes] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES	Each learner individually writes on a sticky note (or index card) their single most important update to their understanding of the driving question: 'Why do cells in our body look so different from each other — and how does the shape of a cell determine the job it can do?' Learners also write one new question that this lesson has raised for them. Sticky notes are placed on the class DQB in two columns: 'What we NOW understand' and 'New questions we have.' The teacher reads 4–5 contributions aloud and facilitates a 5-minute whole-class discussion to identify patterns — what has the class collectively come to understand? What remains	Distribute sticky notes and say: 'Three minutes, individual and silent. Write one thing you now understand about our driving question that you did NOT understand at the start of today's lesson. On the back, write one question this lesson has raised for you.' After 3 minutes, direct learners to post their notes on the DQB. Scan the board quickly, then read 4–5 aloud — choose a mix of strong explanations and interesting new questions. Ask the class: 'Which of these new questions do you think will be answered in our next lesson? Which ones might take us all the way to the end of the unit?' Point to the phenomenon image and say: 'At the start of this	Collaborative Knowledge Tracking via the Driving Question Board. The DQB is a living sensemaking artefact — it externalises the class's growing understanding and makes the trajectory of learning visible to learners themselves. Requiring both an 'update' and a 'new question' on the same card forces metacognitive reflection: learners must assess what they know AND identify the boundary of their current understanding. The teacher's act of reading cards aloud and identifying patterns models the scientific practice of synthesising evidence across multiple sources (Obtaining, Evaluating, and Communicating Information — NGSS SEP 8).	Observable indicator: Every learner has posted a sticky note with both an understanding update and a new question. Read all posted notes after the lesson — sort them into three categories: (1) target understanding achieved (structure-function language present); (2) partial understanding (structure described without functional link); (3) misconception or surface-level response. Use this sort to plan the opening re-anchor of Lesson 9. New questions that reference organs, tissues, or systems (e.g., 'Does every organ have its own special cells?') indicate learners are ready to advance toward the tissue/organ level in Lessons 9–10.

🔍 Search ARES for similar readings	puzzling? The teacher draws explicit attention to how today's lesson has shifted the class from 'cells look different' (observation) to 'cells look different BECAUSE they do different jobs' (explanation).	unit, we asked: how does the body send exactly the right response to heal that cut? Today you built part of that answer. You now know that the right response is possible BECAUSE each cell has a shape that makes it perfect for its job. That is cell specialisation.' Record the 3–4 most generative new questions on the permanent DQB display for the class to track over Lessons 9–12.		
Model Building Phase 📹 **VIDEO:** Professional antigen presenting cells (APC) and MHC II complexes Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative cell model — begun in Lesson 1 and revised in each subsequent lesson. Today's revision task is specific: learners must ADD at least two specialised animal cells to their model, annotating each with (a) 3 structural features, (b) the function each feature enables, and (c) an arrow or label connecting the cell to the healing-cut phenomenon sequence (e.g., 'RED BLOOD CELL → delivers oxygen to healing tissue → biconcave shape + no nucleus = maximum haemoglobin capacity'). Learners also add a new model layer labelled 'Cell Specialisation' and write a one-sentence model claim: 'In our body, cell shape is not random — it is determined by function.' The teacher circulates, stamps or initials completed models, and selects 2–3 models to share under the visualiser in the next lesson as exemplars.	Say: 'Take out your cumulative model. This is Lesson 8 of 12 — your model should now show organelles, the differences between plant and animal cells, and today we are adding a new layer: specialisation. Add at least two specialised cells with full annotations. Connect each one with an arrow back to the healing cut. You have 10 minutes.' Set a timer. Circulate and narrate: 'I can see Baraka has drawn the nerve cell with a long axon — Baraka, add a label explaining WHY the axon is long and connect it to the phenomenon.' At the 8-minute mark, say: 'Two minutes remaining — make sure you have written your one-sentence model claim at the bottom.' After 10 minutes, cold-call one learner to read their model claim aloud. Ask the class: 'Does this claim hold for ALL five cells we studied today? Give me a thumbs up if yes, thumbs sideways if you are not sure.' Address any thumbs sideways before closing. Close the lesson by saying: 'Your model is getting more powerful. Next lesson, we go deeper — we will ask: if individual cells are specialised, what happens when many of the same specialised cell work together? That is the	Cumulative Model Revision (Scientific Modelling — NGSS SEP 2). The cumulative model is the primary sensemaking artefact of the entire 12-lesson sequence. Requiring learners to add to — rather than replace — their model builds a visible record of conceptual change over time. The annotation requirement (structural feature → function → phenomenon connection) forces learners to integrate three levels of explanation in a single diagram, which is a hallmark of systems thinking. The one-sentence model claim is a high-leverage formative tool: it compresses the lesson's core idea into a single falsifiable statement that the class can test against future evidence.	Observable indicator: Each learner's model contains at least two newly added specialised cells with all three annotation components present (structural feature, function, phenomenon connection). During circulation, look for the phenomenon arrow — learners who draw the connection back to the healing cut are demonstrating the highest level of integration. Learners whose annotations only describe structure (e.g., 'biconcave shape') without functional explanation need targeted follow-up at the start of Lesson 9. Collect 5–6 models at the end of class (rotate which learners submit each lesson) for written feedback before Lesson 9.

		question of tissues — and it brings us one step closer to fully explaining the healing cut.'		

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions before planning Lesson 9:

1. PREDICTION QUALITY: Were learners' initial predictions in Phase 1 purely visual (shape descriptions) or functional (shape-to-job links)? If most predictions were visual only, it suggests the organelle lessons (1–7) have not yet been connected to function sufficiently — plan a brief review at the start of Lesson 9.

2. SICKLE-CELL PIVOT: Did the sickle-cell supporting phenomenon land as intended? Could learners articulate the CER independently, or did they need heavy scaffolding? If fewer than 60% of cold-called learners gave a complete CER, consider using a second supporting phenomenon at the start of Lesson 9 (e.g., the effect of dehydration on red blood cell shape in long-distance runners like Eliud Kipchoge's training regime) before advancing to tissues.

3. DQB NEW QUESTIONS: What new questions appeared on the DQB? Sort them and identify whether the class is ready to move to the tissue level (Lesson 9) or whether key structure-function links for individual cells need consolidation. If questions cluster around 'how do cells know what to become?', note that differentiation/gene expression is a Grade 11 concept — acknowledge it but park it explicitly on the DQB as a 'future learning' question.

4. MODEL REVISION DEPTH: Review the 5–6 collected models. Are learners making the phenomenon connection arrow, or treating the model as a stand-alone biology diagram? If the phenomenon connection is absent, open Lesson 9 by projecting one strong model (anonymised or with permission) that shows the connection, and ask the class: 'What makes this model more powerful than one that doesn't have this arrow?'

5. PACING: This lesson is content-rich. If Phase 3 (Explain) ran over time, consider splitting the sickle-cell CER discussion into the opening 10 minutes of Lesson 9 as a 'Do Now' consolidation activity rather than cutting the Model Building phase short.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners examined annotated diagrams and micrographs of five specialised animal cells — red blood cells (biconcave, no nucleus), white blood cells (irregular shape, many lysosomes), nerve cells (long axon, myelin sheath), sperm cells (flagellum, mitochondria-packed midpiece), and muscle cells (many mitochondria, myofilaments) — and compared normal red blood cells with sickle cells using printed micrograph evidence.
What did I learn?	Learners learned that cell specialisation is the process by which cells develop distinct structures that equip them for specific functions; that structural features such as the absence of a nucleus (RBC), a long axon (neuron), or a flagellum (sperm) are not random but are direct adaptations to function; and that when cell shape is disrupted — as in sickle-cell anaemia — function is catastrophically compromised, providing powerful evidence that shape determines function.
How does this explain the phenomenon?	Learners explained how each of the five specialised animal cells contributes to the healing-cut phenomenon sequence — red blood cells deliver oxygen to repair tissue, white blood cells engulf bacteria at the wound site, nerve cells transmit pain signals, muscle cells enable movement that keeps the wound protected, and sperm cells (while not directly involved in healing) demonstrate the principle that extreme specialisation trades generalist survival for singular functional excellence. Learners used the Claim-Evidence-Reasoning framework to argue from the sickle-cell micrograph that cell shape is the physical basis of cell function.

LESSON 9 (80 minutes): Specialised Plant Cells: Form Meets Function in the Bean Seedling	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To examine the structure of specialised plant cells and explain how each cell's shape and internal organisation directly determines its function in the plant body.
Knowledge	Learners will be able to identify and describe the structural features of root hair cells, guard cells, xylem vessels, phloem sieve tubes, and palisade mesophyll cells, and explain how each structure suits the cell to its specific function.
Skills	Learners will practise scientific drawing of specialised plant cells, classify cells by function using evidence from structural features, and construct explanatory links between cell shape and plant physiology.
Attitudes	Learners appreciate that the elegant 'fit' between a cell's shape and its function is not accidental but is the product of millions of years of biological refinement — and that this same principle applies to every cell involved in healing the cut finger.
Key Inquiry Question	How does cell specialisation enable organisms to carry out complex life functions?
Purpose in Storyline	In Lessons 1–8 learners established what cells are, how plant and animal cells differ, and how animal cells specialise (platelets, red blood cells, neutrophils). Lesson 9 pivots to the plant side of the storyline: the bean seedling that has been growing since Lesson 1 is now visibly differentiated — roots, stems, and leaves are distinct. Learners ask: does the same 'shape-determines-job' rule that applied to blood cells also govern the very different cells inside this seedling? By the end of this lesson the driving question expands: the principle of specialisation is not unique to animals healing wounds — it is universal across all living things.
Safety Notes	No sharp instruments or chemicals are used in this lesson. If fresh bean seedlings are handled, learners should wash hands after. Ensure no learner has a legume allergy before handling bean material. Microscope glass slides from previous lessons should be stored away safely before the card-matching activity begins to prevent breakage.

B. LESSON OVERVIEW

Learners begin by predicting whether plant cells also specialise, using the visible bean seedling as evidence. They then observe detailed diagrams and (where available) prepared microscope slides or a teacher-led projected image set of five key specialised plant cells: root hair cells, guard cells, xylem vessels, phloem sieve tubes, and palisade mesophyll cells. A structured 'Match-the-Card' group activity drives sensemaking, requiring learners to link labelled cell diagrams to function cards and habitat-context cards (e.g., dry Turkana vs. humid Kisumu for guard cells). Learners update their cumulative cell-specialisation model and add new question cards to the Driving Question Board, bridging toward the upcoming lessons on plant nutrition and transport.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Intro to exponential	Learners observe the class bean seedling (growing since Lesson 1) displayed at the front. The teacher draws attention to the three visibly	Display the bean seedling prominently. Say: 'Before I show you anything, I want your brain's best guess. Look carefully at this	Prediction Anchoring — by committing predictions to paper and the board before instruction, learners activate prior knowledge	Observable indicator: Every learner has a written sketch and at least one written label in their notebook within the 3-minute

functions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	different regions: the fine white root hairs on the roots, the stiff upright stem, and the flat green leaves. Learners are given 3 minutes of individual silent thinking time to write answers to the prompt: 'We have seen that blood cells specialise — platelets are flat and sticky, red blood cells are disc-shaped. Look at the bean seedling. Do you think plant cells also specialise? If yes, predict: what might a cell near the root tip look like compared to a cell in the green leaf? Sketch one of each and label what you think it does.' Learners share predictions with a shoulder partner for 2 minutes. Teacher cold-calls 4 learners — aiming for at least one who says 'yes, they specialise' and one who is uncertain — to voice predictions aloud before any instruction begins. All predictions are recorded visibly on the board in two columns: 'Root cell ideas' and 'Leaf cell ideas' for later revisitation.	seedling — the root hairs, the stem, the leaf. You have 3 minutes of silence to sketch and write.' Set a visible timer. After 3 minutes say: 'Turn to your partner — share your sketches. 1 minute.' After partner talk, cold-call exactly 4 learners using name cards. Ask each: 'What did you predict the root cell looks like, and why?' Wait 12 seconds after each question before accepting an answer. Record all predictions on the board verbatim — do NOT correct errors yet. Close by saying: 'Hold onto these predictions — we will come back and see who was closest to what biologists actually found.'	about animal cell specialisation (Lessons 7–8) and create a cognitive 'hook'. Revisiting predictions later produces the productive surprise that deepens encoding.	window. Teacher circulates and notes (without correcting) whether learners are drawing elongated vs. round shapes, whether they reference chloroplasts or surface area — these reveal the depth of transfer from animal cell lessons.
Observe Phase 📹 **VIDEO:** Intro to exponential functions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher projects or draws (in sequence) five specialised plant cells, spending approximately 6 minutes per cell. For each cell, learners first observe silently, then record a labelled scientific drawing in their notebooks. The five cells and their key observable features are: (1) Root hair cell — long tubular extension (root hair), large vacuole, thin cell wall, no chloroplasts; (2) Guard cells — kidney-shaped pair, thick inner wall, chloroplasts present, central pore (stoma); (3) Xylem vessel — hollow lumen, thick lignified wall, no end walls, no living cytoplasm; (4) Phloem sieve tube — thin walls, perforated sieve plates at	For each cell, say: 'Study this diagram in silence for 30 seconds before you draw anything — notice the shape, the thickness of the walls, what is inside.' Set 30-second silent observation before each drawing begins. Circulate and prompt with: 'What is the most unusual feature of this cell compared to a basic plant cell?' After the Turkana/Kericho guard cell comparison, ask the whole class: 'Show me on your fingers — 1 for Turkana, 2 for Kericho — in which location are the guard cells more open?' Wait 10 seconds. After the whole observation sequence, cold-call 3 learners and ask: 'Which of these five cells	Structured Observation Table — the three-column table (features seen → function hypothesised) forces learners to move from pure observation to inference on every single cell, preventing passive copying of diagrams. The Turkana/Kericho comparison embeds climate context and makes the guard cell's function tangible and locally meaningful.	Observable indicator: Learners' observation tables show at least two structural features per cell and at least one function hypothesis per cell that uses structural evidence (e.g., 'the hollow xylem lumen means water can flow through easily'). Teacher specifically checks that xylem is not described as 'alive' — a common misconception to address in Phase 3.

	ends, companion cell alongside; (5) Palisade mesophyll cell — tall cylindrical shape, densely packed chloroplasts, large central vacuole. For guard cells specifically, the teacher presents a side-by-side context image: a stomatal image from a maize leaf collected in dry Turkana (stomata mostly closed) vs. a tea leaf from humid Kericho (stomata mostly open) — asking: 'What do you notice about how open the stomata are in each location? What might the guard cells be doing differently?' Learners record observations in a structured table: Cell Name | Key Structural Features I Can See | What I Think This Helps the Cell Do.	surprised you the most, and what specifically surprised you?' Wait 12 seconds per response. Explicitly connect xylem back to the phenomenon: 'Remember the fibroblast — it had lots of rough ER to make collagen. The xylem vessel gave up its entire living contents to become a hollow pipe. Both cells sacrificed something to do a job. What does that tell us about specialisation?'		
Explain Phase 📹 **VIDEO:** Intro to exponential functions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners participate in a structured 'Match-the-Card' group activity in groups of four. Each group receives a set of 15 cards pre-printed or hand-cut by the teacher, comprising: 5 Cell Diagram Cards (unlabelled diagrams of the five cells), 5 Function Cards (e.g., 'Absorbs water and mineral salts from the soil around the bean seedling's roots'), and 5 Context/Location Cards (e.g., 'Found in the outer layer of roots, just behind the root tip' or 'Found in leaves — more open in humid Kericho, more closed in dry Turkana'). Groups match all 15 cards into 5 complete sets of 3. Once matched, each group selects one cell and prepares a 90-second verbal explanation — using the Claim-Evidence-Reasoning (CER) frame — of why that cell's shape suits its function. One spokesperson presents; the rest of the class uses a peer-feedback protocol to agree or challenge. The	Distribute card sets and say: 'You have 8 minutes to match all 15 cards into 5 sets of 3. No card is used twice. Every group member must be able to explain every match — not just the one you personally made.' After 8 minutes, say: 'Freeze. Each group, hold up your set for the root hair cell.' Scan groups for errors. Cold-call one group that has an error: 'Walk me through your reasoning for that match.' Wait 12 seconds. During CER presentations, require the class to use a simple hand signal: thumbs up = agree with the reasoning, flat hand = need more evidence. After all presentations, begin consolidation and ask rhetorically: 'The xylem vessel is dead — no nucleus, no cytoplasm. Is it still a cell? Does it still count as specialised?' Wait 15 seconds. Take 3 responses. Close Phase 3 by returning to the board predictions from Phase 1 and asking: 'Who was closest? What	Claim-Evidence-Reasoning (CER) Frame — by requiring learners to state a claim ('This is a guard cell'), cite structural evidence ('because it is kidney-shaped and has a thick inner wall'), and give reasoning ('which means it can change shape to open or close the stoma'), the CER frame makes the form-function logic explicit and transferable. The card game provides concrete manipulation before the abstract verbal explanation.	Observable indicator: Each group's card matches are checked against the answer key. Teacher specifically listens for whether CER presentations include explicit structural evidence (not just function labels). Misconception probe: after consolidation, teacher asks every learner to write one sentence in their notebook answering: 'Is the xylem vessel alive? Use evidence from today.' This written response is collected as an exit-check for the xylem-is-dead misconception.

	teacher then delivers a 10-minute consolidation mini-lecture clarifying: (a) why xylem cells die to become functional, (b) the role of lignin in xylem walls, (c) how the perforated sieve plates of phloem allow sugar to flow between cells, and (d) the link between palisade cell chloroplast density and photosynthesis rate. Throughout, teacher explicitly links each plant cell back to the driving question and the wound-healing phenomenon.	did we get right, and what surprised us?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Intro to exponential functions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner is given two sticky notes (or small paper slips). On the first, they write one new question that today's lesson raised for them — something they now want to understand that they did not wonder about before (e.g., 'If guard cells control water loss, what happens to a maize plant in Turkana during drought — do the guard cells ever fully close forever?'). On the second, they write one connection: how does today's lesson change or add to their understanding of the wound-healing phenomenon (e.g., 'Just like fibroblasts rebuilt skin, the phloem transports sugars that give fibroblasts the energy to do their job — so plant cells and animal cells are connected in keeping us alive'). Learners place both slips on the class Driving Question Board in designated zones: 'New Questions' and 'Growing Answers'. The teacher reads 3–4 slips aloud and facilitates a 4-minute whole-class discussion on which questions the class most wants to answer in the remaining lessons.	Say: 'Two slips. Two minutes. First slip: a question this lesson made you want to ask. Second slip: one way today's plant cells connect back to the cut finger.' After 2 minutes, collect slips and place them on the DQB. Read three questions aloud and ask: 'By a show of hands — which of these three questions do you most want us to answer before the end of this unit?' Tally hands visibly. Point to the DQB's 'Growing Answers' zone and say: 'Look at the questions we posted in Lesson 1. How many can we now answer? Let's count.' Guide learners to identify 2–3 questions from earlier lessons that today's content resolves. Close by saying: 'Our driving question asks why cells look different and how shape determines job. Today we added five more examples. The pattern is becoming clear — let's see if it holds all the way to Lesson 12.'	Driving Question Board Metacognitive Review — periodically counting how many earlier questions the class can now answer makes learning progress visible and builds scientific confidence. Requiring a cross-domain connection (plant cell → wound healing) strengthens the coherence of the unit storyline and prevents plant biology from feeling like a disconnected topic.	Observable indicator: The 'connection' slip demonstrates whether learners are making genuine cross-domain links or are writing superficial responses. Teacher sorts connection slips into 'strong link' (uses specific cell names and mechanisms) and 'developing link' (general statements) to inform differentiation in Lesson 10.
Model Building	Learners retrieve their cumulative 'Cell Specialisation Model' — a	Distribute or ask learners to open their cumulative models. Say: 'You	Cumulative Model Revision — adding a 'Plant Side' and a 'bridge	Observable indicator: The model shows all five plant cells with

Phase 📹 **VIDEO:** [Intro to exponential functions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plant Cell Structures (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	poster or notebook page begun in Lesson 4 — which currently shows specialised animal cells arranged around the wound-healing phenomenon. Today they add a 'Plant Side' to the model. Using the diagrams from Phase 2, learners draw and label all five specialised plant cells on the plant side, then draw arrows indicating: (a) which cells work together in the root (root hair cells absorb → xylem transports), (b) which cells work together in the leaf (palisade cells photosynthesise → phloem exports sugars → guard cells regulate water loss). Learners then add one 'bridge arrow' connecting their plant model to the animal wound-healing model — labelling what the plant provides that the animal needs (e.g., 'Palisade cells make glucose via photosynthesis → glucose powers fibroblasts to make collagen'). Lesson closes with learners sharing their bridge arrow with a partner and the teacher cold-calling 2 learners to explain their bridge to the class.	have 12 minutes to add the plant side. Every cell must have: a labelled diagram, its key structural feature, and its function. Then draw one bridge arrow to the animal side — what does the plant give the animal?' Circulate and prompt struggling learners: 'Think about what fibroblasts need to do their job — where does that energy come from, all the way back?' After 12 minutes, cold-call 2 learners: 'Show us your bridge arrow and explain it in two sentences.' Wait 12 seconds after each explanation. Close the lesson by saying: 'In Lesson 1 we asked: why does the cut finger heal? Today we discovered that the answer involves not just animal cells but plant cells too — every meal of githeri or sukuma wiki that feeds our body's repair workers is made possible by specialised plant cells doing their jobs perfectly. That is the power of cell specialisation.'	arrow' to the existing animal-cell model makes the unit's central argument visible and spatial: specialisation is a universal biological strategy. The bridge arrow forces systems thinking, connecting plant physiology to animal physiology through the flow of energy and matter.	labelled structural features (not just names), directional arrows showing functional relationships between cells, and at least one bridge arrow that names a specific substance (glucose, water, mineral salts) flowing from plant to animal context. Teacher photographs 4–5 models at the end of class to assess depth of systems thinking before Lesson 10.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) During the Match-the-Card activity, which cell caused the most incorrect matches — and what does this tell you about which structural features learners find hardest to interpret visually? Plan a 5-minute re-teach opener for Lesson 10 if more than one-third of groups mismatched the xylem or phloem cards. (2) Were learners able to draw genuine 'bridge arrows' connecting plant cells to the wound-healing phenomenon, or did most bridges remain abstract? If bridges were weak, consider beginning Lesson 10 with a think-aloud model where you narrate your own bridge arrow construction. (3) Review the DQB 'Growing Answers' slips: are learners making mechanistic connections (naming substances and processes) or surface connections (e.g., 'plants and animals are both living things')? Use this data to set the depth expectation for Lesson 10's writing task. (4) The Turkana/Kericho guard cell comparison is a key Kenya-context moment — note whether learners engaged with this geographical contrast or treated it as decoration. If engagement was low, plan to bring a physical leaf sample from the school compound and hold it under the projector in the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed detailed diagrams (and where available, microscope slides or projected images) of five specialised plant cells

	— root hair cell, guard cell, xylem vessel, phloem sieve tube, and palisade mesophyll cell — recording labelled scientific drawings and completing a three-column observation table of structural features, alongside a contextual comparison of stomatal aperture in dry Turkana vs. humid Kericho.
What did I learn?	Learners learned that plant cells specialise just as animal cells do; that each of the five cells has a specific structural adaptation (e.g., the long extension of root hair cells for absorption, the hollow lignified lumen of xylem for water transport, the dense chloroplasts of palisade cells for photosynthesis) that directly enables its function; and that specialised plant cells work in coordinated systems — absorption → transport → photosynthesis → regulation — mirroring the coordinated cell systems seen in wound healing.
How does this explain the phenomenon?	Learners explained, using the Claim-Evidence-Reasoning frame, why each plant cell's shape suits its function; constructed a 'Plant Side' addition to their cumulative specialisation model with directional arrows showing how root hair cells, xylem, phloem, palisade, and guard cells work together; and drew a 'bridge arrow' articulating how plant cell specialisation (particularly photosynthesis in palisade cells) provides the glucose energy that ultimately powers animal cell processes such as fibroblast collagen synthesis during wound healing.

LESSON 10 (80 minutes): Cell Organisation: Cells → Tissues → Organs → Organ Systems → Organism

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to trace the hierarchical organisation of cells into tissues, organs, organ systems, and the whole organism using the wound-healing phenomenon as a unifying context.
Knowledge	Learners will know that: (1) cells of the same type group together to form tissues; (2) different tissues combine to form organs; (3) organs working together form organ systems; (4) organ systems together constitute the organism — and that this hierarchy is illustrated by the skin's response to a cut finger.
Skills	Learners will be able to: (1) construct a labelled hierarchy diagram (cell → tissue → organ → organ system → organism) using wound-healing examples; (2) draw an analogy between biological organisation and a Kenyan school structure; (3) update their cumulative cell model (Version 4) to incorporate the organisational hierarchy.
Attitudes	Learners appreciate that no single cell type heals a wound alone — interdependence across every level of organisation is what makes complex life functions possible, reflecting the value of collaboration and community.
Key Inquiry Question	How does cell specialisation enable organisms to carry out complex life functions?
Purpose in Storyline	In Lessons 1–9 learners identified cell structures, distinguished plant from animal cells, explored specialised cells (platelets, neutrophils, fibroblasts, epidermal cells), and mapped how each cell's shape suits its wound-healing role. Lesson 10 zooms out: learners now ask HOW these individual specialised cells are organised into larger functional units. This is the conceptual bridge that will carry them into Anatomy and Physiology of Plants (Strand 2) and Animals (Strand 3) — where tissues and organs are the primary units of study. Version 4 of the cumulative model situates every previously drawn cell inside the organisational hierarchy.
Safety Notes	No hazardous materials are used in this lesson. If learners use scissors to cut diagram cards, remind them to cut away from the body and to place scissors flat on the desk when not in use. Supervision required during any cutting activity.

B. LESSON OVERVIEW

Learners begin by predicting how many different cell types are needed to heal the cut finger and whether those cells work in isolation or as part of a larger structure. They then observe a structured reading-and-image evidence set showing the four wound-healing cell types nested inside epithelial tissue → skin → integumentary system → human body. The Kenyan school analogy (student → class → school → county education → Kenyan society) is introduced as a sensemaking scaffold. Learners construct a dual-track hierarchy diagram, update their cumulative model to Version 4, and close by revisiting the Driving Question Board to formally post what the hierarchy explains — and what new questions it raises about how organs and systems work.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Plant Organ	Learners individually examine a projected image showing four labelled cells from the wound-	Display the four wound-healing cell images alongside the prompt. Say: 'Look carefully at these four cells.	Think-Pair-Predict: learners activate prior knowledge of cells and generate a testable prediction	Observable indicator: learners' notebooks contain a written or sketched prediction that

Structure - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	healing phenomenon (platelet, neutrophil, fibroblast, epidermal cell — drawings from their own Version 3 models). The teacher poses a written prompt displayed on the board: 'If all four of these cells are working at the same time inside the cut finger, are they acting alone — or are they part of something bigger? Draw or write your prediction: what structure above the single cell do you think exists?' Learners write/sketch silently in their ARES notebooks for 4 minutes. They then share their prediction with their elbow partner (Think-Pair) for 90 seconds.	We have studied each one. Today I want you to think BIGGER. Before I teach you anything new — draw or write what you predict exists ABOVE the level of a single cell. You have 4 minutes of silent thinking time. Go.' [WAIT — enforce 4 minutes of silence; circulate and observe notebooks without commenting.] After 4 minutes: 'Turn to your elbow partner. Share your prediction. You have 90 seconds.' [WAIT 90 seconds.] Cold-call 3 pairs — choose one pair who drew a tissue-like cluster, one who drew the whole body, and one who was unsure. Ask each: 'What was your reasoning?' Record key prediction words on the board in a 'PREDICT' column (do not confirm or deny): 'tissue', 'group', 'system', 'organ', 'body'. Say: 'Hold these predictions. We are going to test them against evidence right now.'	before instruction. Recording predictions on the board creates a public commitment that will be revisited during the Explain phase, increasing cognitive engagement and accountability.	references at least one structural level above the cell (e.g., 'group of cells', 'layer of skin', 'body system'). Teacher scans 6–8 notebooks during the silent writing period. Flag any learner whose prediction contains no structural reference — pair that learner with a stronger peer during Phase 2 reading.
Observe Phase 📹 **VIDEO:** Plant Organ Structure - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive the ARES Evidence Card Set 10 — a two-sided printed or on-screen resource. Side A contains four annotated microscope images: (1) a single epidermal cell, (2) a cross-section of stratified squamous epithelial tissue (many epidermal cells), (3) a cross-section of skin as an organ (epidermis + dermis + hypodermis layers labelled), (4) a diagram of the integumentary system (skin + hair + nails + sweat glands) within the human body outline. Side B contains a parallel Kenyan school analogy diagram: (1) one Form 2 student at Mwangaza Secondary School, Nairobi, (2) Form 2 Class of 42 students, (3) Mwangaza Secondary School (all classes, all teachers, all departments), (4)	Distribute Evidence Card Set 10. Say: 'On Side A you have real microscope images from biology. On Side B you have a school analogy I want you to think about. Your job in your group is to fill in the Observation Table — four columns. Biological example, school analogy, and the FUNCTION at each level. You have 15 minutes. Use evidence from the cards — not guesses.' [WAIT — circulate actively. At 5 minutes, check whether groups are correctly linking epidermal cell → epithelial tissue → skin → integumentary system. If a group is stuck, ask: 'Look at image 2 on Side A — what has changed compared to image 1? What is made up of many of the same cell?' Do NOT give the answer.] At	Parallel Analogy Mapping: the Kenyan school analogy provides a concrete, culturally familiar scaffold that mirrors the abstract biological hierarchy. By completing both columns simultaneously, learners construct the structural logic of organisation (many similar units → one functional whole) before applying it to biology, reducing cognitive load.	Observable indicator: the completed Observation Table correctly identifies all four biological levels with matching school analogy and a stated function for each level. Mastery threshold: at least 3 of 4 levels correctly completed with a function. Teacher collects one table per group at the end of Phase 2 for rapid review before Phase 3.

	Nairobi County Education System (all schools), (5) Kenyan Society (all counties). Learners work in groups of 4 to complete an ARES Observation Table with four columns: Level of Organisation | Biological Example (from wound healing) | School Analogy | Function at This Level. Groups have 15 minutes.	10 minutes, say: 'Two groups — please be ready to share your Function column. What does the TISSUE do that the single cell cannot do alone?' [WAIT 30 seconds for groups to confirm their answer.] After 15 minutes, cold-call 2 groups to read their completed rows aloud. Correct any factual errors immediately using the board.		
Explain Phase 📹 **VIDEO:** Plant Organ Structure - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class sensemaking discussion anchored to the wound-healing phenomenon. Using the board (or ARES projected slide), the teacher builds a formal five-level hierarchy diagram: Cell → Tissue → Organ → Organ System → Organism, placing the four wound-healing cells (platelet, neutrophil, fibroblast, epidermal cell) at the Cell level, and showing the pathway upward. Learners then complete an individual written explanation in their notebooks responding to the prompt: 'Using the healing cut as your example, explain in 5 sentences why a single epidermal cell cannot heal a wound on its own — and what levels of organisation must work together to restore the finger.' Learners have 8 minutes to write.	Begin by returning to the board's PREDICT column. Say: 'Let's check your predictions. Who predicted the word tissue or a group of the same cells — raise your hand.' [Pause, acknowledge.] 'Who predicted organ or system?' [Acknowledge.] 'Your predictions were strong. Here is what the evidence confirms.' Draw the five-level hierarchy on the board step by step, pausing after each level. Say: 'Epithelial tissue — what do the school analogy and the microscope image have in common? WAIT 10 seconds.' [Cold-call 1 learner.] Add each biological example next to the level as learners contribute. Once the diagram is complete, say: 'Now I want this in YOUR words, in YOUR notebook. Write 5 sentences. Use the healing cut. Explain why one cell is not enough — and what each level adds. 8 minutes, silent, individual writing. Go.' [WAIT — enforce silence. Circulate and read over shoulders. If a learner writes fewer than 3 sentences after 4 minutes, prompt quietly: 'What does the tissue do that one cell cannot? Write that.']. After 8 minutes cold-call 2 learners with strong responses to read aloud. Ask the class: 'Does this explanation match our evidence?	Claim-Evidence-Reasoning (CER) Writing: by requiring learners to make a claim (cell cannot heal alone), cite evidence (levels of organisation), and provide reasoning (function at each level), the teacher embeds NGSS Science and Engineering Practice 6 (Constructing Explanations). The public hierarchy diagram on the board acts as a shared anchor, preventing learners from constructing explanations in isolation.	Observable indicator: the 5-sentence written explanation names at least 3 levels of organisation, references the wound-healing phenomenon explicitly, and includes a stated function for at least one level. Teacher reads 4–6 notebooks during circulation. Any learner who conflates organ with organ system receives a targeted prompt card ('An ORGAN is one structure like the skin. An ORGAN SYSTEM is a group of organs working together — like the integumentary system. Can you revise sentence 3?').

		Thumbs up / thumbs down / thumbs sideways.' Address any thumbs-down.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Plant Organ Structure - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (or ARES DQB digital cards). On the YELLOW note they write: one thing today's evidence ANSWERS about the driving question ('Why do cells look different and how does shape determine job?'). On the BLUE note they write: one NEW question today's lesson has raised — something they now wonder about that they did not wonder about before. Learners post both notes on the class DQB under the columns 'NOW WE KNOW' and 'NEW QUESTIONS'. The teacher reads 5–6 notes aloud and briefly clusters them by theme: structural organisation, function, and questions about how organ systems communicate.	Hand out sticky notes. Say: 'The Driving Question asks why cells look different and how shape determines job. Today we zoomed out. On your YELLOW note — write one thing we can now ANSWER about that driving question because of what we learned today. On your BLUE note — write one NEW question this lesson has opened up for you. You have 3 minutes.' [WAIT 3 minutes — enforce quiet writing.] Direct learners to post notes on the DQB. Read 5–6 notes aloud, grouping similar ones. Say: 'I notice several of you are now asking HOW organ systems communicate with each other — does the integumentary system send signals to the immune system? That is a brilliant question. We will return to that question in Strand 3 when we study animal physiology.' Point to the DQB and say: 'This board is growing. Look at how far our understanding has come from Lesson 1. We started with one cut finger. Now we understand the organisational machinery behind it.'	Driving Question Board Curation: posting both 'answers' and 'new questions' makes learners' conceptual growth visible and models how scientific understanding always generates further inquiry. Clustering questions by theme helps learners see patterns in their collective thinking and primes them for future learning in Strands 2 and 3.	Observable indicator: YELLOW notes contain a specific reference to a level of organisation (e.g., 'Now I know that tissues form because similar cells group together to do one job') rather than a vague statement ('I learned about cells'). BLUE notes contain a question that is generative and forward-looking. Teacher flags any learner whose yellow note is vague for a 1-minute check-in before the next phase.
Model Building Phase 📹 **VIDEO:** [Plant Organ Structure - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:**	Learners retrieve their cumulative cell model from previous lessons (Versions 1–3, which showed individual specialised cells and their structures). Working individually, they create Model Version 4 by constructing a hierarchy diagram that NESTS their previously drawn cells inside the organisational levels. The diagram must show: (a) at least 2 of the 4 wound-healing cell types	Say: 'Take out your cumulative model — Versions 1 through 3. Look at the cells you drew. Today we are going to zoom ALL the way out. Version 4 is a HIERARCHY diagram. It does not replace your earlier work — it organises it. I want to see your cells from Version 3 sitting inside their tissue, inside their organ, inside their system, inside the whole organism.' Display a blank	Cumulative Model Revision: requiring learners to nest previous drawings inside the new hierarchy makes learning incremental and visible. The 'Model Revision Note' prompts metacognitive reflection — learners articulate what the new model explains that the old one could not, directly practising NGSS SEP 2 (Developing and Using Models) and SEP 6 (Constructing Explanations).	Observable indicator: Version 4 diagram contains all five organisational levels correctly ordered, at least 2 named wound-healing cell types at the Cell level, a stated function at each level, and a Model Revision Note that references organisational hierarchy explicitly. Mastery threshold: 4 of 5 levels correctly labelled with function. Learners who only complete 3 levels are

[Cell Transport (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	at the Cell level; (b) the relevant tissue each belongs to; (c) the organ those tissues form; (d) the organ system; (e) the whole organism (the Form 1 student in Nairobi). Learners annotate each level with its function in wound healing. They also write a 'Model Revision Note' at the bottom: 'What does Version 4 show that Version 3 did not?'	template on the board (five labelled tiers, blank boxes at each level). Say: 'You have 12 minutes. Every level must be labelled AND must include the function in wound healing — not just the name. At the bottom, write your Model Revision Note: what does Version 4 show that Version 3 could not show?' [WAIT — circulate and provide targeted feedback.] At 6 minutes, say: 'Check — can I see two cell types at the Cell level? Can I see a tissue label? If not, prioritise those now.' At 12 minutes: cold-call 2 learners to hold up and narrate their Version 4 model. Prompt the class: 'What did they include that you might want to add or revise?' Allow 2 minutes of silent revision. Collect models or photograph them for ARES digital portfolio.		assigned a structured scaffold card for homework ('Complete the missing two levels: what tissue do neutrophils belong to? What organ system do they help protect?').

D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. HIERARCHY UNDERSTANDING — Could learners correctly distinguish between an organ and an organ system without prompting? Were there persistent confusions between 'tissue' and 'organ' in the written explanations or models? If yes, plan a 5-minute warm-up retrieval activity at the start of Lesson 11 using labelled diagram sort cards.

2. ANALOGY EFFECTIVENESS — Did the Kenyan school analogy (student → class → school → county → society) help learners grasp the biological hierarchy, or did it create confusion (e.g., learners thinking a 'school' is the same as an 'organ' rather than being analogous to one)? Note which groups used the analogy productively and which abandoned it.

3. MODEL QUALITY — Review collected Version 4 models. Were learners able to successfully nest their Version 3 cells inside the new hierarchy, or did they redraw cells from scratch? The nesting task assesses whether cumulative model-building is working as a pedagogical strategy. If more than 30% of learners started from scratch, consider introducing a pre-structured Version 4 template in the next lesson.

4. DQB RICHNESS — Were the Blue (new question) sticky notes generative and forward-looking (pointing toward organ systems, physiology, communication between systems), or were they superficial restatements of today's content? Rich questions indicate readiness to move into Strand 2 (Plant Anatomy and Physiology). Thin questions suggest learners need one more consolidation lesson.

5. PHENOMENON CONNECTION — Did learners spontaneously reference the healing cut phenomenon during explanations, or did they explain organisation in the abstract? The phenomenon must remain the anchor. If references were absent, open Lesson 11 by projecting the original phenomenon image and asking: 'Where in the hierarchy does today's new knowledge sit?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Annotated microscope images and diagrams showing one epidermal cell → stratified squamous epithelial tissue → skin organ cross-section → integumentary system → human body; parallel Kenyan school analogy diagram (student → class → Mwangaza Secondary School → Nairobi County Education → Kenyan society).
What did I learn?	Cells of the same type group together to form tissues; different tissues combine to form organs; organs working together form organ systems; organ systems together constitute the whole organism. In the wound-healing phenomenon, epidermal cells form epithelial tissue, which is part of the skin organ, which belongs to the integumentary system, which protects the entire organism.
How does this explain the phenomenon?	No single specialised cell — not the platelet, the neutrophil, the fibroblast, nor the epidermal cell — can heal the cut finger alone. Healing requires all four cell types to be organised into tissues, organs, and systems that coordinate their functions. The organisational hierarchy is what transforms individual specialised cells into a living, self-repairing organism — answering the driving question by showing that shape determines job at the cell level, but ORGANISATION determines survival at the organism level.

LESSON 11 (40 minutes): Synthesis and Model Revision: Putting the Cell Story Together	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners synthesise all prior learning about cell structure, specialisation, and organisation into a single coherent poster-model that explains the wound-healing phenomenon from cell to organism level.
Knowledge	Learners can accurately describe and annotate plant and animal cell organelles; explain structure-function relationships in at least three specialised cells involved in wound healing; and sequence the organisational hierarchy from cell to organism.
Skills	Learners construct and revise scientific models; communicate scientific understanding through annotated diagrams; apply structured peer feedback using a glow-and-grow protocol; and compare their current understanding with their Lesson 1 prior knowledge.
Attitudes	Learners demonstrate intellectual honesty by acknowledging what they did not know in Lesson 1; show collaborative responsibility in group model construction; and appreciate the elegance of how cell structure enables complex life functions.
Key Inquiry Question	How does cell specialisation enable organisms to carry out complex life functions? What are the structural differences between plant and animal cells?
Purpose in Storyline	This is the penultimate lesson in the evidence-gathering sequence. Learners have accumulated knowledge across ten lessons and must now integrate that knowledge into a single coherent model. This lesson serves as the synthesis node before final consolidation and assessment in Lesson 12. The driving question is advanced to near-complete resolution: learners can now explain why different cells look different and why shape determines function, using the wound-healing phenomenon as the unifying context.
Safety Notes	No hazardous materials are used in this lesson. Learners handle poster paper, markers, and scissors where used for card-sorting. Ensure adequate space between groups to allow movement without collision. No specific PPE required.

B. LESSON OVERVIEW

Learners open by revisiting their Lesson 1 sticky-note predictions, held on the Driving Question Board, and compare those naive ideas with their current evidence-based understanding. In groups of four, learners construct a final annotated poster-model integrating: a labelled plant cell, a labelled animal cell, three specialised cells from the wound-healing phenomenon with structure-function annotations, and the full cell-organisation hierarchy. Groups present their posters in a gallery-walk format. Peers provide structured feedback using a one-glow one-grow protocol. The DQB is reviewed as a class and all original wonder questions are addressed, re-answered, or re-categorised as questions for future learning. The lesson closes with a brief individual written comparison showing how each learner's thinking has shifted since Lesson 1.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Thin-layer chromatography (TLC)]	Learners individually retrieve their science notebooks and locate their Lesson 1 entry — their original prediction about what was happening inside the skin when	Teacher retrieves the original DQB poster displayed since Lesson 1 and places it at the front of the room. Teacher reads aloud two or three of the original student sticky-	Metacognitive retrieval and comparison. Learners make their own knowledge change visible by juxtaposing Lesson 1 language with Lesson 10 language. The	Teacher listens to the THEN versus NOW sentence pairs to diagnose whether learners have accurately internalised key concepts — particularly cell

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the Nairobi student cut their finger while slicing onions. Each learner silently re-reads their own words for approximately ninety seconds. Teacher then poses a reflection prompt: Before we share, write one sentence finishing this stem — In Lesson 1 I thought... but now I know... — Learners write without talking for two minutes. Selected learners share aloud. This activates metacognitive awareness and positions the synthesis work that follows as meaningful revision rather than repetition.	note wonder questions verbatim, using a warm and deliberate tone: Here is what Akinyi wrote in Lesson 1 — she asked, Why does the body know which cell to send? Here is what Otieno wrote — he asked, Do all our cells have the same parts? These were great scientific questions ten lessons ago. Today, let us find out how far our answers have come. Teacher pauses for WAIT TIME of five full seconds after each question is read aloud, allowing emotional and cognitive connection to the original curiosity. Teacher then instructs learners to open notebooks to Lesson 1 and write the sentence stem. After writing, teacher cold-calls three learners — selecting students who had a wide range of prior knowledge in Lesson 1 — and records key contrasting phrases on the chalkboard under two columns: THEN and NOW.	DQB serves as a physical artefact of the class inquiry journey, grounding the synthesis in shared history.	specialisation and organelle function. Misconceptions surfaced here (for example, a learner who still conflates cell wall and cell membrane) are noted for targeted clarification during the Model Building Phase.
Observe Phase 📹 **VIDEO:** Thin-layer chromatography (TLC) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher projects or draws on the board a large blank outline of the final poster template — showing four clearly labelled zones: Zone A — Plant Cell, Zone B — Animal Cell, Zone C — Three Specialised Wound-Healing Cells, Zone D — Cell Organisation Hierarchy. Learners observe the template and, in their groups of four, spend three minutes doing a silent planning scan: each member silently reads the zone headings and identifies which zones they feel most and least confident about. Each learner records in their notebook: I feel most confident about Zone... because... and I feel least confident about Zone... and I will need to use my notes for... This observable self-assessment primes learners for	Teacher explains the poster task using precise language: Your group will produce one scientific model — not a decoration, not a copy from the textbook — a model that uses evidence from our own practical work, our diagrams, and our discussions to explain the wound-healing phenomenon. Every label on this poster must connect a structure to a function. Teacher holds up a marker and says: If you can draw a cell without explaining what it does, your poster is incomplete. After explaining the template zones, teacher distributes materials — A2 poster paper, coloured markers, rulers — and sets a visible timer for twenty minutes of construction time. Teacher reminds learners to consult their cumulative cell model	Structured observation of a model template combined with self-assessment of confidence zones. By naming their own uncertainty before beginning, learners engage more deliberately with the evidence they have collected across the unit. The template scaffolds organisation without limiting scientific thinking.	Teacher reads notebook confidence ratings during circulation and uses them to identify groups where all four members have claimed the same strong zone — signalling possible avoidance of difficult content. These groups are given a direct prompt: I notice no one has claimed Zone C — who in this group is going to take ownership of explaining the platelet and the neutrophil?

	equitable division of labour in the group. Teacher circulates, reading notebook entries briefly, and uses this information to assign stretch roles — for example, a learner who is weakest on Zone C (specialised cells) is encouraged, rather than avoided, to take primary responsibility for that zone with peer support.	versions from previous lessons and their practical drawings from Lessons 6 and 7. Teacher circulates during the observe phase using a questioning stance rather than correcting stance — asking What does that organelle do? rather than That label is wrong.		
Explain Phase 📹 **VIDEO:** Thin-layer chromatography (TLC) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups spend approximately twenty minutes constructing their posters. Zone A requires a fully labelled plant cell with at least eight organelles and a one-phrase function note beside each label. Zone B requires a fully labelled animal cell with at least six organelles, noting two structures absent from the plant cell and explaining why their absence matters. Zone C requires annotated diagrams of three specialised cells from the wound-healing sequence — learners most commonly select: platelets (small, no nucleus, packed with clotting proteins — able to aggregate rapidly), neutrophils or macrophages (irregular shape, many lysosomes — engulf and digest bacteria), and fibroblasts or epidermal cells (rich in rough endoplasmic reticulum and Golgi apparatus — secrete collagen or divide rapidly to close the wound surface). Each specialised cell must show a direct arrow linking one structural feature to one function. Zone D requires a hierarchy diagram from cell to organism using the skin as the example — epidermal cell, epithelial tissue, skin organ, integumentary system, whole organism — with a one-sentence explanation of what is gained at	During construction, teacher circulates with a clipboard noting: Are Zone C specialised cells connected to the wound-healing phenomenon or treated as abstract examples? Are structure-function links explicit or implied? Is the hierarchy in Zone D presented as a linear chain or as an explanatory sequence? Teacher intervenes with targeted questions rather than corrections. For example, at one group: You have drawn the red blood cell correctly — it is biconcave and has no nucleus. Now tell me — what does having no nucleus allow the red blood cell to do that a nucleated cell cannot do as efficiently? Teacher allows WAIT TIME of seven seconds before accepting a response, signalling that the question requires genuine thought. At the gallery walk, teacher circulates and validates glow-and-grow feedback that is scientifically grounded: I can see that this feedback note says the axon on the nerve cell should be longer — that is a strong scientific observation, well done. Teacher challenges any purely aesthetic feedback (for example, the drawing is very neat) by redirecting: Can you find one piece of scientific evidence on this poster that either supports or challenges	Collaborative model construction followed by structured peer review. The gallery walk with written feedback transforms peer assessment from a social exercise into a scientific communication act. The glow-and-grow protocol ensures feedback is balanced and specific. By requiring structure-function arrows in Zone C, the task prevents learners from simply drawing cells without connecting morphology to physiology.	Teacher photographs each poster (or sketches key features on the clipboard) during circulation. The structured feedback sticky notes on each poster become a real-time diagnostic of class-wide gaps. Common gaps — for example, most groups showing Zone D as a simple list rather than an explanatory hierarchy — are flagged for whole-class address in the DQB phase. Teacher also listens for oral explanations during gallery walk and probes with cold-call questions to at least five different learners across groups.

	each level of organisation. After construction, groups conduct a four-minute gallery walk: each group leaves their poster on their desk and the group rotates clockwise to view the next group's poster. Viewers use a structured feedback card — one glow (something scientifically accurate and clearly explained) and one grow (one scientific gap or unclear label). Feedback is written on small sticky notes and attached directly to the poster.	what we learned in Lesson 8?		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Thin-layer chromatography (TLC)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups return to their own posters after the gallery walk and spend two minutes reading the feedback sticky notes left by peers. Each group briefly discusses: Do we agree with this grow note? Can we revise this label before the DQB closure? Groups make any final one-minute revisions. Teacher then leads the whole class back to the original DQB poster. Each original wonder question — written in Lesson 1 — is read aloud by the student who originally wrote it (where possible) or by the teacher. The class votes by show of hands whether each question is now: ANSWERED — we have strong evidence; PARTIALLY ANSWERED — we have some evidence but gaps remain; or BEYOND THIS UNIT — this requires further study in later grades or sub-strands. Each question is physically relabelled on the DQB with a green (ANSWERED), yellow (PARTIALLY ANSWERED), or blue (BEYOND THIS UNIT) sticky note. Typical outcomes: questions about organelle function are marked ANSWERED; questions about how DNA controls which	Teacher facilitates the DQB closure with a deliberate ceremonial quality — signalling that this is a significant academic moment. Teacher reads each question slowly and clearly. After each reading, teacher asks: Who wrote this? Can you tell us — do you feel this question has been answered by the work we have done across this unit? Teacher then opens to the class: Does anyone want to add to that? WAIT TIME of five seconds after each student response before allowing another voice. Teacher uses precise language when relabelling: We are not saying BEYOND THIS UNIT because the question is unimportant — we are saying it because great science always generates new questions, and that is a sign of a deep thinker. Teacher ensures that the driving question itself — Why do cells look so different from each other and how does shape determine the job a cell can do? — is explicitly addressed as a class. Teacher cold-calls at least four learners to contribute to a cumulative verbal answer to the driving question, building the response sentence by	DQB closure as collective knowledge consolidation. By returning to the original wonder questions and categorising them using evidence, learners experience the full arc of a scientific inquiry cycle — from curiosity through investigation to provisional explanation. The tricolour relabelling makes the epistemic status of knowledge visible and honest. The cumulative verbal answer to the driving question models collaborative scientific reasoning.	Teacher listens for the quality of justification learners use when voting on question categories. A learner who votes ANSWERED without being able to state the evidence is prompted: What evidence from our lessons, our practical work, or our posters supports that answer? This distinguishes genuine understanding from social conformity. Teacher records any questions that generate significant disagreement for revisiting in Lesson 12.

	genes are expressed in different cells are marked BEYOND THIS UNIT; questions about why some people heal faster than others may be marked PARTIALLY ANSWERED.	sentence on the chalkboard: Learner 1 addresses cell shape; Learner 2 adds organelle function; Learner 3 connects to specialisation; Learner 4 links to the wound-healing phenomenon.		
Model Building Phase 📹 **VIDEO:** [Thin-layer chromatography (TLC)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	In the final five minutes, learners individually open their science notebooks to a fresh page and write a brief personal synthesis entry using the following prompt, written on the board: My cell model started in Lesson 1 as a simple sketch. Today, Version 4 of my model shows... The most important connection I now understand is between... and... because... This connects back to the wound-healing phenomenon because... Learners write for three minutes without discussion. Two learners are cold-called to share their entries aloud. The lesson closes with teacher affirming the intellectual journey: In Lesson 1 you wondered why cells look different. Today your posters and your words show that you can now explain it — using evidence, using the language of science, and using the story of a cut finger healing in Nairobi. That is what it means to think like a biologist.	Teacher circulates during the three-minute individual writing to read entries over shoulders and identify the two most articulate and the two most incomplete responses. The two articulate responses are selected for cold-call sharing — not to embarrass weaker writers but to model the level of specificity expected in Lesson 12 written reflections and in the upcoming sub-strand assessment. After the two shares, teacher explicitly connects to Lesson 12: Tomorrow we return one final time to the cut finger — but this time you will explain it completely, cell by cell, using everything in your notebook and on your poster. Come ready. Teacher does not collect notebooks today — learners need them to prepare overnight. Teacher reminds groups that posters will remain displayed for Lesson 12 reference. Final teacher move: teacher points to the DQB and says — Every green question up there represents something your class figured out together. That is your collective evidence. Be proud of it.	Individual written synthesis as a personal model-revision act. By requiring learners to articulate a structure-function-phenomenon chain in their own words, the prompt prevents surface-level copying from posters and requires genuine integration. The forward-link to Lesson 12 maintains narrative momentum and prevents the synthesis lesson from feeling like a terminal endpoint rather than a penultimate step.	The individual notebook entries serve as the primary formative assessment for this lesson. Teacher reads selected entries during cold-call sharing and identifies: Can the learner name a specific structural feature? Can they link it to a specific function? Can they connect it to the wound-healing phenomenon? Entries that remain abstract (for example, cells are important because they do things in the body) are flagged and the learner is given a quiet verbal prompt to add one specific example before Lesson 12.

D. TEACHER REFLECTION

After the lesson, consider: Did all four zones of the poster model receive roughly equal depth of treatment, or did groups spend disproportionate time on plant and animal cell diagrams while producing superficial Zone C entries? Were the glow-and-grow feedback notes scientifically substantive or primarily social? Could learners articulate the structure-function-phenomenon chain verbally as well as in writing? Were any DQB wonder questions still genuinely unresolved — and if so, is that because the unit has a gap or because the question is legitimately beyond Grade 10 scope? Note which learners gave articulate cold-call responses and which gave vague responses —

use this to plan Lesson 12 cold-call targeting so that learners with gaps are supported to contribute the wound-healing explanation with scaffolded prompts rather than silence.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In this lesson I observed that learners could integrate their cumulative cell knowledge into a four-zone poster model, that peer feedback during the gallery walk was most precise when learners referenced specific organelles and functions rather than general impressions, and that the DQB closure generated genuine discussion about which questions were fully answered versus partially answered.
What did I learn?	Learners consolidated that plant cells and animal cells share core organelles but differ in the presence of a cell wall, large central vacuole, and chloroplasts; that specialised cells in the wound-healing response have structural features directly suited to their functions; and that the organisational hierarchy from cell to organism explains how individual cell functions combine to produce complex responses such as healing.
How does this explain the phenomenon?	Learners explained the driving question — why cells look so different and how shape determines function — by constructing an integrated model that connected organelle structure to cell function to tissue and organ roles to whole-organism response, using the Nairobi wound-healing phenomenon as the unifying explanatory context.

LESSON 12 (80 minutes): Consolidation, DQB Closure, and Final Explanation — The Healing Cut Fully Explained	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate all learning across the sub-strand by constructing a complete, evidence-based explanation of the anchoring phenomenon — wound healing — using knowledge of cell structure and specialisation, and to close the Driving Question Board.
Knowledge	The learner demonstrates understanding of: (a) structural differences between plant and animal cells; (b) how cell specialisation — including platelets, neutrophils, macrophages, fibroblasts, and epidermal cells — enables coordinated wound healing; (c) the relationship between cell shape, organelle composition, and cell function across all specialised cell types studied in this unit.
Skills	The learner is able to: (a) construct a full, sequenced explanation of a complex biological phenomenon using accurate cell biology vocabulary; (b) compare their Lesson 1 mental model with their final revised model, identifying specific conceptual changes; (c) classify Driving Question Board questions as ANSWERED, PARTIALLY ANSWERED, or BEYOND THIS UNIT; (d) write a reflective exit statement articulating personal conceptual growth.
Attitudes	The learner: (a) appreciates that scientific understanding is built incrementally through evidence, revision, and peer collaboration; (b) demonstrates intellectual honesty in acknowledging what remains unknown or beyond the current unit; (c) values the contributions of every member of the class in building a shared explanation.
Key Inquiry Question	Using everything we have learned about cell structure and specialisation — how can we now fully explain, cell by cell, what happens when a student in Nairobi cuts their finger while slicing onions?
Purpose in Storyline	This is the culminating lesson of the sub-strand storyline. Students return to the anchoring phenomenon — The Healing Cut — for the last time, but now armed with 11 lessons of evidence about cell structure, organelle function, and specialisation. The goal is a complete, co-constructed class explanation of wound healing, a side-by-side comparison of Lesson 1 and Lesson 12 models, formal closure of the Driving Question Board, and a sub-strand assessment. The driving question — 'Why do cells look so different, and how does shape determine job?' — is answered fully and publicly.
Safety Notes	No hazardous materials are used in this lesson. If printed sub-strand assessment papers are distributed, ensure scissors or sharp objects are not present on desks. Remind learners that personal reflections are private and will not be read aloud without consent.

B. LESSON OVERVIEW

Lesson 12 is the final lesson of Sub-Strand 1.1: Cell Structure. The teacher opens by returning the class to the anchoring phenomenon — the Form 1 student in Nairobi who cut their finger slicing onions. Through a structured cold-call round (minimum 8 students), the class co-constructs a complete, sequenced, cell-by-cell explanation of wound healing, drawing on all evidence gathered across Lessons 1–11. Learners then place their Lesson 1 cell diagrams alongside their Lesson 11 revised models and articulate, in writing, the specific conceptual changes they made and why. The Driving Question Board is formally closed: each question card is classified aloud by the class as ANSWERED, PARTIALLY ANSWERED, or BEYOND THIS UNIT. Learners complete an individual written exit reflection ("One thing I now understand that I did not understand in Lesson 1 is… because…") before sitting the sub-strand summative assessment. The lesson is designed to produce a sense of intellectual closure and genuine pride in the distance learners have travelled since Lesson 1.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [The force of tension](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their original Lesson 1 cell diagrams and written predictions about wound healing from their notebooks or portfolios. Working individually for 4 minutes, they re-read what they wrote in Lesson 1 without changing anything. They place a sticky note or write in a different colour: 'What I would change today, and why.' They do NOT yet share — this is private preparation for the whole-class explanation that follows.	BEFORE THE LESSON: Ensure every learner has access to their Lesson 1 diagram (returned from teacher files or retrieved from portfolios). Write the anchoring phenomenon on the board in full: 'A Form 1 student in Nairobi cuts their finger slicing onions for supper. Bleeding slows. A clot forms. The wound heals. How — cell by cell?' SAY (slowly, deliberately): 'In Lesson 1, I asked you to explain what happens when you cut your finger. You made your best guess. Today, that same question is back — but this time, you have eleven lessons of evidence behind you. Before we talk, take 4 minutes in silence to re-read what you wrote in Lesson 1.' WAIT: Allow full 4 minutes of silent retrieval. Do not prompt or hint. After 4 minutes, SAY: 'Without changing what you wrote, put a sticky note or write in a new colour — what would you change, and what new word or idea would you now use? You have 3 more minutes.' Circulate quietly, reading over shoulders. Note 2–3 learners who show the most dramatic conceptual shift — you will cold-call them first in Phase 2. Note 1–2 learners who still show partial understanding — plan targeted questions for them.	RETRIEVAL AND CONTRAST: Learners physically return to their Lesson 1 thinking before receiving any new input. This activates prior knowledge, surfaces residual misconceptions, and creates the cognitive 'distance' that will make the Phase 2 co-construction feel meaningful. The contrast between then and now is the central sensemaking engine of this lesson.	Observable indicator: Learners are writing substantively on their sticky notes — naming specific cells (platelets, neutrophils, fibroblasts, epidermal cells) or organelles (nucleus, cell membrane, lysosomes) that were absent from their Lesson 1 writing. Red flag: A learner's sticky note says only 'cells heal the wound' with no specificity — this learner needs targeted cold-calling and scaffolding in Phase 2.
Observe	The class participates in a	DRAW a simple timeline on the	STRUCTURED ACADEMIC	Observable indicators: (a) Called

Phase 📹 **VIDEO:** [The force of tension] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	structured cold-call round. The teacher leads the class through the wound healing sequence — from the moment of the cut to the appearance of new skin — asking each called student to contribute one specific, evidence-based cell claim. Students who are not called listen actively and are expected to agree, build on, or respectfully challenge each contribution. The complete explanation is recorded on the board in a shared 'class story' format. By the end of the round, the board shows a fully sequenced, cell-by-cell explanation of the anchoring phenomenon.	board with 6 stages: (1) Cut occurs → (2) Bleeding begins → (3) Bleeding slows/clot forms → (4) Immune response → (5) Tissue rebuilding → (6) Skin closes. SAY: 'We are going to build the complete explanation together — out loud, in order, using everything we know. I will call on students. When I call your name, give me ONE cell claim: which cell acts at that stage, what its shape is, what its key organelle or feature is, and what job it does. One claim. Precise. Evidence-based. Everyone else — listen. If you agree, nod. If you would add or challenge, raise your hand.' COLD-CALL SEQUENCE (minimum 8 students — use names, not volunteers): — Stage 1–2: Call Student A: 'What is the FIRST cell to respond when the cut happens? Describe its shape and why that shape matters.' WAIT 12 seconds. — Stage 3 (clot): Call Student B: 'Platelets. What is unusual about platelet structure, and how does that unusual feature help them do their job?' WAIT 12 seconds. — Stage 3 (clot continued): Call Student C: 'What protein do platelets help deploy? What organelle in other cells produces proteins like that?' WAIT 10 seconds. — Stage 4 (immune): Call Student D: 'Neutrophils arrive. What do they do, and what organelle makes that job possible?' WAIT 12 seconds. — Stage 4 continued: Call Student E: 'Macrophages. How is their	CONTROVERSY AND CO-CONSTRUCTION: The cold-call round is not a quiz — it is a public knowledge-building exercise. Each contribution is a piece of evidence that advances the shared explanation. By naming specific cells, shapes, and organelles in sequence, learners experience the explanatory power of cell biology as it solves a real, familiar puzzle. The sequential timeline on the board makes the causal chain visible and permanent.	students correctly name at least the cell type and one structural feature without prompting — target: 6 out of 8 students achieve this unaided. (b) Uncalled students are nodding, tracking the board, or raising hands to add — not disengaged. (c) Student H's synthesis response explicitly links DNA → gene expression → specialisation, or at minimum states that different genes are 'switched on' in different cells. Red flag: If fewer than 4 students can name the correct organelle for a given function, pause and run a 2-minute whole-class review of that organelle before continuing.

		shape different from neutrophils? What does that shape allow them to do?' WAIT 12 seconds. — Stage 5 (rebuilding): Call Student F: 'Fibroblasts secrete collagen. What organelle is highly developed in fibroblasts, and why is that logical?' WAIT 12 seconds. — Stage 6 (skin closes): Call Student G: 'Epidermal cells divide and migrate. What cell feature must be active for division to happen?' WAIT 10 seconds. — SYNTHESIS: Call Student H: 'All these cells come from the same organism and carry the same DNA. So why do they look so different? Answer using what you know about specialisation.' WAIT 15 seconds — this is the hardest question. AFTER EACH RESPONSE: Rephrase and write the key claim on the board at the correct stage. If a response is incomplete, SAY: '[Name], you said [X]. Can anyone add the organelle or structure detail that makes that work?' — open to hands. AFTER ALL 8 ROUNDS: SAY: 'Look at the board. That is the complete answer to the question we started with in Lesson 1. Read it. That explanation belongs to this class — you built it together.' Allow 30 seconds of silent reading.		
Explain Phase 📹 **VIDEO:** [The force of tension]() Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Learners place their Lesson 1 cell diagram and their most recent (Lesson 11) revised model side by side on their desks. They complete a structured 'Then vs. Now' written comparison (in their notebooks or on a provided template) with three	Distribute or direct learners to their Lesson 11 revised models. SAY: 'Put your Lesson 1 diagram on the left and your final model on the right. Look at them side by side. In 8 minutes, I want you to write three sentences — the stems are on the	REFLECTIVE MODEL COMPARISON (THEN vs. NOW): Placing the two models physically side by side externalises conceptual growth. The structured sentence stems force learners to name the specific evidence that	Observable indicators: (a) Learners' sentence stems contain specific cell and organelle names rather than general statements. (b) At least one learner references a specific lesson activity (e.g., 'the onion cell microscopy in Lesson 3'

for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	sentence stems: (1) 'In Lesson 1, I thought… but now I know… because…'; (2) 'The biggest change in my model is… because the evidence from [specific lesson/activity] showed me…'; (3) 'One question I still have that this unit did not fully answer is…' After 8 minutes of individual writing, two volunteers share their responses with the class.	board. Use the words from our wall: organelle, specialisation, cell membrane, nucleus, mitochondria, ribosome, chloroplast, cell wall, vacuole, lysosome. Be specific — do not say 'I learned more.' Say what exactly you learned and why the evidence changed your thinking.' WRITE the three sentence stems on the board: 1. 'In Lesson 1, I thought… but now I know… because…' 2. 'The biggest change in my model is… because the evidence from [lesson/activity] showed me…' 3. 'One question I still have is…' WAIT: 8 minutes of silent individual writing. Circulate and read over shoulders. Identify 2 learners with strong, specific responses — ask them in advance if they are willing to share. AFTER 8 MINUTES: Call on the 2 pre-identified volunteers. After each shares, SAY: 'Who else had a similar change in their thinking — hands?' Scan room. SAY: 'Who had a DIFFERENT change — hands?' Call on one. SAY: 'Your third sentence — the question you still have — is not a failure. That is the beginning of the next unit. Scientists in Kenya and around the world are still asking questions exactly like yours. Hold onto it.'	drove each conceptual change — moving reflection from vague ('I learned a lot') to precise ('I now know fibroblasts have an extensive endoplasmic reticulum because we observed that protein-secreting cells need more ER, and fibroblasts secrete collagen'). This is the NGSS Science and Engineering Practice of Obtaining, Evaluating, and Communicating Information.	or 'the specialised cell card sort in Lesson 8') as their evidence source. (c) The 'question I still have' sentences reveal genuine intellectual curiosity — e.g., questions about how genes are switched on/off, how the body knows which cells to send, or how cancer relates to cell specialisation. Red flag: If most learners' 'then vs. now' sentences are identical or vague, the teacher pauses and models one example aloud before continuing.
Driving Question	The teacher displays the full Driving Question Board — all	Bring the physical DQB to the front of the room (or display it on screen	DQB CLOSURE AS METACOGNITIVE AUDIT:	Observable indicators: (a) The class can classify at least 70% of

Board (DQB) Creation 📹 **VIDEO:** [The force of tension](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	question cards posted since Lesson 1. As a whole class, learners call out their classification for each question: ANSWERED (green), PARTIALLY ANSWERED (yellow), or BEYOND THIS UNIT (blue). Students justify their classification with a one-sentence reason. Questions classified as BEYOND THIS UNIT are physically moved to a 'Future Questions Wall' — a permanent display that will seed curiosity for upcoming sub-strands.	if digital). Point to the first question card. SAY: 'For each question on our board, we decide together: ANSWERED — we have strong evidence. PARTIALLY ANSWERED — we have some evidence but gaps remain. BEYOND THIS UNIT — this question needs more biology than we have learned so far. Raise your hand to vote.' WORK THROUGH THE BOARD systematically — spend no more than 45 seconds per card: — For ANSWERED cards: SAY 'Who can give the one-sentence answer? Cold-call one student. WAIT 10 seconds.' — For PARTIALLY ANSWERED cards: SAY 'What do we know, and what is the gap? Cold-call one student.' — For BEYOND THIS UNIT cards: SAY 'Which future topic might answer this? Any guesses?' Accept 2–3 hands. PHYSICALLY mark each card with a coloured sticker or marker: green = ANSWERED, yellow = PARTIALLY ANSWERED, blue = BEYOND THIS UNIT. Move BEYOND THIS UNIT cards to the 'Future Questions Wall.' SAY: 'The questions on our future wall are not failures — they are invitations. Some of them will be answered in Sub-Strand 1.3, some in Strand 2, some you may not encounter until university or research. That is what biology looks like: a discipline that is still	Formally classifying questions as ANSWERED, PARTIALLY ANSWERED, or BEYOND THIS UNIT is a public act of intellectual honesty. It models how scientists work — acknowledging the boundaries of current evidence. Moving BEYOND THIS UNIT questions to a Future Questions Wall transforms unanswered questions from sources of frustration into sources of forward momentum. This directly embodies the NGSS practice of Asking Questions and Defining Problems.	DQB questions as ANSWERED with a correct one-sentence justification. (b) The driving question itself receives a full, accurate, multi-clause answer from the cold-called student — referencing both cell structure AND specialisation AND at least two named cell types. (c) BEYOND THIS UNIT questions generate genuine curiosity responses (students spontaneously name upcoming topics or ask follow-up questions). Red flag: If the driving question receives only a one-word or one-clause answer, the teacher does not correct — instead, opens to the class: 'Who can build on that? Three more words. What is the connection to shape?'

		being written.' POINT to the driving question posted since Lesson 1: 'Why do cells in our body look so different from each other — and how does the shape of a cell determine the job it can do?' SAY: 'Can we answer this now?' Cold-call one student for a final, full-sentence answer. WAIT 15 seconds.		
Model Building Phase 📹 **VIDEO:** The force of tension Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners complete a structured individual exit reflection in writing (5 minutes): 'One thing I now understand that I did not understand in Lesson 1 is… because… The evidence that changed my thinking was…' This reflection is collected by the teacher. Learners then sit the sub-strand summative assessment independently. The assessment tasks learners to: (a) label and describe a diagram of a specialised cell; (b) explain how cell structure supports function using a named Kenyan biological context; (c) compare plant and animal cells; (d) answer structured questions about wound healing and cell specialisation.	SAY: 'Before we close, I want one last individual piece of writing from you. This is for you — it is about your own thinking, not a grade. Complete this sentence in your notebook: One thing I now understand that I did not understand in Lesson 1 is… because… The evidence that changed my thinking was… You have 5 minutes. Be specific. Use the language of cell biology.' WAIT: 5 minutes silent writing. Collect exit reflections (these inform teacher reflection and future planning — see Teacher Reflection section). TRANSITION TO ASSESSMENT: SAY: 'Well done. You have worked hard across twelve lessons. The assessment you are about to do is an opportunity to show what you know — it is not a test of whether you are clever, it is a record of how far you have come. Clear your desks. Work independently and in silence. You have [insert time as per school policy] minutes.' DISTRIBUTE assessment papers. Circulate silently. Do not answer content questions during the	EXIT REFLECTION AS FINAL MODEL REVISION: The written exit reflection is the learner's final, personal model statement. Unlike the group co-construction in Phase 2, this is individual and permanent — it captures where each learner's understanding has genuinely arrived. Collecting these reflections gives the teacher diagnostic data for the next sub-strand. The summative assessment then provides formal evidence of whether the core SLOs have been met. Together, these two artefacts — the reflection and the assessment — constitute the full 'final model' of each learner's understanding.	Observable indicators: (a) Exit reflections name at least one specific cell type, one specific organelle, or one specific evidence activity — not generic statements. (b) During the assessment, learners write continuously without seeking peer help — indicating sufficient independent understanding. (c) In the assessment's wound healing question, learners sequence at least 3 correct cell types in the correct order with accurate structural details. Summative assessment data: Teacher analyses results by SLO — which specific outcomes (cell structure labelling, specialisation explanation, plant vs. animal comparison) show the strongest and weakest performance. This data directly informs the opening of Sub-Strand 1.3.

		assessment. If a learner appears stuck, point to the word wall or their notebook cover vocabulary sheet (if policy permits). AFTER ASSESSMENT: Collect papers. SAY — and mean it: 'In Lesson 1, you saw a cut finger and guessed. Today, you can explain it cell by cell. That is what learning biology looks like. Take that with you into the next sub-strand.'		

D. TEACHER REFLECTION

After collecting exit reflections and marking sub-strand assessments, the teacher should analyse results against three questions: (1) Which SLO showed the greatest growth from Lesson 1 to Lesson 12 — and what instructional moves drove that growth? (2) Which SLO showed the weakest final performance — and should be revisited at the opening of Sub-Strand 1.3? (3) How many learners could fully explain wound healing using 3 or more named cell types with accurate structural details? This number is your 'deep understanding' benchmark. Beyond assessment data, review the exit reflections for the quality of 'evidence that changed my thinking' sentences — if learners cite specific activities (microscopy, card sorts, diagrams) rather than vague statements, your lesson design successfully built cumulative, evidence-anchored understanding. If reflections are generic, revisit how evidence was recorded and revisited across the unit. Finally, note which DQB questions were classified as BEYOND THIS UNIT — these are your strongest bridge into Sub-Strand 1.3 (Chemicals of Life) and Strand 3.2 (Transport in Animals). Post these prominently as the unit transitions.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	The class co-constructed a complete, sequential, cell-by-cell explanation of wound healing on the board — from the moment of the cut to the closure of the skin — naming platelets, neutrophils, macrophages, fibroblasts, and epidermal cells with their shapes and key organelles. A minimum of 8 students contributed publicly via cold-call.
What did I learn?	All specialised cells in the body originate from the same organism and carry the same DNA, but different genes are expressed in different cells, producing different structures — and those structures directly determine function. The shape of a cell is not accidental: it is the physical expression of the job that cell is specialised to do. This is the complete answer to the driving question.
How does this explain the phenomenon?	The anchoring phenomenon — The Healing Cut — is fully explained: platelets (small, anucleate, sticky) aggregate and initiate clotting; neutrophils (multi-lobed nucleus, abundant lysosomes) engulf bacteria; macrophages (irregular, flexible membrane) engulf debris and dead cells; fibroblasts (elongated, extensive rough ER and Golgi) secrete collagen scaffolding; epidermal cells (tightly packed, active nucleus) divide and migrate to close the surface. Every cell's shape is a solution to a specific biological problem — and that is why cells in the same organism look so completely different from each other.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.